

# **DEEP LEARNING BASED AFFORDABLE WATER QUALITY MONITORING AND ACCURATE PREDICTION FOR SMART AQUACULTURE**

**A THESIS**

*submitted by*

**RASHEED ABDUL HAQ K. P.**

*for the award of the degree*

*of*

**DOCTOR OF PHILOSOPHY**



**DEPARTMENT OF ELECTRONICS & COMMUNICATION  
ENGINEERING  
NATIONAL INSTITUTE OF TECHNOLOGY PUDUCHERRY  
KARAIKAL – 609 609**

**MARCH 2024**

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**MARCH 2024**

*Dedicated to*

*Vision of my father and patience of my mother*

*Sacrifice of my wife and kids*

*Support from my brother, sisters, and family*

*Encouragement from my friends*

*Patience and perseverance of my research supervisor*

## **CANDIDATE'S DECLARATION**

I the undersigned hereby declare that the thesis entitled **DEEP LEARNING BASED AFFORDABLE WATER QUALITY MONITORING AND ACCURATE PREDICTION FOR SMART AQUACULTURE** is a bonafide record of the research work done by me in partial fulfilment of the requirements for the award of the degree of **Doctor of Philosophy** in Department of Electronics & Communication Engineering of the National Institute of Technology Puducherry, Karaikal during the period 12.07.2018 to 31.05.2023. The contents of this thesis, in full or in parts, have not been submitted by me to any other Institute or University for the award of any degree or diploma.

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## THESIS CERTIFICATE

This is to certify that the thesis entitled **DEEP LEARNING BASED AFFORDABLE WATER QUALITY MONITORING AND ACCURATE PREDICTION FOR SMART AQUACULTURE** submitted by **Rasheed Abdul Haq K. P. (EC18D1003)** to the National Institute of Technology Puducherry, Karaikal, for the award of the degree of **Doctor of Philosophy** is a bonafide record of research work carried out by him during the period 12.07.2018 to 31.05.2023 under my supervision. The contents of this thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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Date:

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# ABSTRACT

Aquaculture plays an important role in achieving global food security, especially considering the projected population increase to over 9.8 billion by 2050. Traditional harvesting methods have exceeded ecological limits, making aquaculture essential for meeting growing food demands while preserving marine biodiversity. However, effective aquaculture management requires comprehensive ecosystem monitoring and prediction, with water quality (WQ) being a critical factor influencing production and product quality. Traditional water quality monitoring (WQM) methods are expensive and time-consuming, posing challenges for small-scale farms. The advent of technologies such as the Internet of Things (IoT), artificial intelligence (AI), and data analysis has introduced smart aquaculture systems. Nevertheless, accurate water quality prediction (WQP) remains a challenge due to the complex interdependencies and non-linear characteristics of WQ parameters. To address these challenges, this research work investigates the application of deep learning (DL) models to propose cost-effective solutions for WQM and precise WQP in aquaculture. DL models, including long short-term memory (LSTM), gated recurrent unit (GRU), and convolutional neural network (CNN), excel at capturing the non-linear relationships present in aquaculture WQ parameters. The primary objective is to enhance prediction accuracy while reducing computation time, enabling efficient WQ management in aquaculture. Additionally, the proposed models strive to offer economically viable solutions, making advanced technology in WQ management systems accessible and affordable.

This research is motivated by the pressing need for affordable, real-time, and accurate WQM and WQP systems in aquaculture. The research consists of interconnected works, each addressing specific aspects of WQM and WQP using DL methodologies. The first part of the thesis focuses on developing a cost-effective virtual sensing (VS) system for real-time WQM. Traditional methods relying on expensive physical sensors are impractical for small-scale farms. To overcome this challenge, the proposed deep learning enhanced virtual sensing (DLEVS) system leverages DL techniques to significantly improve the accuracy of the VS-based WQM system, reducing reliance on costly sensors. In the second work, LSTM, GRU, and their bidirectional variants are explored for precise WQP. Accurate prediction of WQ is crucial for effective aquaculture management. By leveraging the non-linear modeling capabilities of DL, these models aim to enhance prediction accuracy and provide valuable insights into future WQ conditions.

The third work reported in this thesis focuses on refining the accuracy and compu-

tational efficiency of WQP through the proposed hybrid DL models. Hyperparameter optimization is extensively studied to maximize model performance while minimizing computational requirements, facilitating real-time implementation. The fourth work introduces attention-based DL models to further enhance the accuracy and computational efficiency of WQP. These models employ attention mechanisms to concentrate on the most relevant features and parameters, resulting in improved prediction performance. A comprehensive study of model performance with different hyperparameters is conducted to identify the optimal configuration that balances accuracy and computational resources.

The research works reported in this thesis contribute to the integration of advanced DL techniques into WQ management in aquaculture. By addressing challenges related to cost-effectiveness, prediction accuracy, and computational efficiency, this research work enhances the sustainability of aquaculture.

*Keywords: Aquaculture, IoT, virtual sensing, water quality monitoring, water quality prediction, deep learning, attention-based models.*

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*Rasheed Abdul Haq K. P.*

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## ABBREVIATIONS

|          |                                                                         |
|----------|-------------------------------------------------------------------------|
| A-GRU    | Attention-based GRU                                                     |
| A-LSTM   | Attention-based LSTM                                                    |
| ABD-RNN  | Attention-based Bidirectional Deep Learning Recurrent Neural<br>Network |
| ABi-GRU  | Attention-based Bidirectional Gated Recurrent Unit                      |
| ABi-LSTM | Attention-based Bidirectional Long Short Term Memory                    |
| ADAK     | Agency for Development of Aquaculture                                   |
| AI       | Artificial Intelligence                                                 |
| ANFIS    | Adaptive Neuro Fuzzy Inference System                                   |
| ANN      | Artificial Neural Network                                               |
| ARIMA    | Auto Regressive Integrated Moving Average                               |
| ARMA     | Auto Regressive Moving Average Model                                    |
| BG       | Blood Glucose                                                           |
| Bi-GRU   | Bidirectional Gated Recurrent Unit                                      |
| Bi-LSTM  | Bidirectional Long Short Term Memory                                    |
| CNN      | Convolutional Neural Network                                            |
| DCS      | Distributed Control System                                              |
| DL       | Deep Learning                                                           |
| DL-RNN   | Deep Learning Recurrent Neural Network                                  |
| DLEVS    | Deep Learning Enhanced Virtual System                                   |
| DO       | Dissolved Oxygen                                                        |
| DQN      | Deep Q-Network                                                          |
| ELM      | Extreme Learning Machine                                                |
| GRU      | Gated Recurrent Unit                                                    |
| HWES     | Holt Winter's Exponential Smoothing                                     |
| IMDA     | Integrated Multiple Directed Attention                                  |
| IoT      | Internet of Things                                                      |
| IMDA     | Integrated Multiple Directed Attention                                  |

|         |                                                                              |
|---------|------------------------------------------------------------------------------|
| LS-SVM  | Least Square Support Vector Machine                                          |
| LSTM    | Long Short Term Memory                                                       |
| MAE     | Mean Absolute Error                                                          |
| MAPE    | Mean Absolute Percentage Error                                               |
| ML      | Machine Learning                                                             |
| MLRM    | Multiple Linear Regression Model                                             |
| MSE     | Mean Squared Error                                                           |
| NN      | Neural Network                                                               |
| pH      | Potential of Hydrogen                                                        |
| RMSE    | Root Mean Squared Error                                                      |
| RNN     | Recurrent Neural Network                                                     |
| SARIMA  | Seasonal Auto Regressive Integrated Moving Average Model                     |
| SARIMAX | Seasonal Auto Regressive Integrated Moving Average with<br>Exogenous Factors |
| SVR     | Support Vector Machine                                                       |
| VAE     | Variational Auto Encoder                                                     |
| VS      | Virtual Sensing                                                              |
| WQ      | Water Quality                                                                |
| WQM     | Water Quality Monitoring                                                     |
| WQP     | Water Quality Prediction                                                     |

# **CHAPTER 1**

## **INTRODUCTION**

Aquaculture, an essential component for achieving global food sustainability, has gained increasing prominence as the world's population is projected to exceed 9.8 billion by 2050 [1]. Traditional harvesting from natural resources, especially the ocean, is currently beyond ecological limits and is incapable of meeting escalating food demand [2]. Thus, aquaculture emerges as an efficient method for protein production, contributing to food security while attempting to preserve marine biodiversity [3]. However, the environmental impact associated with aquaculture necessitates comprehensive ecosystem management to ensure sustainability.

Water quality monitoring (WQM) and water quality prediction (WQP) play a vital role in managing aquaculture, as water quality (WQ) is the most influential factor affecting both the production and product quality [4]. Factors such as fish density, quality of the feed, feeding interval, and climate, can affect WQ, potentially triggering problems like stress, reduced food intake, increased vulnerability to diseases, and higher fish mortality rates [5, 6]. Traditionally, WQM involves collecting water specimens for lab testing, a time-consuming process that can delay critical decision-making [7]. Furthermore, most available WQM systems, which often use costly probes, are unaffordable for small-scale farms that significantly contribute to overall production [8, 9]. Some research works have focused on reducing the cost of WQM systems by minimizing sensor costs [10–12], which has proved beneficial for farmers by improving aquaculture production and sustainability.

Recent advancements in technology have introduced smart aquaculture systems by integrating IoT, artificial intelligence, and data analytics [13, 14]. However, accurate WQP remains a challenge due to the non-linear characteristics of WQ parameters and their complex interdependencies [15]. Conventional WQP approaches with machine learning (ML) techniques such as autoregressive moving average model (ARMA), autoregressive integrated moving average (ARIMA), seasonal autoregressive integrated moving average model (SARIMA), seasonal auto-regressive integrated moving average with exogenous factors (SARIMAX), Holt-Winters exponential smoothing (HWES), Markov model, grey model, and support vector regression (SVR) have limitations. These models, assuming a linear relationship with

WQ parameter data, are less accurate and take longer prediction time, making them unsuitable for accurately predicting non-linear aquaculture WQ parameters [16–18].

Deep learning (DL) models like long short-term memory (LSTM), gated recurrent unit (GRU), and convolutional neural network (CNN) overcome these limitations and capture the non-linear nature of aquaculture WQP effectively [19]. These models have achieved success in various applications, making time series prediction more precise and efficient in terms of training time and required processing power [20].

In this context, this research explores various DL models and aims to propose innovative ones, specifically tailored for accurate, low-cost WQM and precise WQP in aquaculture. The focus of this research is to propose DL models which can enhance prediction accuracy, while concurrently reducing the computation time. This directly addresses the urgent necessity for efficient WQ management in aquaculture, facilitating improved performance and reliability of WQM as well as WQP systems. Additionally, these models aim to offer an economically viable solution, reducing the overall cost of the WQM system, thereby ensuring affordable access to advanced and technology assisted WQ management.

## **1.1 MOTIVATION AND MAJOR OBJECTIVES**

The inspiration driving this research is the need for affordable, real-time, and accurate WQM and WQP systems in aquaculture. Traditional monitoring methods are not sustainable, and they need to be more efficient in dealing with the dynamic complexities of the ecosystem. In parallel, the field of DL, a subset of artificial intelligence, has demonstrated potential in managing complex, dynamic, and non-linear systems. It is thus compelling to explore and adapt these techniques to the unique challenges of WQ management in aquaculture.

The objectives of this research, reflecting the primary motivation, are structured as a series of interconnected works:

1. Development of a cost-effective virtual sensing (VS) system for real-time WQM, reducing dependence on costly physical sensors. This involves building a deep learning enhanced virtual sensing (DLEVS) system that harnesses the potential of DL to enhance the accuracy of VS-based WQM systems significantly.
2. Exploration of the potential of LSTM, GRU, and their bidirectional variants in implementing a WQP system, with an emphasis on improving prediction accuracy.
3. Further refinement of the accuracy and computational efficiency of WQP

through the proposal of hybrid DL models. This work extensively studies the impact of hyperparameters ( $h_p$ ) in optimizing these models for peak performance.

4. Proposal of attention-based DL models to further enhance accuracy and computational efficiency of WQP. This work also includes an extensive study of the model performance with tuning of  $h_p$ .

Collectively, these objectives aim to integrate advanced DL methodologies into aquaculture WQ management, striving for improved accuracy, efficiency, and sustainability. The proposed research intends to develop and implement DL models that can accurately predict WQ parameters in real-time, thereby enhancing the sustainability of aquaculture operations. These models are designed to perform efficient and affordable WQM as well as accurate WQP.

## 1.2 CONTRIBUTIONS

The research conducted in this thesis makes several significant contributions to the field of aquaculture WQM and WQP using DL techniques. These major contributions are as follows:

- Development of a cost-effective virtual sensing (VS) system for real-time WQM: A deep learning enhanced virtual sensing (DLEVS) system is proposed to address the challenges of costly physical sensors in WQM. By harnessing the power of DL, the DLEVS system significantly enhances the accuracy of VS-based WQM, providing a cost-effective solution for real-time monitoring of WQ parameters.
- Implementation of LSTM, GRU, and their bidirectional variants for WQP: The thesis investigates the potential of these DL models for accurate WQP in aquaculture. By leveraging the non-linear modeling capabilities of LSTM and GRU, the models improve prediction accuracy, providing valuable insights into future WQ conditions.
- Refinement of WQP accuracy and computational efficiency through hybrid DL models: The research proposes hybrid CNN-LSTM/GRU models for WQP, aiming to optimize the performance of the models. Extensive studies on hyperparameters are conducted to enhance the accuracy and computational efficiency of the models, making them more practical for real-time implementation.
- Proposal of attention-based DL models for enhanced WQP: The thesis presents attention mechanisms into the DL models for further enhancing the prediction

accuracy and computational efficiency in WQP. By focusing on the most relevant features and parameters, the attention-based models improve the prediction performance, ensuring more accurate and efficient WQP.

Overall, the contributions of this research include the development of a cost-effective VS and DL based WQM system, exploration and implementation of DL models for accurate WQP, refinement of prediction accuracy and computational efficiency through hybrid models, proposal of attention-based models, and integration of advanced methodologies for WQ management. This research work integrates the DL models into aquaculture WQ management. These contributions provide valuable insights for the development of sustainable aquaculture practices through smart WQM and WQP.

### **1.3 THESIS ORGANIZATION**

Chapter 1 introduces the thesis by presenting the motivations behind the research and outlining the main objectives aimed to achieve. This chapter also highlights the key contributions of the thesis. With the aforementioned research motivation and objectives, this thesis investigates the development of various approaches to address WQM and WQP in aquaculture using DL techniques. The subsequent chapters of the thesis are organized and presented as follows:

Chapter 2 provides a comprehensive literature review on WQM and WQP in aquaculture. The chapter categorizes existing algorithms and methodologies, discusses their limitations, and analyses performance measures commonly used in this field. Additionally, an overview of relevant databases and the performance metrics used in the research are provided.

Chapter 3 focuses on the development of a VS based WQM system using DL techniques. The proposed methodology, including VS and DL is explained in detail. The chapter describes the experimental setup, including the hardware and software used, and presents the results and discussions based on the ADAK and MAC datasets.

Chapter 4 introduces DL models for WQP in aquaculture. Specifically, the LSTM, GRU, and bidirectional LSTM/GRU DL models are proposed. The chapter presents the experimental setup, including details on the conducted experiments, and provides results and discussions based on the ADAK as well as MAC datasets.

Chapter 5 focuses on the development of hybrid CNN-LSTM/GRU models for WQP. The proposed models and their implementation details are presented in this chapter. The experimental setup, experiments conducted, and results and discussions

based on the ADAK as well as MAC datasets are provided.

Chapter 6 explores attention mechanism-based DL models for WQP in aquaculture. The proposed attention-based bidirectional recurrent neural network (ABD-RNN) models, which include the bidirectional LSTM/GRU DL layer and attention layer, are discussed in detail. The chapter presents the experimental setup, describes the conducted experiments, and provides results and discussions based on the ADAK and MAC datasets.

Chapter 7, summarizes the entire dissertation and provides a comprehensive overview of the contributions made to the research. It also discusses the limitations of the accomplished work and suggests potential future research directions.

By adhering to this organizational structure, the dissertation thoroughly examines DL based WQM and WQP systems for aquaculture. The chapters cover various aspects, including VS, DL based WQP models such as hybrid models, and attention-based models. The dissertation aims to contribute to aquaculture by providing accurate and efficient WQ management solutions.



## **CHAPTER 2**

### **LITERATURE STUDY**

This chapter provides a comprehensive literature survey on WQM and WQP in aquaculture. It emphasizes the significance of WQM and WQP in mitigating environmental effects and optimizing fish growth. This literature survey explains WQM and various prediction models highlighting the limitations of traditional approaches and the potential of DL techniques. Attention mechanisms and hybrid models are discussed for enhancing prediction accuracy. The survey concludes by emphasizing the importance of accurate WQP in high-intensity aquaculture and the potential of attention-based DL models for improved WQP performance. Section 2.1 of this chapter provides literature survey on WQM and Section 2.2 provides literature survey on WQP. Section 2.3 give details on the datasets. The performance metrics are detailed in Section 2.4.

#### **2.1 WATER QUALITY MONITORING IN AQUACULTURE**

Aquaculture is evolving towards intensive controlled environment production, giving a substantial improvement in production. This WQM system in [8] has a set of sensors, gateways and computer. Sensors are used to monitor WQ parameters and water level, effectively reducing the risk of mortality by ensuring continuous WQ monitoring. The system can detect and control temperature, dissolved oxygen (DO) content, pH value and WQ parameters in real-time. Authors used LabView to display information and ZigBee for communication. This work has an extensive study for six months with a study on battery and communication performance. This [21] work shows how a cabled system changes into wireless sensor networks and further studies the challenges faced for energy savings and reliable data transmission across the wireless sensor network. Different topologies are tested and compared to data transmission loss and energy efficiency. Finally, tree topology for WSN is adopted to decrease packet loss rate and improve the reliability of data transmission. Reorganising data frames and sleeping of sensor nodes are the two techniques used for Energy saving in this work.

In [22] a system is developed to maintain aquariums properly with ensured WQ. This system uses sensors and actuators to drive the actuators in real-time. Thus the fish is neither over nor underfed, and at the same time, the fish owner can enjoy watching fish feeding remotely. Another work [23] explains the implementation of a robotic device to monitor WQ and feed crayfish. The robotic device consists of a mobile robot

to hold a multiparameter sensor and six food delivery containers. These works explain the working of a WQ monitoring system for fish farming.

This [24] is a case study of a simple WQM system using open-source hardware (Arduino board) and sensors for WQM in aquaculture. The work demonstrates WSN technology in a distributed monitoring system with Arduino as the communication coordinator. The sensors send the information using the Xbee transmitter to the cloud with Arduino or computer acting as the gateway. The Arduino can decide the time to collect information from each sensor and coordinates transmission through the Xbee module. The information is stored and uploaded to the cloud using web service. The mobile application can take data from the cloud using web service for displaying the data.

Another work [25] introduced a novel low-cost compact and robust autonomous buoy system. Commercial WQM systems are costly and unaffordable for small organizations. This work describes the low-cost WQM system fit for use within a near-shore aquaculture farm or bathing waters. The system includes sensors for detecting variations in near-shore physical, chemical and biological WQ. The design is in such a way that this can be easily used by non-trained staff, and provide quality data. The system collected data from two deployments for fourteen months, one in an estuary and other in a coastal bay. These tests demonstrated the robustness of the system and suitability of these sites for aquaculture.

The comparison of the acquired sensor data and independent measurements showed good agreement between them. The paper [26] shows another application of WSN. The focus is on environmental monitoring using a system based on Raspberry Pi and Arduino. The architecture has sensing nodes which act as WSN and uses the ZigBee protocol for data transmission. The base station coordinates the sensor nodes, collects information and updates the database. This information is then uploaded to the internet and can be accessed using cell phones and computers.

This paper [27] provides a real-time WQM system. The Dong Lake in National Dong Hwa University was chosen as the test field to monitor water temperature, turbidity, conductivity, and pH. Various WQ data is updated to the database using wireless transmission, and the website displays the same information. For the long-range data transmission in the field, it uses LoRaWAN (Long Range Wide Area Network) [28]. Also, to extend system life, they use a solar panel with a rechargeable battery.

This [9] presents an online WQM system for intensive fish culture in China, and it has combined web-server-embedded technology with telecommunication. They base it on historical data to design the system to forecast WQ with artificial neural network

(ANN) and control the WQ in time to reduce catastrophic losses. The model forecast DO for half an hour ahead, and then results were compared with experimental data. The results show that WQ information can be accurately acquired and predicted by using this established monitoring system. A work [29] have used narrowband Internet of Things(NB-IoT) for mealtime monitoring of WQ parameters. The system has stable operation, and data transmission is done real-time with accuracy. Another work [30] implements a WQM system for eel fish farming. In this system, sensors measure DO, pH, and temperature, and all data processing is done on a Raspberry Pi and also uses python for all programming. They have automated aeration in this system.

The existing systems were extensively studied. All these works focus on solving the problems with WQ monitoring and not in reducing the cost of the system for making it affordable to small scale farms. This [10] work focuses on reducing the cost by using low-cost sensors. Reducing the cost by using low-quality sensors will not help with the overall cost of monitoring the WQ as the farmers will have to replace the sensors periodically. This [11] studies the available sensors for WQ monitoring in detail. The performance of sensors for each parameter is studied and compared. The challenges of using various types of sensors were also studied.

Research work in [12] presents a WQM system for drinking water. Thorough research identifies low-cost sensors with reliability. These works on the reduction of cost gave direction for reduction of cost of our system. IoT is one of the rapidly rising fields for social and economic development [31]. The field of IoT is extending its wings in all the fields. In this paper [32], the author is studying progression in integrated on-chip open-source computers like Arduino, Raspberry-Pi. These technologies are reaching people with their utilisation in agriculture and aquaculture. They continuously monitor the WQ with sensors to ensure growth and survival and transfer the data to the farmer's mobile through the cloud. They are hence ensuring on-time precautionary steps to reduce the losses and boost productivity.

WQM is an essential aspect of environmental monitoring and control. Traditionally, WQM systems rely on physical sensors that can be expensive and require regular maintenance. In recent years, there has been an increasing interest in developing virtual sensing techniques that can provide accurate and reliable estimates of WQ parameters using data from low-cost, low-power sensors. Virtual sensing is an emerging concept in WQ monitoring that involves the use of computational algorithms to estimate WQ parameters based on other measurable parameters. This approach reduces the need for physical sensors, which can be costly and require frequent calibration.

In [33], authors have developed a DL based virtual soil moisture sensor for pre-

cision farming applications. The authors use LSTM networks to develop the virtual sensor, which is based on data acquired from other environmental sensors installed on the node. The performance of the proposed virtual sensor is compared to other learning-based approaches, and the results show that the proposed method is effective in estimating soil moisture.

Another work [34] presents a data-driven model based on deep dynamic features extracting and transferring methods for virtual sensing of free calcium oxide (f-CaO) content in clinker, which significantly determines the quality of final cement production. The proposed model uses large-scale unlabeled data collected from the process distributed control system (DCS) for feature extraction and a powerful regression model, the eXtreme Gradient Boosting (XGBoost), for output prediction. The proposed virtual sensor is verified by a dataset acquired from a real cement production process.

In [35], the authors propose a magnetoresistive sensor error compensation method using geometry-constraint contour scaling to improve the accuracy of sensitive geomagnetic signals. The proposed method uses a benchmark ellipsoid to determine the shortest distance between any measured single point and the benchmark ellipsoid. The calibration parameters are extracted to complete the single-point error compensation. Simulation and experiment results show that the compensation accuracy of the proposed method is improved by two orders of magnitude compared with the traditional method. This work shows the possibility of using an existing method to improve accuracy of another existing model or method.

In their work [36], the authors present the development of a novel low-cost sensor node for measuring air pollution in real-time using electrochemical and infrared sensors. The sensor node utilizes wireless communication systems such as LoRaWAN and Wi-Fi for long-range, low power communication. The authors also implemented a calibration method using multiple linear regression and ANN, which resulted in significant improvement in the accuracy of the sensor data. This work demonstrates the potential of using low-cost sensor technology and wireless communication systems for air quality monitoring in urban environments.

In [37], authors propose a concept of virtual sensing for hydrogen sulfide ( $H_2S$ ) gas detection that can augment a single sensing platform into a virtual array, thus increasing the detection accuracy. The authors use a new feature extraction method based on fractional derivatives to enhance the model's learning capabilities. An array of four gas sensors is fabricated to record and analyse the signature of  $H_2S$  at various concentration levels, and a shallow neural network (NN) model is trained and tested

on the recorded data. The proposed model is highly sensitive and reliable with the ability to detect the gas instantaneously.

In [38], the authors propose a novel methodology for solving the problem of a water distribution network contamination event, which includes determining the exact source of contamination, the contamination start and end times, and the injected contaminant concentration. The methodology is based on coupling a ML algorithm for predicting the most probable contamination sources in a water distribution network with an optimization algorithm for determining the values of contamination start time, end time, and injected contaminant concentration for each predicted node separately. Two slightly different algorithmic frameworks were constructed, which utilize the Random Forest algorithm for classification of top source contamination node candidates. Both algorithmic frameworks perform well and show robustness in determining the true source node, start and end times, and contaminant concentration.

In summary, the literature survey shows that virtual sensing techniques based on sensors and ML algorithms can provide low cost solutions for aquaculture WQM.

## **2.2 WATER QUALITY PREDICTION IN AQUACULTURE**

Aquaculture is an integral part of the agricultural economy and has undeniably established its crucial role in global food security [39]. Intelligent and precise aquaculture is essential in improving sustainability and reducing the carbon footprint of food production. In intensive aquaculture, WQ is one of the critical factors affecting fish production. The aquaculture WQ is influenced by many factors, such as pollution, climate, quality of fish feed, feeding interval, fish density, etc. Due to the extreme differences in WQ, bacterial and algae balances may be disrupted. This makes WQ a significant factor in affecting aquaculture production. Many aquaculture farms have established WQM systems to monitor WQ in real-time. This approach has helped in managing the WQ in aquaculture to some extent. Further improvement in WQ control can only be realized through prediction, allowing users to take action ahead of time to minimize the occurrence of adverse events. Therefore, aquaculture WQP research is fundamental and has always been crucial in managing the aquaculture WQ. A precise and real-time WQP is necessary to help farmers take measures required to regulate the WQ to make it suitable for high-density aquaculture. Also, with an accurate forecast and the appropriate WQ management, the usage of antibiotics and other chemicals can be minimized, making high-intensity precision aquaculture green, sustainable, and much more profitable [40, 17].

WQM and WQP play a critical part in mitigating the environmental effects of aquaculture production. The WQ parameters need to be maintained in the optimum condi-

tion to ensure healthy fish growth giving good yield, leading to economic benefits [5]. Traditional techniques require collecting samples and transporting them to the lab for further analysis or using test kits. These methods are inaccurate, consume time, require human resources, and make continuous monitoring impossible. However, WQ parameters can be monitored continuously using sensor-based systems, which have advantages like remote 24 x 7 monitoring, control, and real-time data availability. These advantages can incorporate data analysis and application of ML to find optimal WQ parameters for fish growth, optimal use of the resources and predicting the WQ parameters. Thus, maintaining the aquaculture WQ parameters in the optimum range and reducing the impact of aquaculture on the environment.

WQ is the most influential and critical factor that affects production as well as product quality in aquaculture. The vital WQ parameters are salinity, pH, DO, and temperature [4]. Aquaculture species are prone to variations in the WQ parameters. High and unexpected variations in DO, temperature, salinity, pH and other factors can directly result in the mortality of aquaculture species. A slight fluctuation from the ideal environment may trigger stress such as lack of food intake and vulnerability to infections [6, 41]. WQM and WQP are essential to maintain and manage aquaculture [15]. WQM and WQP allow preemptive measures, averting ecological imbalances and ensuring optimal conditions for fish growth. By doing so, aquaculture can witness enhanced productivity, leading to increased profitability and sustainability.

The WQP models earlier used are known as mechanism models, which consider physical, chemical, and biological factors impacting the WQ. The QUAL and QUAL2E based models are two types of mechanism models [42]. But they require an extensive collection of data, including hydrological and meteorological data. The ML-based models are the non-mechanism model that uses past data to predict future data. The classical time series forecasting has methods like ARMA, ARIMA, SARIMA, SARIMAX, HWES, Markov model, grey model, SVR [43–45]. But they have drawbacks such as insufficient accuracy, low generalization, and high computation time. These models assume linear interrelationships between the parameters in WQ data. They cannot handle data with non-linear relation between parameters like aquaculture WQ parameters [18].

The variations in aquaculture WQ parameters are non-linear with respect to time. This non-linear nature of aquaculture WQ makes accurate WQP by using statistical and shallow ML methods a challenging task. There are many conventional time series models for WQP, including auto regressive models and statistical models. [46–50]. Statistical models have strong predictive abilities for stationary time series but need

larger datasets [51–54]. However, all these models have some drawbacks in non-linear prediction, such as lower predictive accuracy, weak generalization capability, and increased time complexity. Hence, these models are not suitable for high-intensity aquaculture WQP [55].

The non-linear nature of the aquaculture WQ parameters has to be handled differently from the mechanical and statistical models. Recently several AI-based models have studied the forecasting of WQ parameters. The authors in [56] have used the SVR model, and the multiple linear regression model (MLRM) for the predicting river water temperature in a river in Karnataka, India. In another work, the authors have used SVR optimised using genetic algorithm to predict the DO and water-temperature [57]. ANN and adaptive neuro-fuzzy inference system (ANFIS), along with its wavelet models are used to predict salinity concentration of a river in Iran by the authors [58]. In another paper, Extreme Learning machine (ELM) models and their improved versions were applied for predicting DO [59]. The ANN, SVR and ELM based models showed promising results in WQP, but there were shortcomings. These models also exhibited poor learning of the complex non-linearity of WQ parameters.

Aquaculture WQ is affected by meteorological conditions and complex interdependency relations between different WQ parameters. Hence the change of WQ parameters exhibit non-linear characteristics, which results in low prediction accuracy. The time-series data shows periodic variations depending on the seasons and climatic conditions. Time series data is used for analysis and prediction in various fields such as the stock market, medical field, energy consumption, weather forecasting, solar radiation etc. [60–65]. A WQP model could describe the relationship between monitored data and trends in WQ parameters based on available data gathered by the WQM system and forecast changes in WQ for a specified period. There are also many shortcomings in conventional prediction approaches, such as weak forecasting precision, least stability etc. Advancements in computer processing power and efficiency have created new prediction models focused on artificial intelligence (AI). Traditional methods for estimating WQ mainly include: the Markov model based on fuzzy-set theory [66, 67], the Grey Markov chain model method [68, 69], the method of prediction of regression [70, 71], the method of time series [72, 73], methods of WQP model [74, 75], the ANN methods [76, 77], the least square support vector machine (LS-SVM) prediction method [78, 79], the combination prediction method [80]. These approaches offer an efficient method for forecasting WQ but are not suitable for the aquaculture environment. The aquaculture environment is affected by many factors such as complex inter-relationship between WQ parameters, non-linearity in the WQ changes etc. Hence these models result in poor prediction accuracy.

Recent advancements in artificial intelligence have made DL one of the most proficient methods, and DL is popularly applied in different fields. DL models consider the non-linear characteristics of WQ changes. DL can determine the essential features that can efficiently provide concise and reliable results [81]. The LSTM model is a widely used DL method for WQP for its remarkable performance in time series prediction [82, 83]. LSTM is an enhanced version of the recurrent neural network (RNN) [84]. Some reported research works using the LSTM model resolves the issues in conventional prediction methods. The authors have used LSTM to predict drinking WQ in [85] and in [86] for mariculture WQP. To solve the problems in aquaculture WQP mentioned above, along with LSTM [87] based DL-RNN, preprocessing and correlation analysis is applied to the data. Correlation analysis uses Pearson's correlation coefficient method. The LSTM based model is used to train and predict WQ parameters. To further enhance the performance, an efficient stochastic optimisation method [88, 89] is used. Adam combines advantage of AdaGrad [90] and RMSProp [91], which are two popular methods. Therefore an Adam optimised LSTM based accurate WQP system for aquaculture is designed.

DL-RNN models have received significant attention due to their ability to capture non-linear inter-dependencies. LSTM, GRU, and CNN models are the DL-RNN models with the ability to capture the non-linearity in the aquaculture WQ. More recently, DL has been widely applied to sequence data processing, time series prediction, and various fields involving the processing of text, image, and speech [20, 92–96]. Even though LSTM and GRU DL-RNN models have overcome the limitations of simple RNN models, they have issues maintaining long-term memory. The forgetting mechanism results in the exponential decay of information, which limit their capacity to capture long-timescale information [97–99]. Past information has the most influence on the prediction at the present step. In certain circumstances, the observation that had a substantial impact may have come long before the current time step. To store memory for the learned long-term dependencies, NNs must be able to keep memories for an extended period. According to recent research [100, 101], the capacity of LSTM models to extract long-term dependency data from past observations remains a crucial performance barrier. The study reported in [97] has theoretically shown that the typical LSTM does not have a long memory. Therefore, a critical issue that must be solved is creating a time-series prediction model capable of collecting and remembering these complex dependencies.

The bidirectional LSTM and GRU (Bi-LSTM and Bi-GRU) models process the data in both ways, as opposed to the standard DL-RNN models, which only process the data in one direction. Therefore, by conducting bidirectional non-linear mapping of the sequence, Bi-LSTM, and Bi-GRU accurately capture the sequence correlation

of data to provide more precise prediction results [102]. Additionally, Bi-DL-RNN models are prevalent in many domains like bio-medical, financial forecasting, electricity consumption and production, weather forecasting, solar radiation, air quality, etc and even for depression identification [102–109]. The use of deep Q-networks in solving optimization problems has been explored in previous research, such as in the case of the flexible job shop scheduling problem and multiobjective FJSP-CS [110]. These studies have demonstrated the effectiveness of deep Q-network (DQNs) [111] in improving efficiency in production and optimizing various objectives. Similarly, the application of DQNs in predicting WQ has shown promising results, indicating its potential in enhancing the accuracy and speed of WQ monitoring and management.

The DL models like LSTM, GRU, and CNN have the flexibility in capturing the non-linear nature of the aquaculture WQP [19]. These DL models have overcome the limitations of RNNs and achieved great success in various applications. Recent advancements in artificial intelligence have made DL one of the most proficient methods, and DL is popularly applied in different fields such as image processing, speech recognition, text prediction, etc. [94–96]. DL makes time series prediction more precise and efficient in terms of training time and required processing power. In addition, DL models consider the non-linear characteristics of WQ changes. LSTM is the widely used DL method for WQP due to its remarkable performance in time series prediction [20].

LSTM and GRU models are not very capable of maintaining long-term memory for time series prediction, especially for long sequences [97–99]. In time series forecasting, the prediction value at the current time step is impacted mainly by past observations. In some cases, the ones with a significant influence might have appeared at the time step long back from the current one. Hence the neural networks (NNs) must have the prolonged memory holding capacity to hold memory for the learned long-term dependencies. The ability of LSTM models to capture long-term dependencies information from historical observations is still considered a critical performance bottleneck, as indicated in recent studies [100, 101]. The work [97] has theoretically proven that standard LSTM does not have long memory from a statistical perspective. Hence, building a time-series prediction model capable of capturing and remembering complex dependencies is a crucial problem that needs to be addressed. Their inability to retain information from distant past steps hampers accurate long-term predictions.

In [99, 112], the authors have discussed how incorporating an attention module can improve the prediction accuracy of LSTM. Different works have combined the attention model to LSTM to capture the relevant data from long time series data, helping to improve its prediction performance. LSTM network with an attention mechanism has an adaptive decay rate of long-term memory. This decay rate is much lower than

the polynomial or exponential decay rate. In the work [113], the reading of the geo-sensor is predicted for a few hours by using multi-level attention-based RNN by feeding data from multiple sensors and meteorological data along with spatial information of these sensors. This work uses the attention mechanism to model the dynamic spatio-temporal relations within the sensors. A fusion module is designed to input the effects of external factors from different domains. The authors have tested their model on a WQ and air quality dataset.

The hybrid CNN-LSTM models also work similarly, helping the LSTM model capture the relevant data. In hybrid models, CNN with maxpooling is used to reduce the length of the input sequence, which is fed into the LSTM and GRU models. The hybrid models are not affected by exponential decay, as the CNN with maxpooling will input the key features and helps LSTM in forecasting with better accuracy. Thus the hybrid models can achieve higher accuracy in prediction. Hybrid models aim to build neither overfit nor underfit models with strong generalization ability and require less training time for fitting the models. The convolutional layer with maxpooling enhances the capacity of the LSTM and GRU models to learn and store the critical features in the non-linear WQ parameters. Thus convolution layer with maxpooling reduces exponential decay in LSTM and GRU, bypassing selective information to the LSTM/GRU layer using convolution and maxpooling [92, 114]. CNNs are faster by design since the computations in CNNs can happen in parallel, while RNNs need to be processed sequentially since the subsequent steps depend on previous ones. In [115], authors introduced the Quasi-Recurrent NNs that use some of the CNN components to imitate RNNs while speeding them up. CNN leads to various complexity reductions by focusing on key features, which consequently reduces the size of tensors. The incorporation of Maxpooling contributes to an additional decrease in training time. As a result, the training time of the hybrid models is effectively reduced.

CNN and its hybrid variations are used in many research work for time series prediction. DL models have been extensively used in many fields of time series prediction, including finance (stock price, cryptocurrency, and precious metal prediction) [116, 117]. The authors [118] have used the CNN-LSTM model for gold price prediction. They have proposed two simple hybrid models and compared the results with other DL models. The authors also compared LSTM models with the hybrid models and observed that hybrid models perform better. In [119] to predict blood glucose (BG) levels, authors have used multiple layers of CNN along with LSTM. This hybrid model has shown superior performance in BG prediction. In another work, the authors have used Bi-LSTM with CNN for air quality prediction [120]. They have done an extensive study with two datasets and shown results proving the model's capability for accurate air quality prediction. There have

been multiple works based on hybrid DL models for the prediction of air quality (PM<sub>2.5</sub>) based on CNN and LSTM models [121, 122].

The authors of [112] and [123] have described the impact of adding an attention module to increase the performance of the LSTM DL-RNN model. Various studies have used the attention-based LSTM DL-RNN model to extract the relevant information from large time-series data, enhancing prediction accuracy. An attention-based LSTM network features an adaptive long-term memory decay rate. This is much less than the memory's exponential decay rate in the LSTM DL-RNN model. The attention mechanism reduces exponential decay in LSTM and GRU by passing selective information to the LSTM/GRU layer. The attention mechanism effectively highlights the impact of the key features in the prediction model by assigning different weights to hidden layer elements of the NN. Attention mechanisms selectively focus on and extract the essential features of the input data, thus improving model performance.

Attention-based DL-RNN models have also been used for time-series prediction. The authors of the work [124] want to create a unified model to correct the imbalance and enhance the predictability of specific times. The authors have proposed a self-attention-based time-varying prediction model with a two-stage training method [124]. Extensive testing demonstrates that this model has better performance than the baseline DL-RNN models. In [125], authors suggest a versatile and practical DL-RNN model for predicting air pollution. To create the forecasting model based on the novel integrated multiple directed attention (IMDA) DL architecture, the authors first developed the variational autoencoder (VAE) and the attention mechanism. Results show that this model outperforms other DL models, including VAE, LSTM, GRU, Bi-LSTM, Bi-GRU, and ConvLSTM. Results also show significant improvement in the performance of air pollution prediction. In [126], authors suggest SAIFlu-Net, a self-attention-based forecasting model for regional influenza forecasting. The model uses a self-attention mechanism to identify similarities between the occurrence patterns and an LSTM network to extract time-series patterns for each location. The detailed comparisons of the proposed model to other forecasting models utilizing weekly regional influenza datasets reveal that it outperforms other baseline DL-RNN models.

In order to meet the requirement of high prediction accuracy for high-intensity aquaculture, attention-based bidirectional DL-RNN (ABD-RNN) models are proposed in this work. Bi-LSTM and Bi-GRU incorporate additional training layers that allow input data to be trained twice to improve the accuracy of predictions. In this work, the prediction accuracy and training time of the proposed attention-based LSTM and attention-based GRU (ABi-LSTM and ABi-GRU) models are compared

with LSTM, GRU, Bi-LSTM, Bi-GRU, attention-based LSTM (A-LSTM) and attention-based GRU (A-GRU) DL-RNN models for WQP in aquaculture.

**Table 2.1:** Water Quality Monitoring Approaches in Aquaculture

| Approach                                  | Results                                                                        | Strengths                                                       | Weaknesses                                                        |
|-------------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------|
| Sensor-Based Monitoring System [8]        | Effective monitoring of WQ parameters and water level, reducing mortality risk | Real-time monitoring and comprehensive 6-month study            | Dependency on battery life and communication performance          |
| Wireless Sensor Networks [21]             | Improved reliability of data transmission in aquaculture systems               | Tree topology decreases packet loss and energy-efficient design | Challenges in energy saving and data transmission reliability     |
| Aquarium Maintenance System [22]          | Real-time management of aquariums, ensuring proper WQ                          | Automated feeding and remote fish watching capability           | Limited to aquarium settings, not scalable for larger aquaculture |
| Robotic Device for WQ Monitoring [23]     | Effective monitoring and feeding in crayfish farming                           | Integrated multiparameter sensor and food delivery system       | High complexity and cost for small scale applications             |
| Open-Source Hardware System [24]          | Distributed WQM using Arduino and sensors                                      | Affordable, Scalable and cloud integration                      | Relies on stable internet connectivity and power supply           |
| Autonomous Buoy System [25]               | Long-term data collection in near-shore environments                           | Low-cost robust design and easy for non-trained staff           | Limited to specific environmental conditions                      |
| Raspberry Pi Based System [26]            | Environmental monitoring with Raspberry Pi and Arduino                         | Versatile, updatable system and internet connectivity           | Requires technical expertise for setup and maintenance            |
| LoRaWAN WQM System [27]                   | Real-time monitoring of various WQ parameters                                  | Long-range wireless transmission and solar-powered              | Limited to areas with LoRaWAN coverage                            |
| Online WQM for Intensive Fish Culture [9] | Real-time WQ monitoring and DO prediction                                      | Accurate WQ acquisition with ANN-based forecasting              | High initial setup cost, requires historical data                 |
| NB-IoT for WQM [29]                       | Real-time and accurate WQ parameter monitoring                                 | Stable operation and real-time data transmission                | Dependency on IoT network availability                            |
| Low-Cost Sensor Monitoring [10]           | Cost-effective WQM for small scale farms                                       | Reduced costs with low-quality sensors                          | Frequent sensor replacement, reduced overall reliability          |

**Table 2.2:** Water Quality Prediction Approaches in Aquaculture

| <b>Approach</b>                                          | <b>Results</b>                                          | <b>Strengths</b>                                     | <b>Weaknesses</b>                                                       |
|----------------------------------------------------------|---------------------------------------------------------|------------------------------------------------------|-------------------------------------------------------------------------|
| SVR and MLRM Models [56]                                 | Prediction of river water temperature                   | Suitable for specific parameters like temperature    | Limited in handling complex interdependencies in WQ parameters          |
| Genetic Algorithm Optimized SVR [57]                     | Predicts DO and water temperature                       | Enhanced accuracy through optimization               | Requires genetic algorithm knowledge for effective implementation       |
| ANN and ANFIS Models [58]                                | Prediction of salinity concentration in a river         | Handles non-linear data effectively                  | Complex model structures, requiring substantial computational resources |
| ELM Models [59]                                          | Prediction of Dissolved Oxygen                          | Improved prediction accuracy                         | May not generalize well to different types of WQ parameters             |
| LSTM Models [85, 86]                                     | Prediction of drinking and mariculture water quality    | Effective in time series prediction                  | Challenges in maintaining long-term memory                              |
| Adam Optimized LSTM [88, 89]                             | Accurate WQP system for aquaculture                     | Combines advantages of AdaGrad and RMSProp           | Complexity in implementation and optimization                           |
| Bidirectional LSTM and GRU Models [102]                  | Accurate sequence correlation capture for WQ prediction | Processes data in both directions for precision      | Higher computational demands compared to unidirectional models          |
| Hybrid CNN-LSTM Models [92, 114]                         | Enhanced time series prediction accuracy                | Reduction in training time, focusing on key features | May require fine-tuning to balance CNN and LSTM components              |
| Attention-Based LSTM Models [99, 112]                    | Improved prediction accuracy with attention mechanism   | Adaptive decay rate of long-term memory              | Increased model complexity and computational needs                      |
| Self-Attention-Based Time-Varying Prediction Model [124] | Enhanced predictability for specific times              | Addresses imbalance in conventional models           | May not be suitable for all types of WQ prediction scenarios            |

The research in this thesis effectively addresses the key challenges in WQM and WQP, as identified in previous studies through a literature survey, as presented in Tables 2.1 and 2.2. Traditional WQM methods are costly and labor-intensive, unsuitable for small-scale farms, and existing WQP approaches often fail to accurately predict

complex, non-linear water quality parameters. This thesis introduces a Deep Learning Enhanced Virtual Sensing (DLEVS) system for WQM, offering a cost-effective and accurate solution by reducing reliance on expensive physical sensors. For WQP, it utilizes advanced DL models such as LSTM, GRU, their bidirectional variants, and CNN to precisely capture non-linear water quality relationships. The research further innovates with hybrid DL models and attention-based DL models, enhancing prediction accuracy and computational efficiency. This strategic approach not only addresses the limitations of past methodologies but also significantly advances aquaculture technology, promoting sustainable and efficient management practices.

## **2.3 DATASETS**

### **2.3.1 ADAK water quality dataset**

The first dataset used in this research work is the aquaculture WQ data collected from aquaculture farms under ADAK, Kerala. The dataset includes WQ parameters such as salinity, pH, DO, and temperature. As part of this work, the data underwent preprocessing and cleansing. Spanning a three-year period from January 2016 to December 2018, this dataset consists of daily observations collected at a consistent time, resulting in a total of 1096 samples. In terms of size, it is considered a medium-sized dataset when compared to the MAC dataset employed in this research.

### **2.3.2 MAC water quality dataset**

The second dataset is sourced from a publicly available research work [17]. This dataset is acquired from the marine aquaculture base located in Xincun Town, LingShui County, Hainan Province, China. Like the previous dataset, it includes essential WQ parameters such as salinity, pH, DO, and temperature. Data collection occurs at 5-minute intervals. In total, this dataset comprises 23,200 samples gathered over a span of 80 days. In comparison to the ADAK dataset, this is classified as a large dataset.

## **2.4 PERFORMANCE METRICS**

The performance of the prediction models are evaluated using mean absolute error (MAE), mean square error (MSE), root mean squared error (RMSE) and mean absolute

percentage error (MAPE), computed by the set of Eq. (2.1) given below:

$$\begin{aligned}
MAE &= \frac{1}{n} \sum_{i=1}^n |A_i - Y_i| \\
MSE &= \frac{1}{n} \sum_{i=1}^n (A_i - Y_i)^2 \\
RMSE &= \sqrt{\frac{1}{n} \sum_{i=1}^n (A_i - Y_i)^2} \\
MAPE &= \frac{1}{n} \sum_{i=1}^n \frac{|(A_i - Y_i)|}{A_i}
\end{aligned} \tag{2.1}$$

where  $A_i$  is the actual value of  $i_{th}$  sample,  $Y_i$  is the predicted value of  $i_{th}$  sample and  $n$  is the number of samples.

The chosen metrics, namely MAE, MSE, RMSE, and MAPE, indeed provide different interpretations of model performance. MAE and MSE are measures of average error, with MSE placing greater emphasis on larger errors due to its squared term. RMSE, being the square root of MSE, maintains this emphasis but in the same unit as the data, making it more interpretable. MAPE, expressed as a percentage, offers a scale-independent perspective, making it particularly useful for comparative analysis across different datasets or models.

Regarding their ability to represent both global and local performance, these metrics primarily provide a global summary of the model's performance. However, they can indirectly indicate local discrepancies. A high MSE, for example, may suggest not only an overall inaccuracy but also the presence of significant errors at specific points in the time series.

## 2.5 LOSS FUNCTION

The primary loss function utilized in this study was the MSE, a widely adopted metric for regression problems, including those within the domain of water quality prediction. MSE proves effective in emphasizing and penalizing larger prediction errors more significantly than smaller ones. This characteristic ensures that models exhibit heightened sensitivity to substantial deviations in predictions, a critical aspect for achieving high accuracy in forecasting water quality parameters.

The selection of MSE as the loss function aligns with the objective of minimizing large errors, thereby enhancing the model's predictive accuracy and reliability. MSE quantifies the average of the squares of the errors, representing the average squared difference between the estimated values and the actual value. This feature makes MSE particularly suitable for models where mitigating large errors is of utmost importance.



## **CHAPTER 3**

# **DEEP LEARNING BASED VIRTUAL SENSING FOR AFFORDABLE WATER QUALITY MONITORING IN AQUACULTURE**

This chapter presents an approach that uses VS and DL methods for real-time WQM. The fusion of the cost-effectiveness of VS with the accuracy of DL results in a cost-efficient yet precise solution for real-time WQM. This methodology is rigorously tested using two separate datasets, ADAK and MAC, demonstrating its efficacy and promising capabilities. In this approach, the intent is to replace high-priced DO sensor with virtual reducing the overall cost of the WQM system. Section 3.1 of this chapter serves as an introductory section, outlining the key contributions of this work. Section 3.2 of this chapter explains the proposed DL enhanced VS methodology for WQM. Section 3.3 explains in details the experimental setup for testing the proposed methods. Section 3.4 presents the final results and comparisons of the VS and DLEVS. The chapter is summarized in Section 3.5.

### **3.1 INTRODUCTION**

Effective management of aquaculture's impact on the environment requires a comprehensive understanding of the ecosystem. WQM provides a holistic view of the ecosystem and is essential for managing aquaculture's impact. Traditional manual inspections may not provide all the required information, and continuous monitoring using sensors is necessary to obtain extensive data that includes all factors affecting the farm [127]. Continuous monitoring provides extensive data that farmers cannot assess manually, enabling the control of the system to reduce the impact of aquaculture on the environment while improving product quality and production quantity.

WQM optimizes resource utilization, enhances sustainability, and reduces wastage, thereby increasing profit margins for farmers. The salinity, pH, DO, and temperature are crucial WQ parameters in aquaculture, and their variation can cause production problems [4]. Fluctuations in any of these parameters can result in significant production losses [128]. Additionally, water contamination due to urbanization and industrialization poses further challenges. DO affects the growth

rate and health of aquatic animals, feed utilization, and can lead to stress and disease when temperature changes abruptly or is chronically near the animals' tolerance levels [129–133].

The traditional means of WQM involve collecting water specimens and sending them to a lab, which can consume time and delay decision-making, impacting productivity [7]. Small-scale farms account for the majority of aquaculture production and can benefit significantly from becoming sustainable and efficient. However, available WQM systems are often not affordable to small-scale farms that contribute significantly to overall production. While some systems for WQM in aquaculture exist, many use probes that are costly [8, 9]. A few works have focused on reducing the cost of WQM systems, including reducing sensor costs [10] and conducting extensive studies on sensors to lower overall system costs [11, 12]. After a careful study, some parameters that can be calculated using other parameters have been identified, enabling the replacement of costly sensors. This system uses solubility equations to calculate DO in water and is modular, portable, and low-cost. It shares information with farmers promptly, enabling them to manage the farm efficiently. Reducing the cost of WQM systems will encourage more farmers to use them and improve overall aquaculture production and sustainability.

This work presents a VS based accurate low-Cost WQM System for Aquaculture, with the following major contributions:

1. Development of a WQM system with improved functionality and efficiency through the incorporation of solar energy harvesting and an energy-efficient scheduling approach. This WQM system served as the foundation and test bed for subsequent research on development of affordable WQM and accurate WQP systems.
2. The application of a novel DLEVS mechanism for aquaculture WQM is introduced, incorporating the strengths of LSTM DL-RNN and VS techniques.
3. LSTM model is developed to predict and compensate potential inaccuracies in DO estimation, which is a crucial component of the aquaculture industry.
4. The performance and efficiency of the DLEVS model is meticulously evaluated with the help of two WQ datasets. It is revealed that this model has the potential to significantly improve the accuracy and reliability of DO estimation in aquaculture environments.

## 3.2 PROPOSED METHODOLOGY OF AFFORDABLE WATER QUALITY MONITORING IN AQUACULTURE

### 3.2.1 Virtual sensing based water quality monitoring

VS techniques, also called soft sensing, proxy sensing, inferential sensing, or surrogate sensing, are used to provide feasible and economical alternatives to costly or impractical physical measurement instrument. This technique predicts parameters using mathematical models, which are not measured using physical sensors. A VS system uses information available from other measurements and process parameters to calculate an estimate of the quantity of interest. In this work, expensive DO sensor is replaced with solubility equations. In this study, three solubility equations for estimation of DO is investigated.

Weiss Equation [134]: Derived in 1970, Weiss introduced a set of equations to calculate DO as a function of temperature and salinity using the method of least squares to fit data collected on the solubility of nitrogen, oxygen, and argon in water and seawater. The DO at zero salinity, and one atmosphere is given in Eq. (3.1). The Weiss salinity factor is given in Eq. (3.2). The equation for calculating DO is given in Eq. (3.3).

$$DO_o = 1.42905 \exp \left[ -173.4292 + 249.6339 \left( \frac{100}{T} \right) + 143.3483 \left( \ln \frac{T}{100} \right) - 21.8492 \left( \frac{T}{100} \right) \right] \quad (3.1)$$

$$F_S = \exp \left[ S \times \left( -0.033096 + 0.014259 \left( \frac{T}{100} \right) - 0.0017 \left( \frac{T}{100} \right)^2 \right) \right] \quad (3.2)$$

$$DO = DO_o \times F_S \quad (3.3)$$

Benson and Krause Equation [135], [136]: Benson and Krause derived equations based on thermodynamic principles to compute the solubility of oxygen in the water. Their work resulted in equations that provide oxygen solubility as a function of temperature in freshwater and adjustments based on barometric pressure and salinity. Eq. (3.4) is used to compute the DO in mg/L at zero salinity. The Eq. (3.5) gives the salinity factor which is used to estimate the DO when salinity is more than zero. The equation for calculating DO is given in Eq. (3.6).

$$DO_o = \exp \left[ -139.34411 + \frac{1.575701 \times 10^5}{T} - \frac{6.642308 \times 10^7}{T^2} + \frac{1.243800 \times 10^{10}}{T^3} - \frac{8.621949 \times 10^{11}}{T^4} \right] \quad (3.4)$$

$$F_S = \exp \left[ -S \times \left( 0.017674 - \frac{10.754}{T} + \frac{2140.7}{T^2} \right) \right] \quad (3.5)$$

$$DO = DO_o \times F_S \quad (3.6)$$

Garcia and Gordon Equations [137]: Garcia and Gordon re-examined and refitted the data used in many previous works to a higher order polynomial to estimate the DO when salinity is more than zero. The Eq. (3.7) gives scaled temperature which is used in Eq. (3.8) and Eq. (3.9). The Eq. (3.9) gives the salinity factor which is used to

estimate the DO when salinity is more than zero. The equation for calculating DO is given in Eq. (3.10).

$$T_s = \ln \left[ \frac{298.15 - t}{273.15 + t} \right] \quad (3.7)$$

$$DO_o = 1.42905 \exp [2.00907 + 3.22014T_s + 4.05010T_s^2 + 4.94457T_s^3 - 0.256847T_s^4 + 3.88767T_s^5] \quad (3.8)$$

$$F_S = \exp [(-0.00624523 - 0.00737614T_s - 0.0103410T_s^2 - 0.00817083T_s^3) S - 4.88682 \times 10^{-7} S^2] \quad (3.9)$$

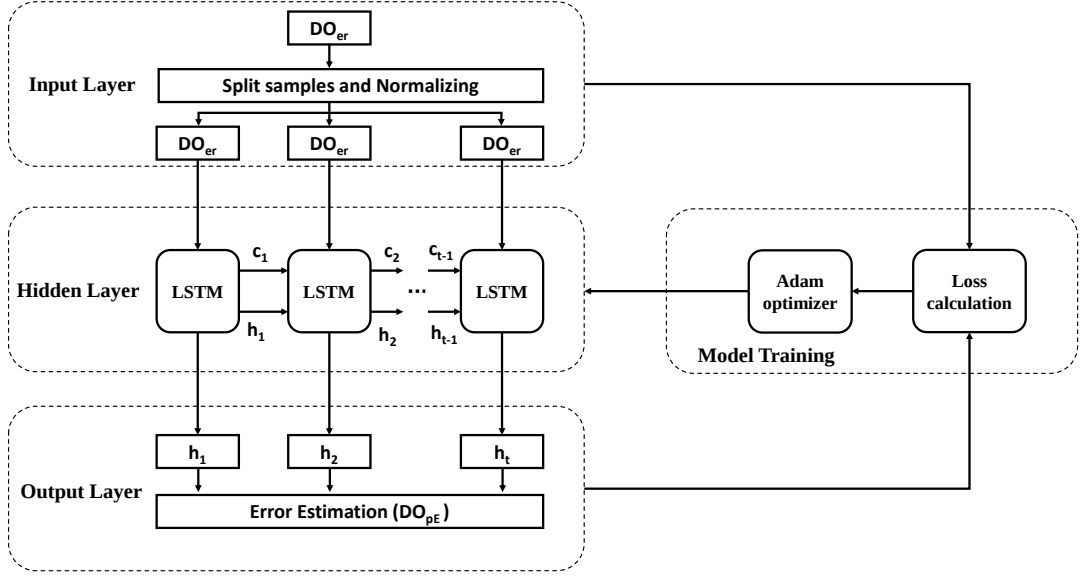
$$DO = DO_o \times F_S \quad (3.10)$$

### 3.2.2 Deep learning enhanced virtual sensing for water quality monitoring

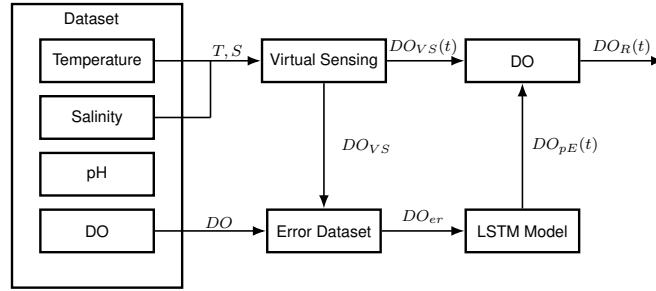
This section explains the use of DL to reduce the error in the VS based estimation of the DO using the solubility equation. DL, specifically an LSTM model, is utilized to improve the accuracy of the VS-based estimation of DO. This is achieved by predicting the error in VS using the LSTM model, then subtracting this error from the DO value estimated using VS, thus enhancing the accuracy of DO estimation. An error prediction model is initially developed, which predicts the error using WQ time sequence data collected via the WQM system. The predicted error is then used to reduce the error in VS and improve the accuracy of VS. The error at any point in time is calculated as the difference between the actual value and the VS values, as shown in Eq. (3.11).

$$DO_{er}(t) = DO_{vs}(t) - DO(t) \quad (3.11)$$

Eq. (3.11)  $DO_{vs}(t)$  is the estimated DO using solubility equations with temperature and salinity data at time  $t$ .  $DO(t)$  is the original value of DO in the data which is measured using the DO sensor at time  $t$ .  $DO_{er}(t)$  is the estimated error at time  $t$ . The model uses the error dataset to train the LSTM model and predict the error. Figure 3.1 shows how LSTM model is used predict the error in VS of DO. Here the error time sequence WQ dataset is used for training the LSTM model. For making the error prediction, the error in VS is estimated using the datasets and this error data is used to train the LSTM model. After training the LSTM model using the time sequence error data, the model predicts the error which is used to reduce the error in VS. The error predicted by the LSTM model is used for reducing the error in VS. Here the error in VS is minimized to make the system more accurate and reliable. LSTM model predict the error in the system, which can be feedback to the system to reduce the error during estimation of DO. Figure 3.2 shows the block diagram of the error reduction in VS using LSTM model. Here  $T$ ,  $S$  is the temperature and salinity values which are used for VS.  $T$  is the temperature dataset  $T(0)$  to  $T(t-1)$ ,  $S$  is the salinity dataset  $S(0)$  to  $S(t-1)$ ,  $DO$  is the DO dataset  $DO(0)$  to  $DO(t-1)$ ,  $DO_{er}$  is the error dataset  $DO_{er}(0)$  to  $DO_{er}(t-1)$ ,  $DO_{er}$  is given by the Eq. (3.11). The LSTM model predicts



**Figure 3.1:** Error prediction system based on LSTM model



**Figure 3.2:** Block diagram of DLEVS

the error in VS and is denoted as  $DO_{pE}$ . Predicted error  $DO_{pE}$  improve the accuracy of the VS by deducting the same from  $DO_{VS}$  to give  $DO_R$  as in Eq. (3.12).  $DO_R$  is the DO which is computed finally after using LSTM to reduce the error of the VS.

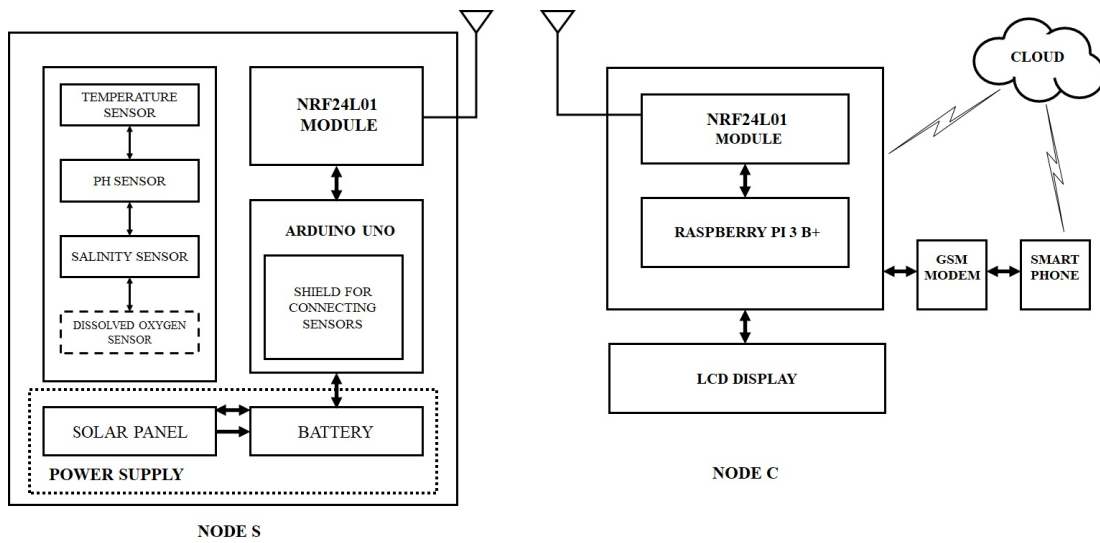
$$DO_R(t) = DO_{vs}(t) - DO_{pE}(t) \quad (3.12)$$

### 3.3 EXPERIMENTAL SETUP FOR TESTING THE PROPOSED METHODS

This section presents the characteristics of the proposed IoT-based WQM system for aquaculture. The entire system is a WSN-based system where Node S collects data from a distributed network of sensor nodes deployed on a wide field. Node S is a solar-powered sensing unit that collects different WQ parameter data and transmits them to a central node, Node C. The proposed virtual sensing is implemented in Node S, a low-cost and low-power device. The design of this system requires the selection of important WQ parameters since using more sensors will make the system costly. The key parameters are temperature, pH, and salinity. Using them, DO can be estimated by

virtual sensing.

A WQM system requires the development and integration of hardware and software components. Figure 3.3 shows the overall architecture of the developed WQM system. The first part of the system is Node S. Node S comprises the sensors responsible for detecting WQ parameters and converting them into electrical signals. Another component of Node S is the Microcontroller, which converts the electrical signals or voltage levels from sensors into data. Following Node S, there is Node C, tasked with coordinating all Node S units. It gathers data from each Node S, stores the collected data, and facilitates information communication. It serves as the central hub of the system, hence its designation as Node C.



**Figure 3.3:** WQM system architecture

### 3.3.1 Hardware

Node S and Node C constitute the two fundamental modules of the hardware. This system can accommodate a variable number of Node S units and one Node C responsible for controlling and gathering data from all the Node S units.

**Node S:** Node S is comprised of four subsystems: the sensor part, the microcontroller (Arduino Uno), the communication subsystem (NRF24L01), and the power supply. The WQM sensors incorporated in this node encompass temperature, pH, and salinity. To estimate DO, the Weiss equation is employed within the microcontroller. Each sensor can detect specific WQ parameters and convert them into electrical signals. The microcontroller executes programs that translate these electrical signals into corresponding WQ parameters. The microcontroller chosen for this purpose is the ARDUINO UNO REV3, which is built around the ATmega328P microcontroller chip, providing ease of use and compatibility with numerous shields

designed for Arduino Boards. The power supply is equipped with a 20 ampere-hour battery and an 18-watt solar panel to ensure continuous operation of the sensor node throughout the year, eliminating the need for frequent battery recharging or replacement. Table 3.1 lists the sensors employed in Node S.

**Table 3.1:** Sensors used for measuring the water quality parameters

| Sensor      | Units | Range     | Sensor Type                 |
|-------------|-------|-----------|-----------------------------|
| pH          |       | 0 - 14    | Gel-filled(Ag/AgCl)         |
| Temperature | °C    | -40 - 135 | Thermistor                  |
| Salinity    | ppt   | 0 - 50    | Parallel platinum electrode |
| DO          | mg/L  | 0 - 20    | Luminescent                 |

**Node C:** Serving as the core of the system, Node C handles the task of aggregating data from Node S units (retrieved via sensors, the Arduino microcontroller, and the NRF24L01 transceiver). It archives data for prolonged periods, analyses data, disseminates information to farmers, and uploads data to the cloud as required. The Raspberry Pi 3 B+ is utilized as the controller within Node C. It is interconnected with the NRF24L01 transceiver to retrieve data from Node S and to manage Node S operations.

### 3.3.2 Software

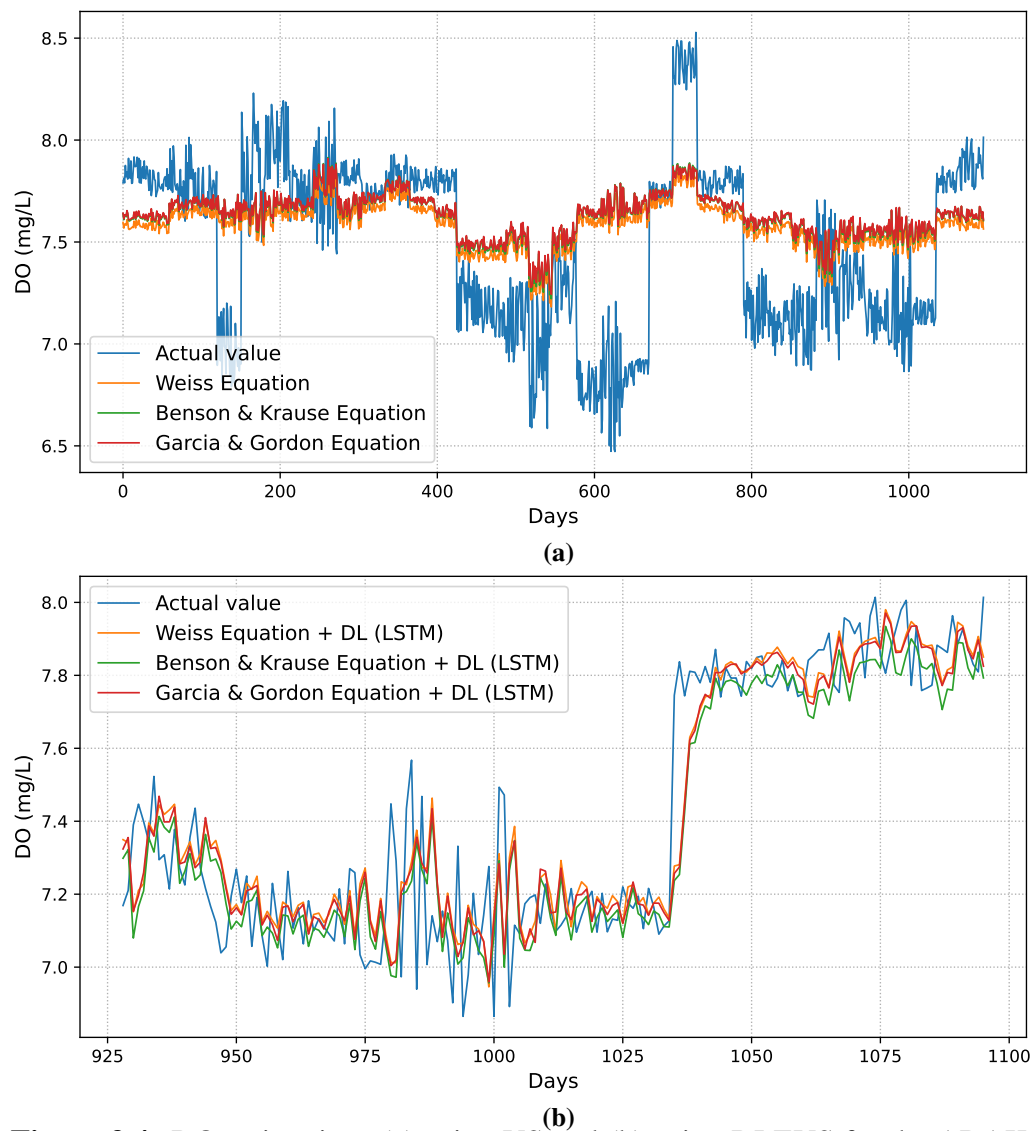
Once the hardware of the system is prepared, the configuration and installation of the necessary software, along with the upload of codes for proper operation of Node S, require the utilization of the Arduino IDE. Node C functions as a central unit, gathering all the data. The software aspect encompasses two components: one for Node S and the other for Node C. A dedicated custom shield is employed with the Arduino Uno to capture analogue signals from the sensors. A monitoring program, referred to as a sketch, identifies each probe by comparing its signature voltage range and subsequently converts the analogue electrical signal into meaningful data. The Arduino IDE facilitates the uploading of these sketches onto the Arduino Uno, enabling the sketches to read the sensors and collect data. Multiple sensors can be connected to a single board, enhancing versatility. The Raspberry Pi (RPi) board accumulates data from various Arduino boards. As the central node, the RPi is an optimal platform to implement decision support mechanisms. For instance, it determines critical DO levels that require immediate farmer notification, in addition to generating periodic reports.

The Arduino retrieves information from the sensor node and interacts with farmers via Node C, which serves as a communication gateway for various forms of communication. Each sensor gathers data at scheduled intervals, facilitated by Arduino. The Arduino establishes direct communication with the sensor through its analogue pins,

while communication with Node C is achieved via the NRF24L01 transceiver. Using data from temperature and salinity sensors, the Arduino employs the Weiss Equation to estimate the DO content in water. It processes the electrical signal received from the sensors to generate the necessary data. After acquiring data from Node S at specific intervals, the Arduino relays the information to farmers through Node C. The timing of these alerts is predefined within Node C. Acting as a bridge between Node C and sensors, the Arduino operates as a central communication hub. The architecture follows a modular design, facilitating system expansion. This adaptability makes the system suitable for deployment in both small and large farms.

### 3.4 RESULTS AND DISCUSSIONS

#### 3.4.1 ADAK water quality dataset



**Figure 3.4:** DO estimations (a) using VS and (b) using DLEVS for the ADAK water quality dataset

Figure 3.4a presents a comparison of DO values obtained from the proposed VS model with those gathered from the actual sensor for the ADAK dataset. The comparison demonstrates a close alignment between the predicted and actual values. Furthermore, Fig. 3.4b showcases how integrating the VS model with DL (resulting in a DLEVS system) can further enhance the predictions, bringing them into even closer proximity with the actual values. Table 3.2 offers a comprehensive comparison between the VS and DLEVS methods for various equations. It shows that the DLEVS method significantly enhances the accuracy of predictions across all metrics, outperforming the standalone VS method.

**Table 3.2:** DO estimation using VS and DLEVS for the ADAK water quality dataset

| Equation        | VS     |        |        |        | DLEVS  |        |        |        |
|-----------------|--------|--------|--------|--------|--------|--------|--------|--------|
|                 | MAE    | MSE    | RMSE   | MAPE   | MAE    | MSE    | RMSE   | MAPE   |
| Weiss           | 0.3038 | 0.1445 | 0.3802 | 0.0417 | 0.1163 | 0.0235 | 0.1534 | 0.0158 |
| Benson & Krause | 0.3050 | 0.1542 | 0.3927 | 0.0421 | 0.1157 | 0.0244 | 0.1561 | 0.0156 |
| Garcia & Gordon | 0.3071 | 0.1564 | 0.3955 | 0.0425 | 0.1143 | 0.0230 | 0.1516 | 0.0155 |

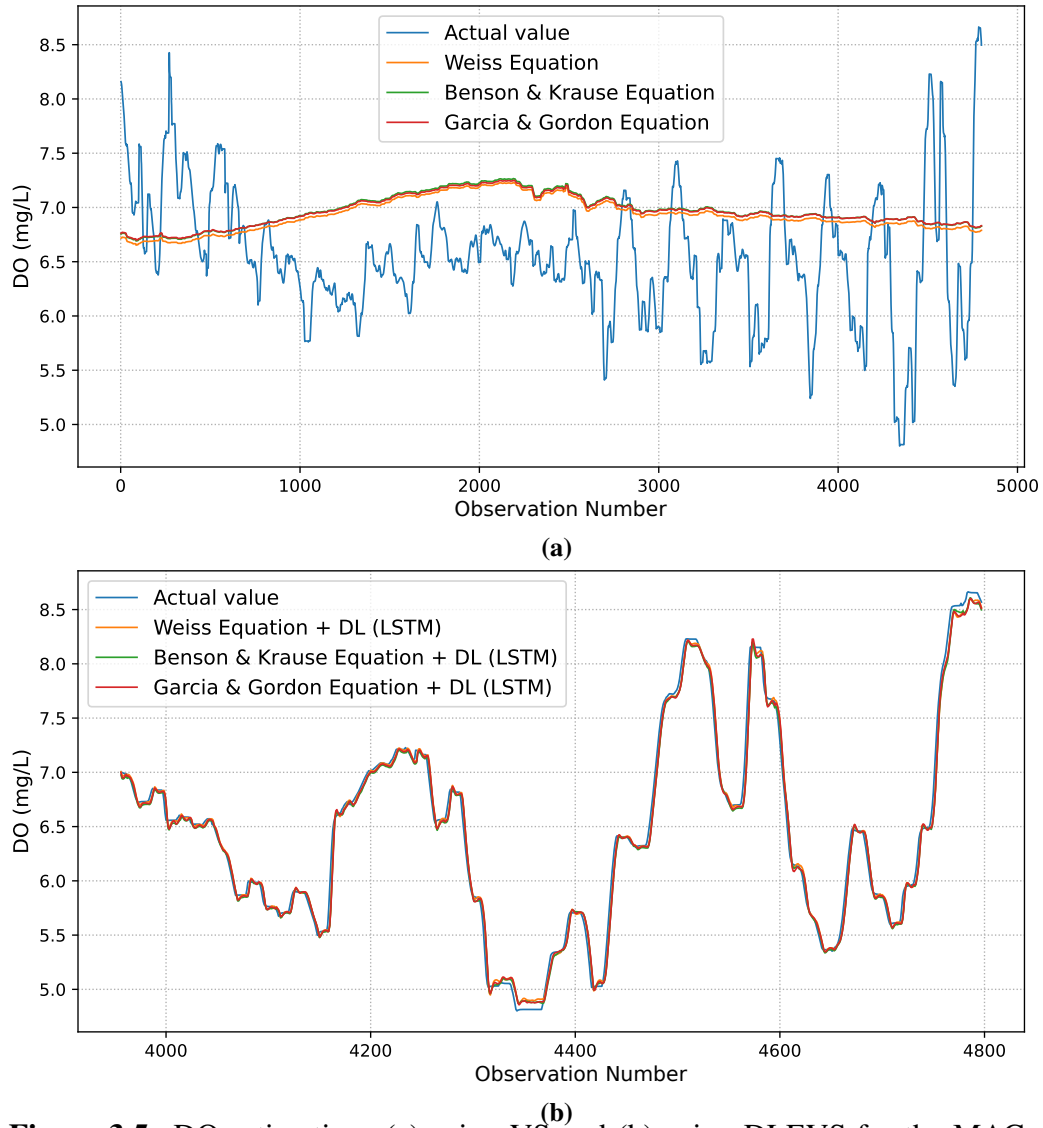
### 3.4.2 MAC water quality dataset

Figure 3.5, show the comparison of the DO values from the proposed VS and DLEVS methods with the actual sensor's readings for the MAC dataset. Specifically, Fig. 3.5a displays the comparison for the VS method, whereas Fig. 3.5b presents the comparison for the DLEVS method. From these figures, it can be seen that the proposed methods align well with the actual DO values. Moreover, the DLEVS method demonstrates a closer agreement with the actual values than the VS method does, as evidenced by the smaller errors presented in Table 3.3.

The results for both datasets indicate that the proposed method of combining VS with DL models (DLEVS) outperforms the traditional VS for estimating DO levels in water. The DLEVS method consistently achieved lower MAE, MSE, RMSE, and MAPE values across all three DO equations for both the ADAK and MAC datasets. The results suggest that incorporating DL models into the virtual sensor framework can lead to more accurate and reliable estimates of WQ parameters.

**Table 3.3:** DO estimation using VS and DLEVS for the MAC water quality dataset

| Equation        | VS     |        |        |        | DLEVS  |        |        |        |
|-----------------|--------|--------|--------|--------|--------|--------|--------|--------|
|                 | MAE    | MSE    | RMSE   | MAPE   | MAE    | MSE    | RMSE   | MAPE   |
| Weiss           | 0.6316 | 0.5633 | 0.7505 | 0.1002 | 0.0567 | 0.0072 | 0.0848 | 0.0089 |
| Benson & Krause | 0.6531 | 0.5936 | 0.7705 | 0.1038 | 0.0590 | 0.0074 | 0.0862 | 0.0092 |
| Garcia & Gordon | 0.6474 | 0.5855 | 0.7652 | 0.1029 | 0.0587 | 0.0076 | 0.0870 | 0.0091 |



**Figure 3.5:** DO estimations (a) using VS and (b) using DLEVS for the MAC water quality dataset

### 3.5 SUMMARY

In this research work, a low-cost WQM system is proposed for aquaculture, which utilizes a virtual sensing approach based on the LSTM DL-RNN. The virtual sensor estimates the DO value using various solubility equations, and the LSTM DL-RNN is used to reduce the error in DO estimation, thus improving the accuracy of the virtual DO sensor. The results demonstrate that the proposed LSTM-based DO virtual sensor can accurately estimate DO levels, providing a cost-effective and reliable solution for monitoring WQ parameters in aquaculture. This research work paves the way for the development of low-cost and efficient WQM systems for aquaculture, which are very important for ensuring the sustainability and productivity of Aquaculture industry.

## **CHAPTER 4**

### **DEEP LEARNING MODELS FOR WATER QUALITY PREDICTION IN AQUACULTURE**

This chapter delves into applying DL models, such as LSTM, GRU, and their bidirectional counterparts, for WQP in aquaculture. The primary focus of this study lies in comparing these DL models in terms of accuracy. DL models can consider the non-linear characteristics of WQ changes, enabling them to capture intricate patterns in the data. This unique trait allows DL models to identify essential features that efficiently provide concise and reliable results [81]. Among the DL models, LSTM and GRU are widely used for time series prediction due to their remarkable performance [82]. In this chapter, Section 4.1 provides an insightful introduction, highlighting the significant contributions of this work. Section 4.2 of this chapter comprehensively explains the ADAK dataset, offering detailed insights into the data characteristics and pre-processing. Moving on to Section 4.3, the DL models employed for WQP in this study are explained. The section outlines the architecture and underlying mechanisms of each DL-RNN model, focusing on LSTM, GRU, and their bidirectional counterparts (Bi-LSTM and Bi-GRU). These DL models have been selected for their proficiency in capturing temporal patterns and sequences, making them suitable candidates for WQ time series prediction. Section 4.4 is dedicated to result analysis, a critical component of this study focusing on comparing the performances of the different DL models, including LSTM, GRU, Bi-LSTM, and Bi-GRU. The chapter is summarized in Section 4.5.

#### **4.1 INTRODUCTION**

A WQP model could describe the relationship between monitored data as well as trends in WQ parameters based on available data gathered by the WQM system and forecast changes in WQ for a specified period. There are many shortcomings in conventional prediction approaches, such as weak forecasting precision, least stability etc. Novel prediction methods focused on artificial intelligence have emerged with the development of computational intelligence and bionic technology. Traditional methods for estimating WQ mainly include: the Markov model based on fuzzy-set theory [66, 67], the Grey Markov chain model method [68, 69], the method of prediction of regression [70, 71], the method of time series [72, 73], methods of WQP

model [74, 75], the ANN method [76, 77], the LS-SVM prediction method [78, 79], the combination prediction method [80]. These approaches offer an efficient method for forecasting WQ but are not suitable for aquaculture. WQ is affected by many factors such as complex inter-relationship between the parameters, non linearity in the WQ changes etc. Hence these models results in poor prediction accuracy.

LSTM was introduced to enhance long sequence learning and avoid the vanishing-gradient issue of RNN [138]. GRU DL-RNN was proposed in [139] and it has performance comparable with LSTM in terms of accuracy and lower demand for processing resources. Some reported research works based on LSTM model resolves the issues with conventional prediction methods. The authors have used LSTM to predict drinking WQ in [85], and in [86] for mariculture WQP. In order to further improve the performance Adam an efficient stochastic optimisation method is used [88, 89]. Adam combines advantage of AdaGrad [90] and RMSProp [91], which are two popular methods.

WQP is a typical time series problem, which can be considered to be a sequential regression problem. For training the model, this sequential time series data is converted into a set of feature values and the corresponding target value using the sliding window procedure since the regression is a supervised learning problem. The predicted output depends on the current input and the entire length of the time sequence with different levels of importance or weightage. Many prediction models map relationships with the input and output data at a particular time, ignoring the features in the lengthy sequence. To capture the features from this whole sequence data, different DL models can be used in the DL layer. Bidirectional DL models can be used for the utilization of the information from the sequence data in the forward and backward directions. In the bidirectional NNs, with their non-linear mapping of the sequence, the DL layer accurately captures the sequence correlation of data to provide more precise prediction results. LSTM and GRU are two types of DL-RNN models used for sequence modeling tasks.

The significant contributions of the work presented in this chapter can be highlighted as follows:

1. This work conducts a comprehensive evaluation of standalone DL models, including LSTM, GRU, Bi-LSTM, and Bi-GRU, for accurate WQP. The models are trained and tested on diverse WQ parameters and datasets, providing a comprehensive understanding of their performance.
2. An in-depth analysis of the models' performance is carried out, considering metrics such as MAE, MSE, RMSE, and MAPE. The analysis reveals the strengths

and limitations of each model in predicting different WQ parameters.

## 4.2 DATA COLLECTION AND ANALYSIS

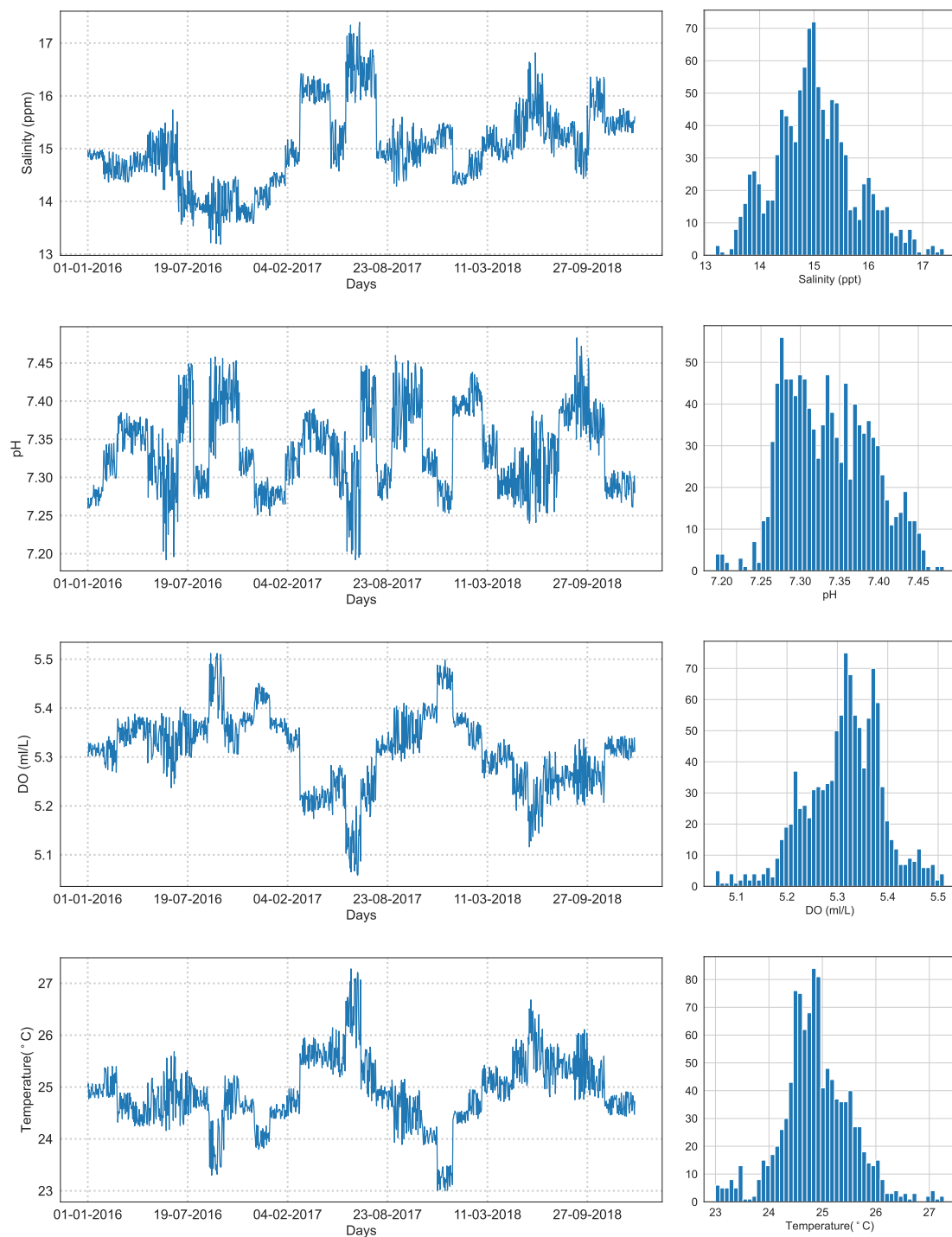
### 4.2.1 Acquisition of data

The data of aquaculture WQ parameters used in this work is collected from an aquaculture farm under ADAK at Kollam, Kerala, India. The WQ parameter data for three years, from January 2016 to December 2018, is collected on a daily basis from the pond. The trends in variations of WQ parameters of the aquaculture pond are studied. Figure 4.1 shows the annual trends of salinity, pH, DO, and temperature. From the plots, it is clear that all WQ parameters are non-stationary in nature. Table 4.1 shows

**Table 4.1:** Average water quality parameters monitored in 2016, 2017 and 2018

| Water quality parameter | Year | Mean    | Min    | Max    |
|-------------------------|------|---------|--------|--------|
| Salinity (ppt)          | 2016 | 14.3791 | 13.186 | 15.736 |
| pH                      |      | 7.3338  | 7.192  | 7.458  |
| DO (ml/L)               |      | 5.3581  | 5.237  | 5.512  |
| Temperature (°C)        |      | 24.6277 | 23.299 | 25.685 |
| Salinity (ppt)          | 2017 | 15.4084 | 14.261 | 17.395 |
| pH                      |      | 7.3326  | 7.192  | 7.46   |
| DO (ml/L)               |      | 5.2995  | 5.058  | 5.499  |
| Temperature (°C)        |      | 24.9244 | 22.999 | 27.278 |
| Salinity (ppt)          | 2018 | 15.2589 | 14.307 | 16.818 |
| pH                      |      | 7.3411  | 7.24   | 7.483  |
| DO (ml/L)               |      | 5.2856  | 5.116  | 5.391  |
| Temperature (°C)        |      | 25.1092 | 24.294 | 26.681 |

the average of the WQ parameter data of each year. The salinity in 2016 was between 13.18 ppt and 15.73 ppt; in 2017, it was between 14.26 ppt and 17.39 ppt, and in 2018 it was between 14.30 ppt and 16.82 ppt. The pH in 2016 was between 7.20 and 7.45, and in 2017, it was between 7.19 and 7.46. Moreover, in 2018, it was between 7.24 and 7.48. The DO in 2016 was between 5.23 ml/L and 5.51 ml/L; in 2017, it was between 5.05 ml/L and 5.49 ml/L, and in 2018 it was between 5.11 ml/L and 5.39 ml/L. The temperature value in 2016 was between 23.30 °C and 25.69 °C; in 2017 it was between 22.99 °C and 27.27 °C, and in 2018 it was between 24.29 °C and 26.68 °C. The mean DO minimum in 2018, where the mean temperature is also highest, indicates the relation between DO and temperature. When temperature increases, DO will decrease. In all three years, the average salinity value is coming down in June - July, coinciding with the monsoon season in Kerala.



**Figure 4.1:** Water quality parameters variation trend of ADAK water quality dataset

#### 4.2.2 Correlation analysis

Correlation analysis is performed on data from January 2016 and December 2018 to analyse the relationship between these WQ parameters (salinity, pH, DO and temperature). Pearson's correlation coefficient shows the correlation between the variables. A correlation ratio of more than 0.5 indicates a direct correlation with the parameter and a significant interdependence between the parameters. A value between 0.5 to 0.2 demonstrates a correlation with interdependence between them to some extent. A value of less than 0.1 suggests a poor correlation with the parameter and an insignificant interdependence between the parameters. The data is standardised to eliminate dimensional influence, and then the Pearson correlation coefficient matrix is applied. Table 4.2 shows the correlation coefficient matrix. The correlation between salinity, pH, DO, and temperature is given in Table 4.2. The findings in Table 4.2 show that temperature and salinity have a significant influence on the DO. Moreover, pH does not have any impact on the other WQ parameters monitored here.

**Table 4.2:** Correlation coefficient matrix of ADAK water quality dataset

|             | Salinity | pH      | DO      | Temperature |
|-------------|----------|---------|---------|-------------|
| Salinity    | 1.0000   | -0.0880 | -0.7203 | 0.5278      |
| pH          | -0.0880  | 1.0000  | 0.0083  | 0.0185      |
| DO          | -0.7203  | 0.0083  | 1.0000  | -0.9692     |
| Temperature | 0.5278   | 0.0185  | -0.9692 | 1.0000      |

Temperature and DO show the most correlation. Whereas temperature and pH show the minimum correlation. The pH has an insignificant correlation with salinity, DO and temperature. DO has a higher correlation with temperature, followed by salinity and an insignificant correlation with pH. Similarly, the temperature has a high correlation with DO and a moderate correlation with salinity. However, the correlation of temperature with pH is insignificant.

#### 4.2.3 Data pre-processing

Aquaculture WQ parameters data measured from the ponds may have anomalies such as missing data or abnormal data. This can be due to problems with sensors or possible errors while storing the data. These variations will lead to excessive deviation of predicted values from actual monitored values. In order to improve the accuracy of prediction, the predictive models must be provided with clean, reliable and succinct data. Unavailable data can no longer be retrieved, and missing data can only be computed as accurately as possible. Throughout this work, the missing part of the WQ parameters data is first filled using LIN algorithm [140], [141], and then this data is used to forecast the WQ parameters. As illustrated in Fig. 4.1, the WQ parameters

show continuous seasonal variation and also exhibit correlation with time.

Any data taken at definite time intervals forms time-series data. The dataset of WQ parameters acquired from a source also constitutes time-series data. It is an organized collection comprising measured values of WQ parameters at regular intervals. WQ parameters are measured at specific times every day.

$$TS_{i,n} = ((y_{i,1}, T_1), \dots, (y_{i,n}, T_n)) \quad (4.1)$$

In Eq. (4.1)  $TS_{i,n}$  is defined as  $n$ -length time-series with  $i$ th WQ parameter. The sampling interval for all parameters is one day i.e., all parameters are measured at the same time on a daily basis. If the value  $y_{i,v}$  at  $T_v$  is missing, then approximated value can be obtained using the LIN algorithm. The LIN function can be constructed as:

$$Y_{i,v} = y_{i,u} + \frac{y_{i,u} - y_{i,w}}{T_u - T_w}(T_v - T_u) \quad (4.2)$$

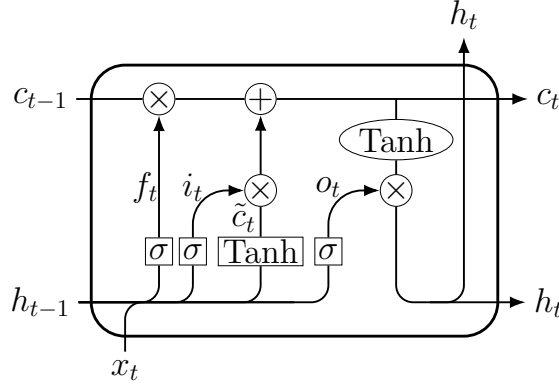
When a WQ data is missing at any point, the LIN algorithm first finds the two closest moments represented as  $T_u$  and  $T_w$ . Then it calculate the missing value at that moment  $T_v$  by utilizing the parameter values  $y_{i,u}$  and  $y_{i,w}$  at the moments  $T_u$  and  $T_w$  respectively based on Eq. (4.2). Here  $Y_{i,v}$  represents the estimated value of missing value  $y_{i,v}$ .

### 4.3 DEEP LEARNING MODELS

#### 4.3.1 LSTM DL-RNN

LSTM DL-RNN was proposed in 1997 by authors of [138] to avoid long-term dependency problems in RNN. In LSTM, a control unit is introduced to store information, unlike a hidden layer in RNN. This hidden state is divided to memory cells  $c_t$  and working memory  $h_t$ . The  $c_t$  is responsible for sequence features retention, and previous sequence memory is controlled by the forgetting gate  $f$ . The portion of the current memory  $c_t$  is controlled by output gate  $o_t$  and  $h_t$  is used as the output. The current state  $h_{t-1}$  and current input  $x_t$  written to memory cells are responsibility of the input gate  $i$ . The parameters associated with the LSTM model are collectively denoted as  $\theta_{LSTM}$ , representing the set of weights and biases. The LSTM architecture [142] is shown in Fig. 4.2, and memory units are defined as follows in Eq. (4.3):

$$\begin{aligned}
h_t^{LSTM} &= H^{(L)}(h_{t-1}^{LSTM}, x_t, \theta_{LSTM}) \\
&= \begin{cases} f_t = \sigma(W_f \times [h_{t-1}, x_t] + b_f) \\ i_t = \sigma(W_i \times [h_{t-1}, x_t] + b_i) \\ \tilde{c}_t = \tanh(W_c \times [h_{t-1}, x_t] + b_c) \\ c_t = f_t \times c_{t-1} + i_t \times \tilde{c}_t \\ o_t = \sigma(W_o \times [h_{t-1}, x_t] + b_o) \\ h_t^{LSTM} = o_t \times \tanh(c_t) \end{cases} \quad (4.3)
\end{aligned}$$



**Figure 4.2:** Forward LSTM cell structure

In this architecture  $f_t$ ,  $i_t$ , and  $o_t$  are forget, input and output gate layer respectively.  $\tilde{c}_t$  and  $c_t$  are new and final memory cell,  $w$  is weight matrices,  $b$  is bias vectors,  $\sigma$  is the sigmoid activation function.

#### 4.3.2 GRU DL-RNN

GRU DL-RNN was proposed by authors of [139] in 2014. It similar to LSTM, but required less computing power. GRU is an improved version of RNN with only two gates, an update gate and a reset gate. There are no additional memory cells to store information; GRU can control information inside the unit. The update gate decides whether to pass the previous output  $h_{t-1}$  to the next cell. The reset gate reads the input sequences when the gate is set to zero and forgets the previously calculated state. As a result, GRU has fewer tensor operations than LSTM and runs typically faster than LSTM. The parameters associated with the GRU model are collectively denoted as  $\theta_{GRU}$ , representing the set of weights and biases. The GRU architecture [142] is shown in Fig. 4.3 and memory units are defined as follows in Eq. (4.4):

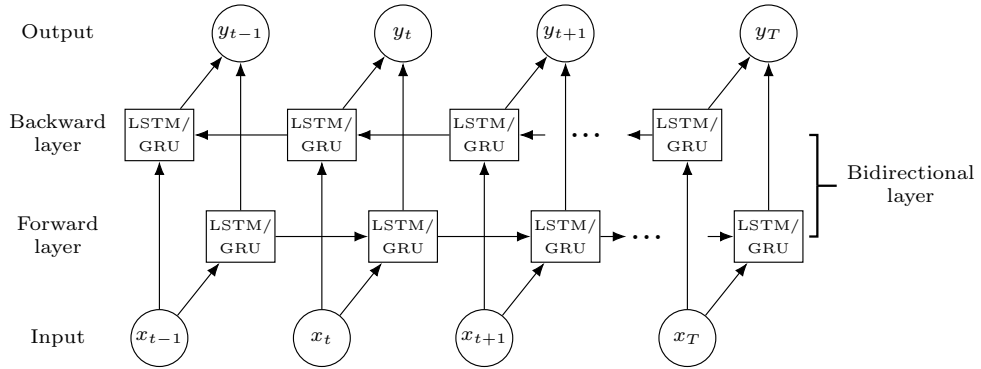
$$h_t^{GRU} = H^{(G)}(h_{t-1}^{GRU}, x_t, \theta_{GRU})$$

$$= \begin{cases} z_t = \sigma(W_z \times [h_{t-1}, x_t]) \\ r_t = \sigma(W_r \times [h_{t-1}, x_t]) \\ \tilde{h}_t = \tanh(w \times [r_t \times h_{t-1}, x_t]) \\ h_t^{GRU} = (1 - z_t) \times h_{t-1} + z_t \times \tilde{h}_t \end{cases} \quad (4.4)$$

**Figure 4.3:** Forward GRU cell structure

In this architecture  $z_t$  is the reset and  $r_t$  is the update gate,  $\tilde{h}_t$  is process input, and  $h_t$  is hidden state update,  $w$  is weight matrices and  $\sigma$  is the sigmoid activation function.

#### 4.3.3 Bidirectional LSTM/GRU DL-RNN



**Figure 4.4:** Structure of Bi-LSTM/Bi-GRU

The bidirectional DL models consist of two processes in both forward and backward directions, as depicted in Fig. 4.4. The processing in both directions helps the model capture patterns that the unidirectional LSTM and GRU models cannot capture. The bi-directional DL-RNNs are an extension of LSTM and GRU. Bi-directional DL-RNNs can fit data in both forward and backward directions and concatenate the prediction, while LSTM/GRU can only fit data in one direction [143]. A Bi-LSTM/Bi-GRU model first feeds data into one of the LSTM/GRU models for training the given data sequence, i.e., feedback layer and re-training with another LSTM/GRU model, but

with the opposite order of the input data sequence. Bi-LSTM/Bi-GRU model can capture patterns ignored by LSTM/GRU because it has added a reverse directional layer. Double LSTM/GRU application improves long-term study and thereby improves accuracy [144]. Bi-directional networks are the extension of the LSTM/GRU model, which can learn about the relation of past and future factors of load and current load, which helps in extracting the in-depth features of loaded data [145]. The bi-directional LSTM/GRU structure is shown in Fig. 4.4.

#### 4.3.4 Model implementation

The WQ dataset is used to train the DL models to predict each WQ parameter. First, the dataset is divided into two; 80% for the training and 20% for the testing. The training dataset is used to develop the models, while the testing dataset is used to validate and compare the performance of models developed. All input and output variables were scaled between 0 and 1 via normalization and minimum-maximum scaling techniques, using the MinMaxScaler in the scikit-learn preprocessing library using Python [146]. Moreover, the Adam optimizer is employed for training the models due to its adaptive learning rate mechanism and ability to handle sparse gradients effectively [147]. The training and testing datasets are transformed into supervised learning frames with various array manipulation techniques. The time series sequences are converted to input-output equations using the sliding window procedure. This transformed data is used to train the models for WQP.

### 4.4 RESULTS AND DISCUSSIONS

The objective is to analyse the performance of DL models, LSTM, GRU, Bi-LSTM, and Bi-GRU to predict aquaculture WQ parameters (salinity, ph, DO and temperature). In this section, the performance of DL models is studied with a fixed set of  $h_p$  on the two datasets.

#### 4.4.1 ADAK water quality dataset

Figure 4.5 compares the predicted values with the true data on the ADAK WQ dataset using different DL models. All these models are trained using 80% of the ADAK WQ dataset and tested with 20% of the data. The predicted values are compared with the true data. Here a fixed set of  $h_p$  (epochs = 100, learning rate = 0.0008, batch size = 32, window size = 30 and Adam optimizer) is used. The models are evaluated using performance metrics MAE, MSE, RMSE, and MAPE in the training and testing periods. The computation time required for each of the models is also calculated.

The performance of standalone DL models, such as LSTM, GRU, Bi-LSTM, and

**Table 4.3:** Prediction accuracy of LSTM, GRU, Bi-LSTM and Bi-GRU for ADAK water quality dataset

| Water Quality Parameter | Model  | Training Period |        |        |        |          | Testing Period |        |        |        |          |
|-------------------------|--------|-----------------|--------|--------|--------|----------|----------------|--------|--------|--------|----------|
|                         |        | MAE             | MSE    | RMSE   | MAPE   | Time (s) | MAE            | MSE    | RMSE   | MAPE   | Time (s) |
| Salinity (ppt)          | LSTM   | 0.2574          | 0.1135 | 0.3368 | 0.0172 | 25.44    | 0.2388         | 0.0939 | 0.3065 | 0.0154 | 0.06     |
|                         | GRU    | 0.2315          | 0.0952 | 0.3085 | 0.0155 | 26.65    | 0.2436         | 0.1019 | 0.3191 | 0.0159 | 0.06     |
|                         | BiLSTM | 0.2486          | 0.1089 | 0.3300 | 0.0165 | 33.66    | 0.2877         | 0.1268 | 0.3561 | 0.0185 | 0.06     |
|                         | BiGRU  | 0.2231          | 0.0903 | 0.3005 | 0.0149 | 35.36    | 0.2309         | 0.0874 | 0.2956 | 0.0149 | 0.06     |
| pH                      | LSTM   | 0.0249          | 0.0012 | 0.0339 | 0.0034 | 27.81    | 0.0239         | 0.0010 | 0.0314 | 0.0033 | 0.05     |
|                         | GRU    | 0.0302          | 0.0015 | 0.0391 | 0.0041 | 28.30    | 0.0311         | 0.0015 | 0.0391 | 0.0042 | 0.06     |
|                         | BiLSTM | 0.0244          | 0.0011 | 0.0331 | 0.0033 | 34.12    | 0.0239         | 0.0010 | 0.0319 | 0.0033 | 0.06     |
|                         | BiGRU  | 0.0327          | 0.0017 | 0.0408 | 0.0044 | 35.15    | 0.0330         | 0.0017 | 0.0406 | 0.0045 | 0.06     |
| DO (ml/L)               | LSTM   | 0.0333          | 0.0018 | 0.0420 | 0.0063 | 26.67    | 0.0305         | 0.0014 | 0.0373 | 0.0058 | 0.06     |
|                         | GRU    | 0.0255          | 0.0011 | 0.0338 | 0.0048 | 27.00    | 0.0263         | 0.0012 | 0.0348 | 0.0050 | 0.06     |
|                         | BiLSTM | 0.0235          | 0.0010 | 0.0309 | 0.0044 | 34.93    | 0.0246         | 0.0010 | 0.0313 | 0.0047 | 0.06     |
|                         | BiGRU  | 0.0263          | 0.0012 | 0.0340 | 0.0050 | 38.10    | 0.0252         | 0.0011 | 0.0325 | 0.0048 | 0.06     |
| Temperature(°C)         | LSTM   | 0.3030          | 0.1495 | 0.3866 | 0.0123 | 26.69    | 0.3275         | 0.1748 | 0.4181 | 0.0131 | 0.05     |
|                         | GRU    | 0.2822          | 0.1325 | 0.3640 | 0.0113 | 28.13    | 0.2784         | 0.1318 | 0.3630 | 0.0110 | 0.06     |
|                         | BiLSTM | 0.2582          | 0.1225 | 0.3501 | 0.0104 | 36.69    | 0.2469         | 0.1012 | 0.3181 | 0.0098 | 0.07     |
|                         | BiGRU  | 0.3269          | 0.1806 | 0.4249 | 0.0131 | 36.29    | 0.3093         | 0.1535 | 0.3918 | 0.0122 | 0.09     |

$h_p : L_r = 0.0008, E_p = 100, W_z = 30, B_z = 32$ , and Adam optimizer.

**Table 4.4:** Prediction accuracy of LSTM, GRU, Bi-LSTM and Bi-GRU for MAC water quality dataset

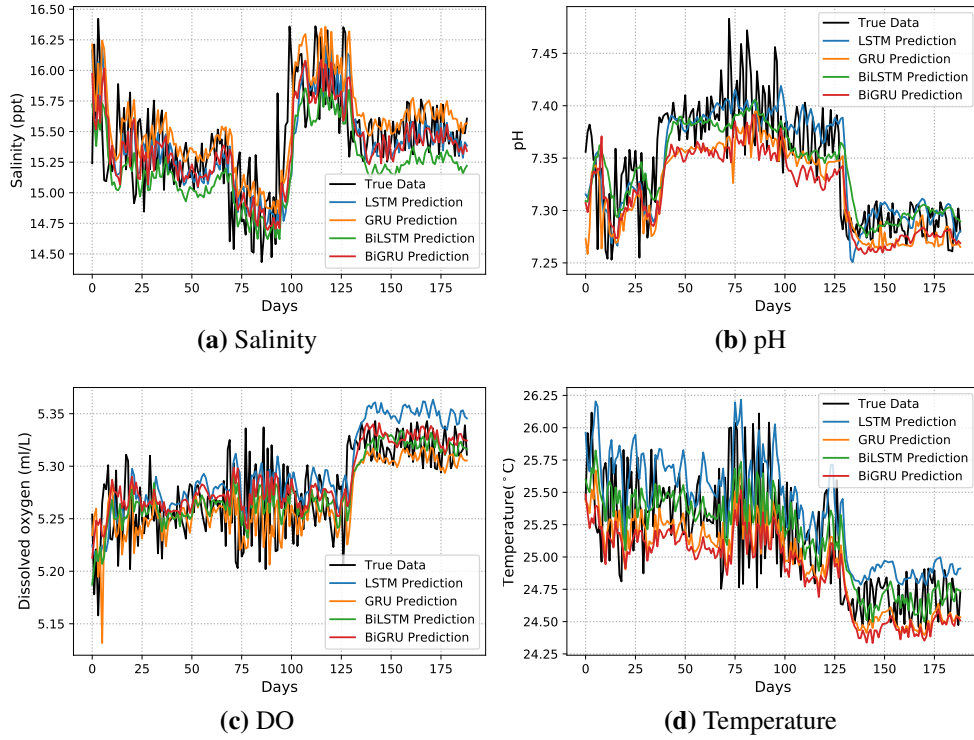
| Water Quality Parameter | Model  | Training Period |        |        |        |          | Testing Period |        |        |        |          |
|-------------------------|--------|-----------------|--------|--------|--------|----------|----------------|--------|--------|--------|----------|
|                         |        | MAE             | MSE    | RMSE   | MAPE   | Time (s) | MAE            | MSE    | RMSE   | MAPE   | Time (s) |
| Salinity (ppt)          | LSTM   | 0.1216          | 0.0198 | 0.1407 | 0.0035 | 131.11   | 0.2590         | 0.9396 | 0.9693 | 0.0078 | 0.44     |
|                         | GRU    | 0.0156          | 0.0005 | 0.0217 | 0.0005 | 135.46   | 0.0905         | 0.5058 | 0.7112 | 0.0029 | 0.45     |
|                         | BiLSTM | 0.0224          | 0.0011 | 0.0338 | 0.0007 | 167.55   | 0.0394         | 0.0121 | 0.1101 | 0.0012 | 0.52     |
|                         | BiGRU  | 0.0453          | 0.0024 | 0.0489 | 0.0013 | 180.59   | 0.1234         | 0.3936 | 0.6274 | 0.0038 | 0.47     |
| pH                      | LSTM   | 0.0078          | 0.0001 | 0.0090 | 0.0009 | 188.08   | 0.0093         | 0.0001 | 0.0111 | 0.0011 | 1.03     |
|                         | GRU    | 0.0521          | 0.0030 | 0.0550 | 0.0063 | 203.74   | 0.0570         | 0.0077 | 0.0878 | 0.0067 | 0.63     |
|                         | BiLSTM | 0.0071          | 0.0001 | 0.0092 | 0.0009 | 226.54   | 0.0118         | 0.0004 | 0.0198 | 0.0014 | 0.70     |
|                         | BiGRU  | 0.0266          | 0.0007 | 0.0270 | 0.0032 | 225.32   | 0.0582         | 0.0152 | 0.1233 | 0.0067 | 0.59     |
| DO (ml/L)               | LSTM   | 0.0648          | 0.0081 | 0.0898 | 0.0204 | 166.33   | 0.1114         | 0.0527 | 0.2295 | 0.0209 | 0.56     |
|                         | GRU    | 0.1213          | 0.0224 | 0.1495 | 0.0380 | 131.55   | 0.2427         | 0.6086 | 0.7801 | 0.0391 | 0.38     |
|                         | BiLSTM | 0.0792          | 0.0103 | 0.1014 | 0.0206 | 170.38   | 0.1096         | 0.0295 | 0.1717 | 0.0259 | 0.52     |
|                         | BiGRU  | 0.0981          | 0.0151 | 0.1231 | 0.0241 | 173.14   | 0.4637         | 2.1555 | 1.4682 | 0.0534 | 0.47     |
| Temperature(°C)         | LSTM   | 0.6193          | 0.3871 | 0.6222 | 0.0246 | 138.20   | 0.6209         | 0.3955 | 0.6289 | 0.0227 | 0.45     |
|                         | GRU    | 0.6364          | 0.4208 | 0.6487 | 0.0252 | 134.23   | 0.7345         | 0.5432 | 0.7370 | 0.0268 | 0.40     |
|                         | BiLSTM | 0.1039          | 0.0129 | 0.1136 | 0.0040 | 173.03   | 0.1413         | 0.0212 | 0.1454 | 0.0052 | 0.51     |
|                         | BiGRU  | 0.3063          | 0.0953 | 0.3086 | 0.0121 | 174.85   | 0.3437         | 0.1196 | 0.3459 | 0.0125 | 0.57     |

$h_p : L_r = 0.0008, E_p = 100, W_z = 30, B_z = 32$ , and Adam optimizer.

Bi-GRU are evaluated for WQP. Figure 4.5 and Fig. 4.6 illustrate the comparative analysis of predicted values versus actual data from the ADAK and MAC WQ datasets, respectively, using these DL models.

#### 4.4.2 MAC water quality dataset

Figure 4.6 compares the prediction of all DL models with the true data on the MAC WQ dataset. All these models are trained using 80% of the ADAK WQ dataset and predicted the 20% values. The predictions are compared with the true data. Here a fixed set of  $h_p$  (epochs = 50, learning rate = 0.0008, batch size = 64, window size = 30 and Adam optimizer) is used. The models are evaluated using MAE, MSE, RMSE, and MAPE in the training and testing periods. The computation time required for each



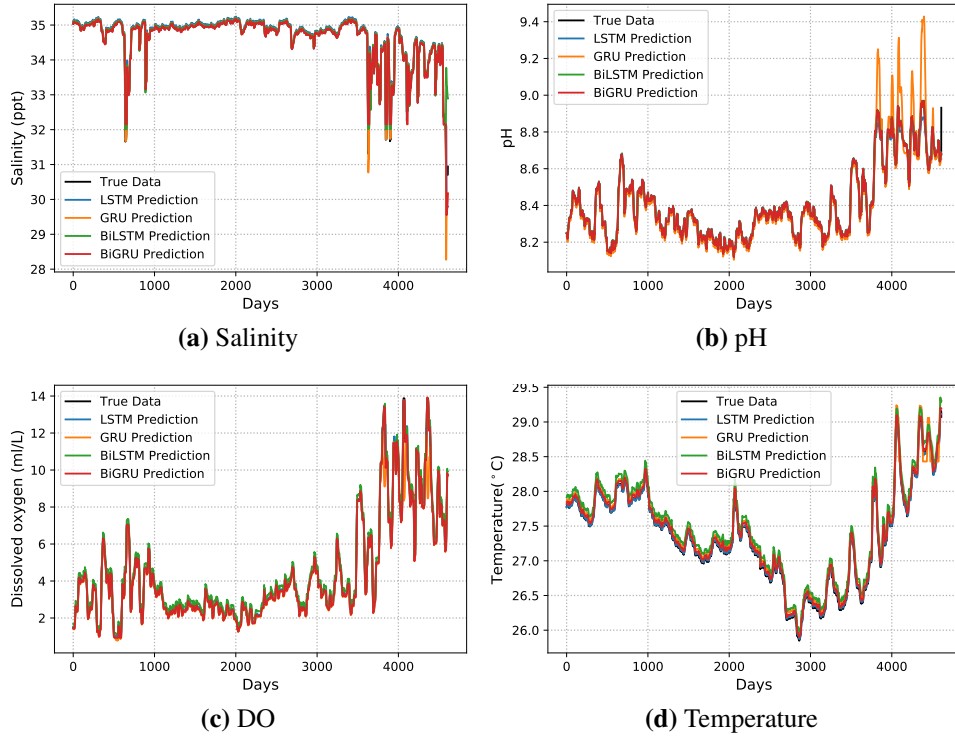
**Figure 4.5:** Comparison of predicted values using LSTM, GRU, Bi-LSTM, and Bi-GRU predictions and ADAK water quality observed data. (a) Salinity, (b) pH, (c) DO, (d) Temperature

of the models is also calculated.

As demonstrated in Table 4.3 and Table 4.4, these models have shown varying degrees of performance in predicting different parameters of the ADAK and MAC WQ datasets. All models exhibit a level of competence, yet distinct strengths and limitations are observed. The LSTM model, for example, offers a satisfactory performance across both datasets, although it tends to be slightly less accurate compared to the BiLSTM and GRU models in certain parameters. The BiLSTM model, despite its bidirectional learning mechanism, does not consistently outperform its unidirectional counterparts (LSTM and GRU), suggesting that the additional complexity may not always translate to superior performance.

It's important to note that while these models show adequate performance, none of them significantly outperforms the others in predicting all parameters for distinct datasets. This indicates the need for more sophisticated models to better predict WQ parameters, and serves as a prelude to the introduction of hybrid models in the next chapter, which aim to combine the strengths of these individual models for superior predictive performance.

Table 4.5 provides a comprehensive performance comparison of DL-RNN models, encompassing LSTM, GRU, Bi-LSTM, and Bi-GRU, provides an insightful



**Figure 4.6:** Comparison of predicted values using LSTM, GRU, Bi-LSTM, and Bi-GRU predictions and MAC water quality observed data. (a) Salinity, (b) pH, (c) DO, (d) Temperature

**Table 4.5:** Comprehensive Performance Comparison of DL-RNN Models

| Model   | ADAK Dataset |        |        |        |          | MAC Dataset |        |        |        |          |
|---------|--------------|--------|--------|--------|----------|-------------|--------|--------|--------|----------|
|         | MAE          | MSE    | RMSE   | MAPE   | Time (s) | MAE         | MSE    | RMSE   | MAPE   | Time (s) |
| LSTM    | 0.1547       | 0.0665 | 0.1998 | 0.0098 | 26.65    | 0.2034      | 0.1038 | 0.2154 | 0.0124 | 155.93   |
| GRU     | 0.1424       | 0.0576 | 0.1864 | 0.0089 | 27.52    | 0.2064      | 0.1117 | 0.2187 | 0.0175 | 151.25   |
| Bi-LSTM | 0.1387       | 0.0584 | 0.1860 | 0.0087 | 34.85    | 0.0532      | 0.0061 | 0.0645 | 0.0066 | 184.38   |
| Bi-GRU  | 0.1523       | 0.0685 | 0.2001 | 0.0094 | 36.22    | 0.1191      | 0.0284 | 0.1269 | 0.0102 | 188.48   |

Values represent the average of the respective metric across all parameters.

overview of their effectiveness in predicting water quality parameters across two distinct datasets: ADAK and MAC. The evaluation metrics employed, including MAE, MSE, RMSE, MAPE values, and computation Time, offer a holistic view of each model's performance.

#### 4.5 SUMMARY

In this chapter, the focus is on the application of DL-RNN models with Adam optimization for aquaculture WQP. The chapter explores the use of DL-RNN models, including LSTM, GRU, and their bidirectional counterparts for WQP. The main objectives are to compare the performance of these DL models and identify the most accurate and efficient one for the task at hand. The ADAK and MAC datasets are used for experimental results and comparisons. The chapter provides valuable insights into the performance of various DL-RNN models for aquaculture WQP. While all models

show competence in predicting WQ parameters, no single model significantly outperforms the others in all cases. This suggests the need for more sophisticated approaches, leading to the introduction of hybrid models in subsequent chapters.



## CHAPTER 5

### HYBRID DEEP LEARNING MODELS FOR ACCURATE AND COMPUTATIONALLY EFFICIENT WATER QUALITY PREDICTION IN AQUACULTURE

This chapter advances the research by developing hybrid CNN-LSTM/GRU models for WQP. This chapter introduces the implementation of these hybrid models and evaluates their performance on the ADAK and MAC WQ datasets. The results highlight the potential of these hybrid models in enhancing prediction accuracy and computational efficiency. CNN can effectively fetch the aquaculture WQ characteristics, whereas GRU and LSTM can learn long-term dependencies in the time series data. In Section 5.1 of this chapter, a comprehensive introduction is provided, highlighting the significant contributions of this work. Section 5.2 explains the architecture of the proposed CNN-LSTM/GRU models. A critical aspect of building an accurate DL WQP model is the identification of the suitable  $h_p$ . Section 5.3 presents an extensive study analysing the impact of various  $h_p$  variations on the performance of the proposed DL-RNN models. Different experiments are conducted using the two different WQ datasets and present an extensive study on the impact of  $h_p$  on the performance of the proposed hybrid DL models. Furthermore, the performance of hybrid CNN-LSTM and CNN-GRU models are compared with different baseline LSTM, GRU, and CNN DL models. Two different WQ datasets are utilized for the experimental evaluations to facilitate a comprehensive comparison, considering existing models such as LSTM, GRU, and CNN. Section 5.4 delves deeper into the performance analysis of all the DL-RNN models for a selected set of  $h_p$ , ultimately presenting the final results and comparisons. The results show that the hybrid CNN-LSTM outperformed all other models in terms of prediction accuracy and computation time. The chapter is summarized in Section 5.5.

#### 5.1 INTRODUCTION

WQP plays an essential role in WQ management for aquaculture to make aquaculture production profitable and sustainable. It can be viewed as a sequential regression problem, aligning it with the structure of a typical time series challenge. Time series data has applications across fields like stock markets, healthcare, energy consumption, and weather forecasting [60–65]. Traditional methods for WQP, like ARMA, ARIMA,

SARIMA, and SVR, are unsuitable for aquaculture due to their linear assumptions and inefficiency in predicting non-linear water quality parameters [16–18].

The adaptability of DL models like LSTM, GRU, and CNN to the non-linear characteristics of aquaculture WQP is crucial [19]. These DL models surpass traditional RNNs and excel in various applications. Recent AI advancements have established DL as a proficient methodology, applied extensively in diverse fields [95, 96, 94]. DL enhances time series prediction precision and efficiency while accommodating non-linear water quality trends. Among DL models, LSTM stands out for its exceptional WQP performance [20]. However, LSTM and GRU models might struggle with extended memory retention in time series forecasting [97–99], hindering their capacity to capture prolonged dependencies.

Hybrid CNN-LSTM models function similarly to assist the LSTM model in capturing pertinent data. Utilizing a convolutional layer with maxpooling, the input sequence is compressed prior to utilization by the LSTM and GRU models. This integrated approach counters exponential decay; CNN with maxpooling inputs essential features, bolstering LSTM's forecasting accuracy for higher precision. These hybrid models prioritize accuracy, seeking to avoid overfitting or underfitting while demonstrating robust generalization. Moreover, their training demands reduced time. The convolutional layer, coupled with maxpooling, enhances the capacity of LSTM and GRU to learn and retain pivotal features within non-linear water quality parameters, mitigating exponential decay through convolution and maxpooling [92, 114]. Notably, CNNs inherently operate faster, processing computations in parallel, providing an advantage over RNNs requiring sequential processing. Quasi-Recurrent Neural Networks [115] emulate RNN behaviour using CNN components, effectively boosting speed. Furthermore, CNNs focus on crucial features, leading to complexity reduction. By incorporating convolution layers with Maxpooling, the tensor size is diminished, which accelerates training for hybrid models. These hybrid CNN-LSTM models find applications in various research endeavors involving time series prediction. DL models, widely adopted across different time series domains, encompass finance (stock, cryptocurrency, precious metal prediction) [116, 117]. For instance, the CNN-LSTM hybrid, utilized for gold price prediction [118], exhibits efficacy relative to other DL models. The employment of Bi-LSTM with CNN for air quality prediction [120] achieves high accuracy in air quality estimation. Hybrid DL models are prevalent in air quality (PM<sub>2.5</sub>) prediction, leveraging CNN and LSTM [121, 122].

In this work, a hybrid DL models, CNN with the LSTM and GRU for aquaculture WQP is proposed. CNN can effectively fetch the aquaculture WQ characteristics,

whereas GRU and LSTM can learn long-term dependencies in the time series data. Different experiments are conducted using the two different WQ datasets and present an extensive study on the impact of hyperparameters on the performance of the proposed hybrid DL models. Furthermore, the performance of hybrid CNN-LSTM and CNN-GRU models are compared with different baseline LSTM, GRU and CNN DL models.

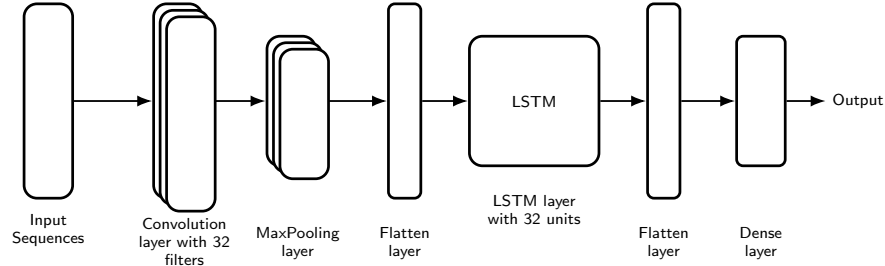
The significant contributions of the research work presented in this chapter are as follows:

1. This work pioneers the application of hybrid CNN-LSTM and CNN-GRU models in the realm of WQP. These hybrid models fuse the feature extraction prowess of CNNs with the long-term dependency learning capabilities of LSTM/GRU networks, yielding a novel approach to WQP.
2. Developed hybrid models, meticulously designed to predict critical WQ parameters. These parameters, including salinity, pH, DO, and temperature, are vital indicators in aquaculture.
3. Extensive experimental evaluations are conducted to investigate the impact of  $h_p$  on the performance of the proposed hybrid CNN-LSTM, CNN-GRU, and all DL-RNN models, using two different aquaculture WQ datasets. Determining the right set of  $h_p$  plays a significant role in building an accurate DL-RNN WQP model.
4. The hybrid models' performance is rigorously evaluated using real-world sensor data from two different WQ datasets. The results underline the models' potential to significantly improve the accuracy of WQP, thereby enhancing the reliability of monitoring systems in aquaculture settings.
5. The performance of hybrid CNN-LSTM and CNN-GRU models are compared with baseline DL models LSTM, GRU, and CNN. Results show that the hybrid models significantly improve accuracy compared to the baseline models for aquaculture WQP. Also, the hybrid models outperform all the DL models in terms of computation time.

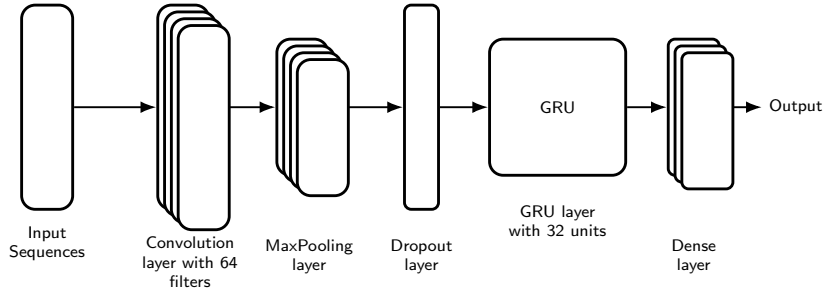
## 5.2 PROPOSED HYBRID CNN-LSTM/GRU DL-RNN MODELS

The Hybrid CNN-LSTM and CNN-GRU DL-RNN structures are shown in Fig. 5.1 and Fig. 5.2. The two models are the hybrid of CNN with LSTM and CNN with GRU. The first part of the model is CNN, to which the data is fed, and it extracts the

features. There is a dropout layer after the convolution layer (Conv1D) and pooling layer (MaxPooling1D). The second part has the LSTM or GRU, followed by a dense layer to give the output.



**Figure 5.1:** Proposed Hybrid CNN-LSTM DL-RNN model structure



**Figure 5.2:** Proposed Hybrid CNN-GRU DL-RNN model structure

In the CNN-LSTM model, Conv1D with 32 filters is used, a kernel size of 3 and ReLU is used as the activation function. A pooling layer MaxPooling1D follows this with a pooling size of 2. The CNN layer extracts all the features and then feeds them to the LSTM layer. The output of the CNN layer is fed to the LSTM layer after pooling through the flatten layer. Then, the LSTM layer outputs its output to the dense layer through the flatten layer, and finally, the prediction is output at the dense layer.

The CNN-GRU models use Conv1D with 64 filters, and a kernel size of 5 and ReLU is used as the activation function. A pooling layer MaxPooling1D follows this with a pooling size of 4. The CNN layer extracts all the features and then feeds them to the GRU layer. The output of the CNN layer is fed to the GRU layer after pooling and 0.2 dropout. After that, the GRU layer outputs its output to the dense layer, and finally, the prediction is output at the dense layer. These models use a learning rate of 0.0008 and Adam for optimisation [89]. Here, MSE is used as the loss function. The algorithm is based on CNN, LSTM, and GRU DL-RNN implemented using Python, Keras, and TensorFlow.

### 5.2.1 Model implementation

The WQ dataset forms the basis for training the hybrid CNN-LSTM/GRU DL-RNN models for WQP. The dataset is initially split into two sections: 80% for training and 20% for testing. The training dataset aids in model development, while the testing dataset is essential for validating and comparing the performance of the developed DL-RNN models. To ensure effective training, a key preprocessing step involves scaling all input and output variables within the range of 0 to 1, achieved through normalization and minimum-maximum scaling techniques. This scaling is facilitated using the MinMaxScaler from the scikit-learn preprocessing library in Python [146]. In addition, for model training, the Adam optimizer is strategically chosen due to its adaptive learning rate mechanism and adeptness in handling sparse gradients [147]. This choice of optimizer contributes to the efficiency of training the DL-RNN models. The training and testing datasets then undergo a transformation into supervised learning frames through various array manipulation techniques. Subsequently, the time series sequences are translated into input-output equations using the sliding window procedure. This transformed data is used for effectively training the DL models for WQP.

## 5.3 EXPERIMENTS: HYPERPARAMETER TUNING

This research is to analyse the performance of hybrid models CNN-LSTM and CNN-GRU. Two WQ data sets are used to conduct experiments, analyse and evaluate the performance of the proposed models. Furthermore, by comparing proposed models with other DL models, the prediction performance and effectiveness of the proposed models are validated. All experiments are conducted on Microsoft Azure Virtual Machines with specifications: Inter(R) Xeon (R) 8272CL CPU @2.60GHz, 32 GB RAM. The open source software used are Python and Keras based on Tensorflow.

The hybrid models are compared with CNN, LSTM, and GRU DL models. In the data set, 80% of the data is used to train the model, and the remaining 20% to test the model's prediction accuracy. The most critical part of building a DL model is tuning and optimization of the  $h_p$ . For the model to be most effective and efficient, tuning as many  $h_p$  as possible is necessary. In this work, four  $h_p$  are selected for tuning: learning rate, epochs, window size, and batch size. In each of the models, the number of layers and neurons in each layer are fixed. A dropout of 0.2 is applied after the primary layers to avoid overfitting, and *ReLU* is applied as the activation function. Additionally, the optimizer adopted in all the DL models in this work is Adam.

This work aims to improve the accuracy of aquaculture WQP and reduce computation time. Four sets of experiments are conducted involving varying of each  $h_p$ . Learning rate, epoch, window size, and batch size are plotted against RMSE. Additionally,

each hyperparameter is plotted against computation time. The experiments encompass both datasets (ADAK and MAC). This aids in analysing DL model performance on medium-sized and large datasets. The ADAK WQ dataset, despite spanning three years, only comprises 1096 data points, recorded once daily. In contrast, the MAC WQ dataset has 23200 data points, representing 80 days of recordings, taken every 5 minutes.

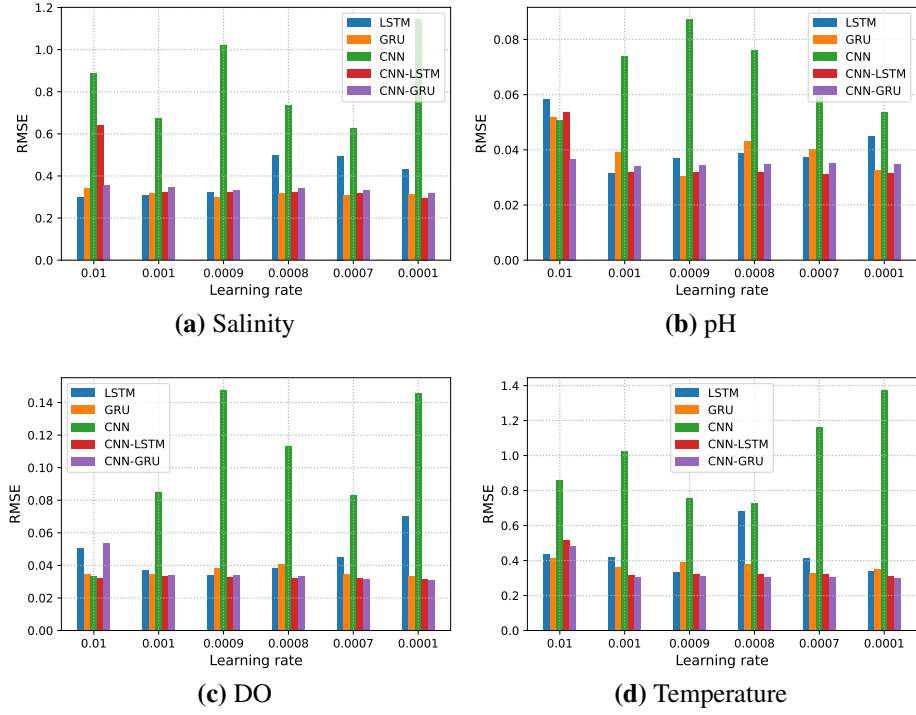
### 5.3.1 Learning rate ( $L_r$ )

The impact of learning rate  $\{0.01, 0.001, 0.0009, 0.0008, 0.0007, 0.0001\}$  on the performance for each of the DL models is analysed. In Fig. 5.3, RMSE vs learning rate performance is compared between proposed hybrid DL models and baseline models using the ADAK WQ dataset. The proposed hybrid DL models perform consistently better than baseline DL models. However, the performance of baseline DL models is not consistent for various learning rates and different WQ parameters. Hence using the baseline DL models for WQP is not practical.

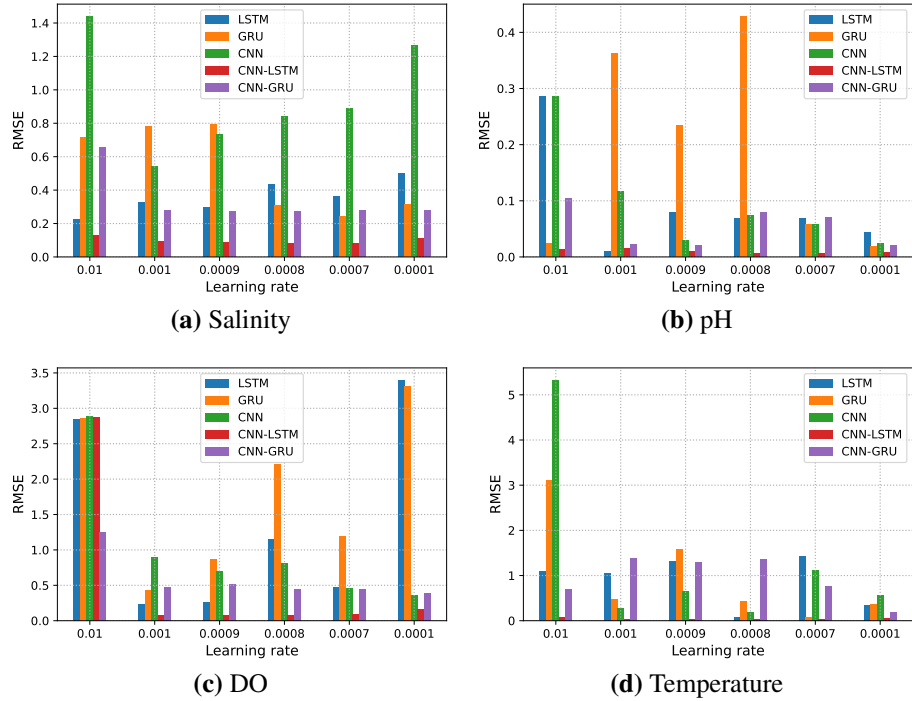
To further analyse the performance of these models, the experiment is repeated with the MAC WQ dataset. In Fig. 5.4, RMSE vs learning rate performance of proposed hybrid DL models is compared with baseline models using the MAC WQ dataset. LSTM, GRU and CNN-GRU show moderate performance at some learning rates. However, the performance of CNN does not come close to the performance of hybrid models at any point. Here, it can be observed from Fig. 5.4 that the hybrid CNN-LSTM model has the best and consistent performance compared to other models.

### 5.3.2 Epoch ( $E_p$ )

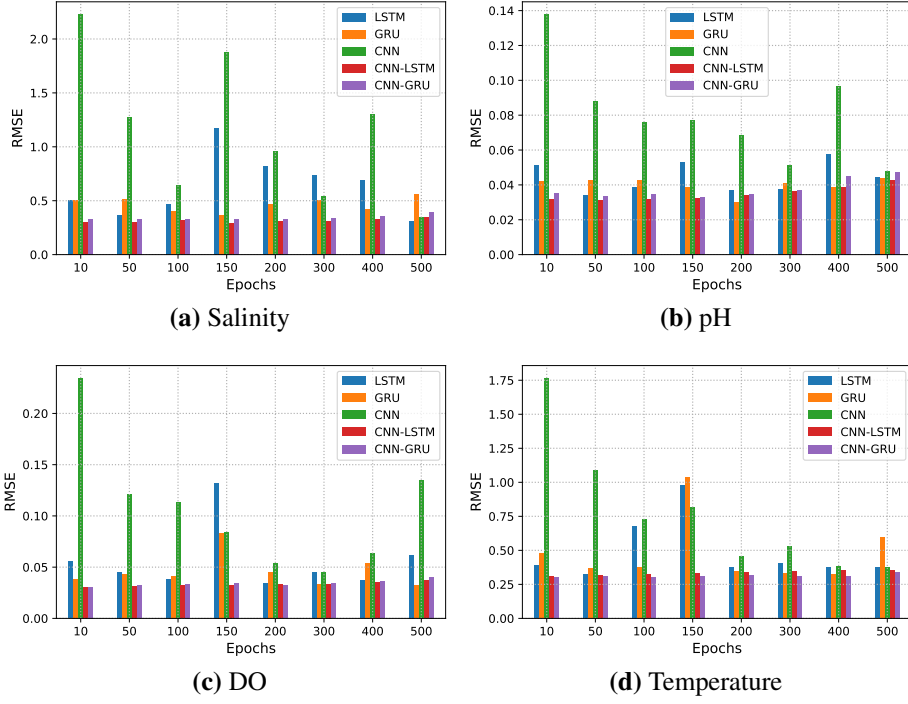
The outcome of choosing different epochs  $\{10, 50, 100, 150, 200, 300, 400, 500\}$  for each DL model is studied. Figure 5.5 shows the RMSE vs epoch for different WQ parameters using ADAK WQ dataset for the proposed hybrid DL models and LSTM, GRU as well as CNN DL models. Here the proposed hybrid DL models CNN-LSTM and CNN-GRU maintain better performance than baseline DL models LSTM, GRU and CNN. Hybrid models achieve good performance within 50 to 100 epochs, and the generalization is attained within 100 epochs. Prediction accuracy performance is maintained for all four WQ parameters for epochs 10 to 500. Also, the prediction performance of our hybrid DL models is higher compared to the baseline DL models. As well as baseline DL models are taking at least 400 epochs to achieve generalization and is under-fitting training data for epochs less than 400. The baseline DL models are close to the performance of the proposed models at 500 epochs. The baseline models require more computation resources to achieve similar results. The hybrid models start



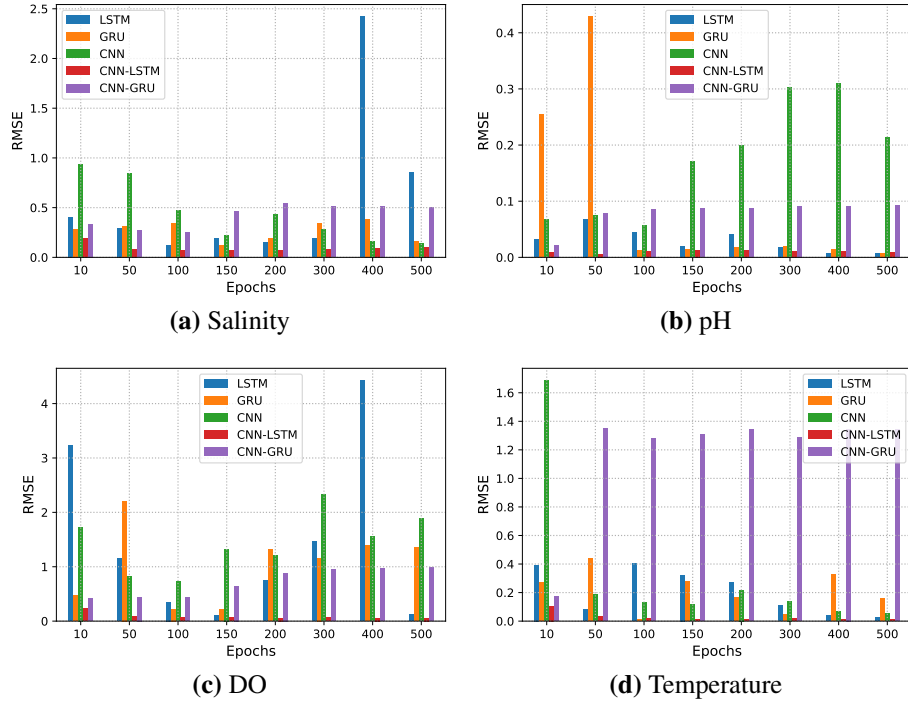
**Figure 5.3:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $L_r$  and fixed  $h_p$  ( $E_p = 100$ ,  $W_z = 30$ ,  $B_z = 32$ ) on the ADAK water quality dataset



**Figure 5.4:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $L_r$  and fixed  $h_p$  ( $E_p = 50$ ,  $W_z = 30$ , and  $B_z = 64$ ) on the MAC water quality dataset



**Figure 5.5:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $E_p$  and fixed  $h_p$  ( $L_r = 0.0008$ ,  $W_z = 30$ ,  $B_z = 32$ , and  $O_p$  is Adam) on the ADAK water quality dataset



**Figure 5.6:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $E_p$  and fixed  $h_p$  ( $L_r = 0.0008$ ,  $W_z = 30$ ,  $B_z = 64$ , and  $O_p$  is Adam) on the MAC water quality dataset

to overfit a little after 150 epochs for the ADAK WQ dataset. Still, the accuracy is maintained, indicating the adaptability of the proposed DL model to different datasets.

To further evaluate the performance, the experiment is repeated with the MAC WQ dataset. Figure 5.6 shows the RMSE vs epoch for all four WQ parameters on the MAC dataset of the proposed hybrid DL models, LSTM, GRU and CNN. Here the proposed hybrid DL models CNN-LSTM maintain better performance than the other DL models for predictions of WQ parameters. In this dataset, our model attained the required accuracy and generalization in 50 epochs. Furthermore, the accuracy loss is minimal after increasing epochs, and the CNN-LSTM model is neither over-fitting nor under-fitting the training data as the epochs is increased from 10 to 500 epochs.

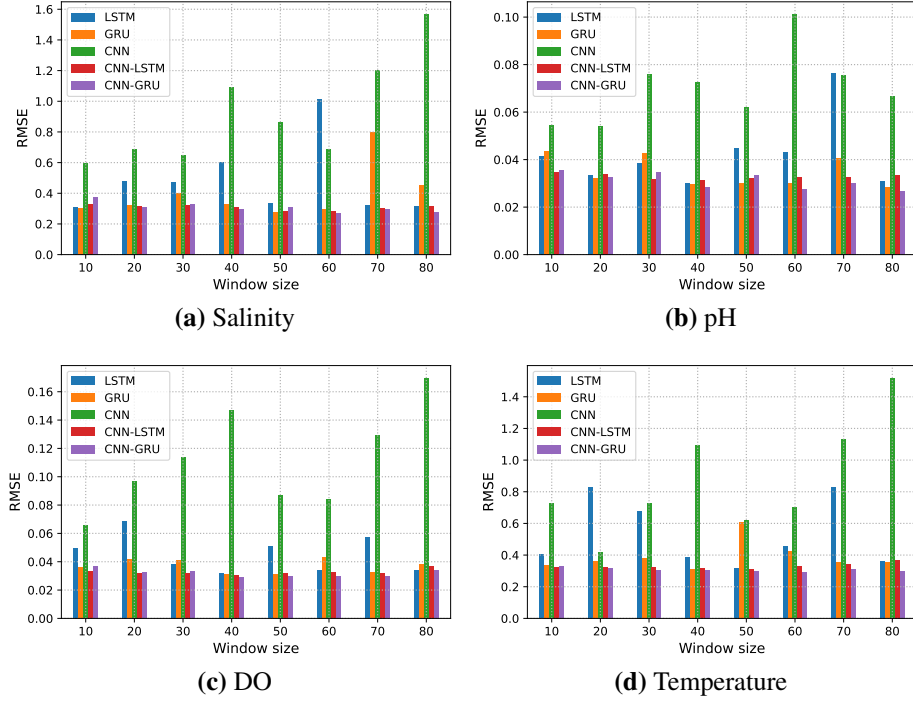
### 5.3.3 Window size ( $W_z$ )

The influence of choosing different window size  $\{10, 20, 30, 40, 50, 60, 70, 80\}$  for each DL model is analysed. Figure 5.7 shows the RMSE vs window size for different WQ parameters on the ADAK WQ dataset with the proposed hybrid DL models and the baseline DL models. The proposed hybrid DL models perform better than baseline DL models. As the window size is increased from 10 to 80, the error remains constant for hybrid DL models. In this experiment, it can be seen that the baseline models are comparable in performance with hybrid DL models for some window sizes for each WQ parameter. However, no baseline models are performing consistently well for all four WQ parameters. For example, the performance of LSTM is slightly better than CNN-LSTM at window size 40 and 80 for pH. However, for the other three WQ parameters, the performance of LSTM is not good. In comparison, proposed hybrid models consistently show good performance for the four WQ parameters and the window sizes experimented with.

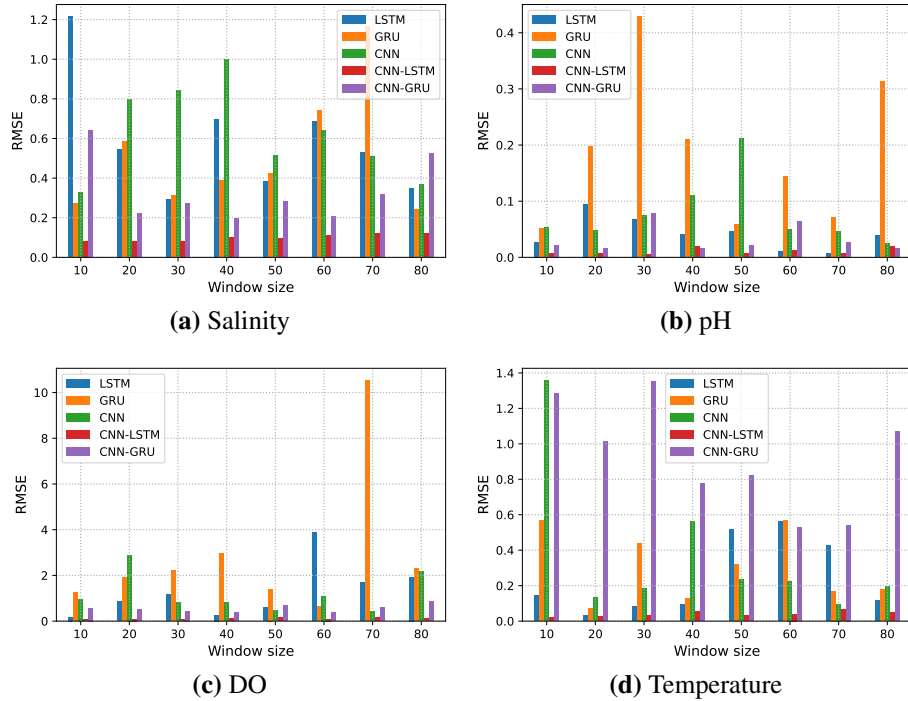
To further analyse the performance of these models, the experiment is repeated with the MAC WQ dataset. Figure 5.8 shows the RMSE vs window size for all four WQ parameters on the MAC dataset with the proposed hybrid DL models, LSTM, GRU as well as CNN. Furthermore, it can be observed from Fig. 5.8 that the hybrid CNN-LSTM model is having good performance consistently when compared to the other models.

### 5.3.4 Batch size ( $B_z$ )

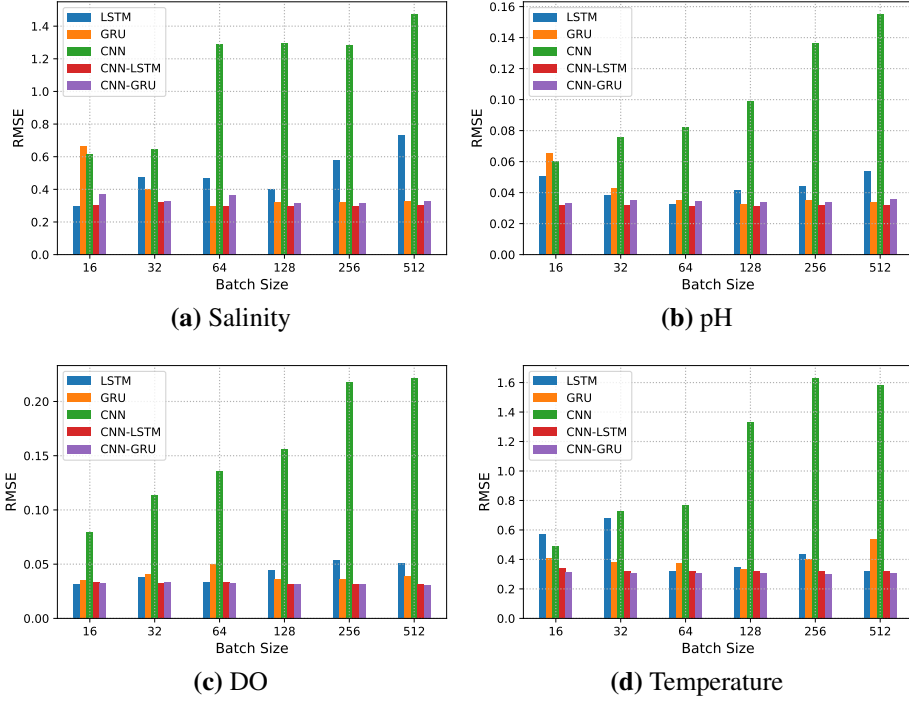
In this experiment, the performance of each DL model is analysed on different batch sizes  $\{16, 32, 64, 128, 256, 512\}$ . Figure 5.9 shows the RMSE vs batch size for different WQ parameters on the ADAK WQ dataset for the proposed hybrid DL models with the baseline DL models. Compared to the proposed hybrid DL models, it can be



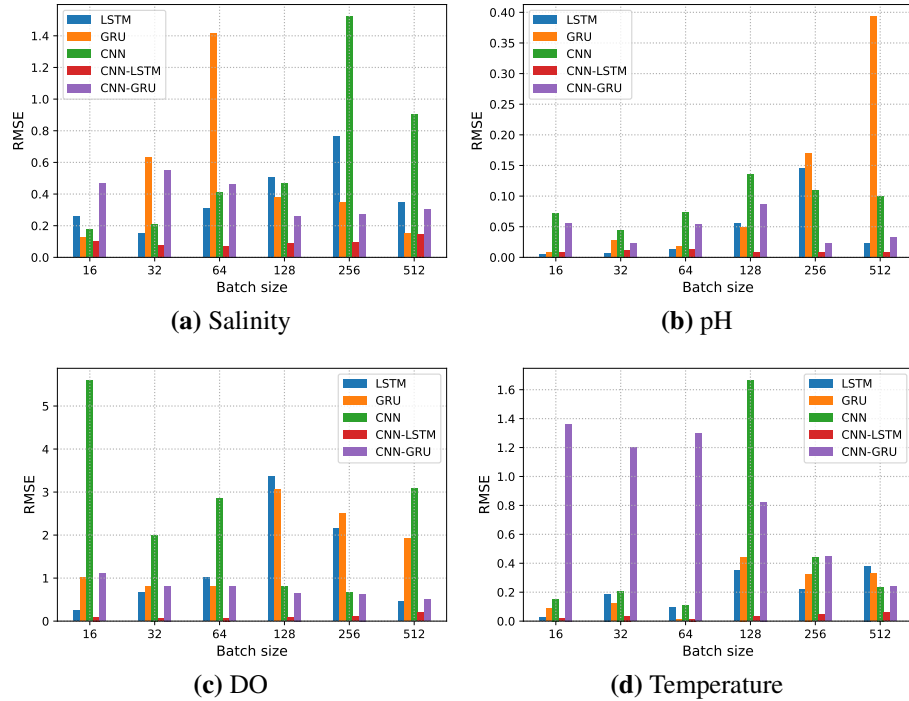
**Figure 5.7:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $W_z$  and fixed  $h_p$  ( $L_r = 0.0008$ ,  $E_p = 100$ ,  $B_z = 32$  and  $O_p$  is Adam) on the ADAK water quality dataset



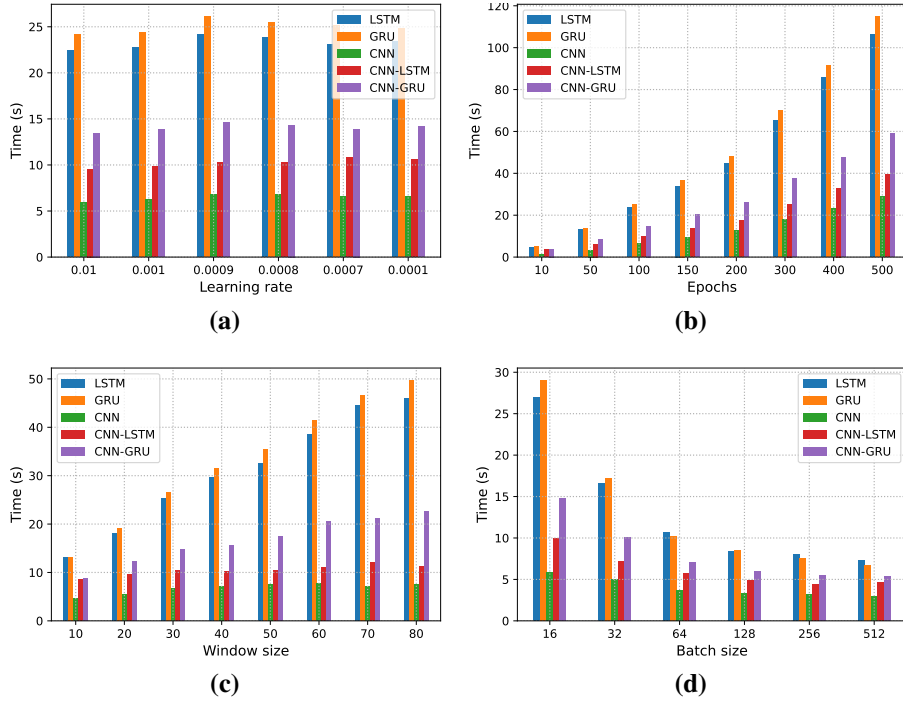
**Figure 5.8:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $W_z$  and fixed  $h_p$  ( $L_r = 0.0008$ ,  $E_p = 50$ ,  $B_z = 64$  and  $O_p$  is Adam) on the MAC water quality dataset



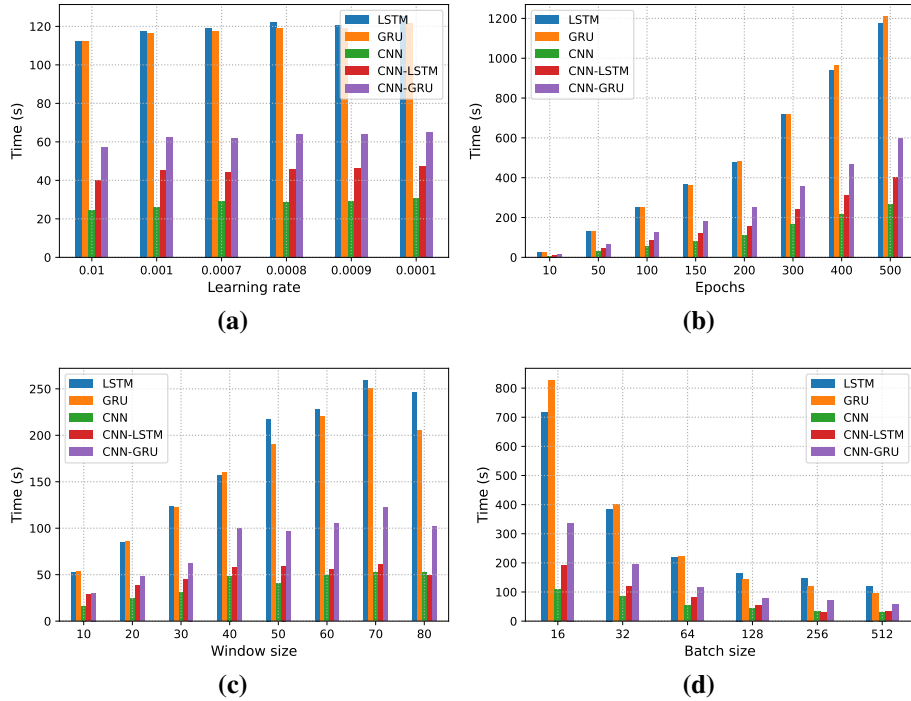
**Figure 5.9:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $B_z$  and fixed  $h_p$  ( $L_r = 0.0008$ ,  $E_p = 100$ ,  $W_z = 30$ , and  $O_p$  is Adam) on the ADAK water quality dataset



**Figure 5.10:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $B_z$  and fixed  $h_p$  ( $L_r = 0.0008$ ,  $E_p = 100$ ,  $W_z = 30$ , and  $O_p$  is Adam) on the MAC water quality dataset



**Figure 5.11:** Comparison of computation time for prediction using all DL-RNN models. (a)  $L_r = \{0.01, 0.001, 0.0009, 0.0008, 0.0007, 0.0001\}$ , (b)  $E_p = \{10, 100, 150, 200, 300, 400, 500\}$ , (d)  $W_z = \{10, 20, 30, 40, 50, 60, 70, 80\}$ , and (e)  $B_z = \{16, 32, 64, 128, 256, 512\}$  on ADAK water quality dataset



**Figure 5.12:** Comparison of computation time for prediction using all DL-RNN models. (a)  $L_r = \{0.01, 0.001, 0.0009, 0.0008, 0.0007, 0.0001\}$ , (b)  $E_p = \{10, 100, 150, 200, 300, 400, 500\}$ , (d)  $W_z = \{10, 20, 30, 40, 50, 60, 70, 80\}$ , and (e)  $B_z = \{16, 32, 64, 128, 256, 512\}$  on MAC water quality dataset

observed that the performance of the baseline DL models is inconsistent and inferior. Furthermore, it can be seen that the performance of CNN is reducing with the increase of the batch size.

Figure 5.10 shows the RMSE vs batch size for different WQ parameters using the MAC WQ dataset with the proposed hybrid DL models and the baseline DL models. As this dataset has more data points, it can be seen from the plots the performance difference of each model. From the results, it can be observed that the proposed hybrid CNN-LSTM models have better performance compared to other baseline models and the hybrid CNN-GRU model. Here also, the baseline DL models are inconsistent and underperform compared to the proposed hybrid DL models.

### 5.3.5 Computation time

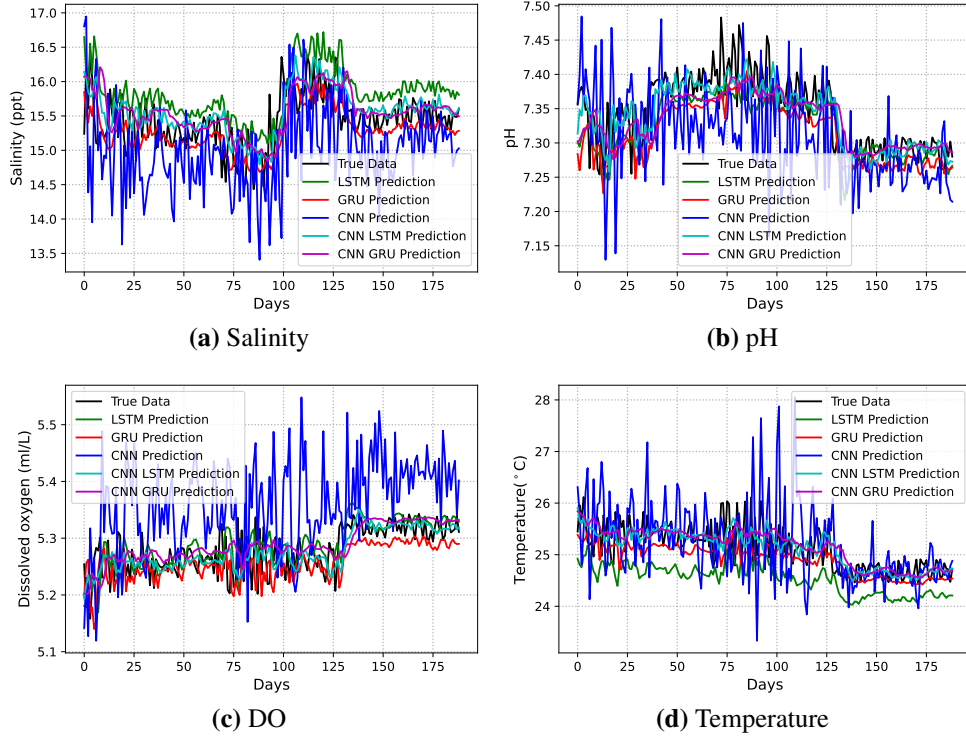
In the above four experiments, the computation time was also simultaneously measured and stored. Figure 5.11 and Fig. 5.12 plot the computation time for each hyperparameter for the ADAK WQ dataset and MAC WQ dataset, respectively. The computation time required by each DL model is different. Here some common trends from the plots can be observed. One among them is that the computation time increases with an increase in epochs and window size. But computation time is reduced with an increase in batch size. Significant changes in computation time based on different learning rates are not observable due to the constrained range of learning rates applied consistently across all models for both datasets.

In summary, for WQP, the proposed CNN-LSTM model maintains the superior performance in various experimental conditions in terms of accuracy and computation time, indicating the adaptability of the proposed CNN-LSTM DL model.

## 5.4 RESULTS AND DISCUSSIONS

The aim is to assess how effectively hybrid DL models, CNN-LSTM and CNN-GRU, perform compared to LSTM, GRU, CNN DL models to predict aquaculture WQ parameters. In this section, the comparison involves assessing the performance of proposed hybrid DL models against the baseline DL models. The evaluation employs a fixed set of  $h_p$  across the two datasets.

The  $h_p$  is selected by studying the performance of the DL models for various hyperparameters (epochs, window size, learning rate and batch size) in terms of prediction accuracy and computation time. The performance of each model is studied through various experiments by varying the hyperparameters. The optimal set of hyperparameters in terms of prediction accuracy and computation time is selected to run the final



**Figure 5.13:** Comparison of predicted values using LSTM, GRU, CNN, CNN-LSTM, and CNN-GRU predictions and ADAK water quality observed data. (a) Salinity, (b) pH, (c) DO, (d) Temperature

experiment for comparing the performance of all the DL models.

#### 5.4.1 ADAK water quality dataset

Figure 5.13 compares the predicted values with the true data on the ADAK WQ dataset using different DL models. All these models are trained using 80% of the ADAK WQ dataset and tested with 20% of the data. The predicted values are compared with the true data. Here a set of  $h_p$  (epochs = 100, learning rate = 0.0008, batch size = 32, window size = 30 and Adam optimizer) is fixed. The models are evaluated using performance metrics MAE, MSE, RMSE, and MAPE in the training and testing periods. The computation time required for each of the models is also calculated.

For the ADAK WQ dataset, the prediction performance of all models is shown in Table 5.1. From the results its clear that the proposed CNN-LSTM model (MAE = 0.2509 ppt, MSE = 0.1035 ppt, RMSE = 0.3217 ppt and MAPE = 0.0164 ppt) outperforms all baseline DL models in the training and testing period. For the salinity prediction, the LSTM model (MAE = 0.3942 ppt, MSE = 0.2230 ppt, RMSE = 0.4722 ppt and MAPE = 0.0257 ppt) outperforms the CNN model (MAE = 0.4800 ppt, MSE = 0.4180 ppt, RMSE = 0.6466 ppt and MAPE = 0.0312 ppt). When the performance of WQP is compared with other works [148], a similar trend can be observed where

**Table 5.1:** Prediction accuracy of LSTM, GRU, CNN, CNN-LSTM and CNN-GRU for ADAK water quality dataset

| Water Quality Parameter | Model    | Training Period |        |        |        |          | Testing Period |        |        |        |          |
|-------------------------|----------|-----------------|--------|--------|--------|----------|----------------|--------|--------|--------|----------|
|                         |          | MAE             | MSE    | RMSE   | MAPE   | Time (s) | MAE            | MSE    | RMSE   | MAPE   | Time (s) |
| Salinity (ppt)          | LSTM     | 0.2741          | 0.1287 | 0.3588 | 0.0184 | 23.47    | 0.3942         | 0.2230 | 0.4722 | 0.0257 | 0.05     |
|                         | GRU      | 0.3550          | 0.1843 | 0.4293 | 0.0237 | 25.02    | 0.3270         | 0.1614 | 0.4017 | 0.0212 | 0.05     |
|                         | CNN      | 0.4180          | 0.2911 | 0.5396 | 0.0282 | 05.55    | 0.4800         | 0.4180 | 0.6466 | 0.0312 | 0.04     |
|                         | CNN-LSTM | 0.2274          | 0.0917 | 0.3028 | 0.0153 | 10.27    | 0.2509         | 0.1035 | 0.3217 | 0.0164 | 0.04     |
|                         | CNN-GRU  | 0.2434          | 0.1066 | 0.3264 | 0.0164 | 14.00    | 0.2475         | 0.1100 | 0.3317 | 0.0161 | 0.04     |
| pH                      | LSTM     | 0.0321          | 0.0017 | 0.0409 | 0.0044 | 22.72    | 0.0300         | 0.0015 | 0.0386 | 0.0041 | 0.05     |
|                         | GRU      | 0.0341          | 0.0018 | 0.0427 | 0.0046 | 25.32    | 0.0353         | 0.0018 | 0.0429 | 0.0048 | 0.05     |
|                         | CNN      | 0.0498          | 0.0037 | 0.0608 | 0.0068 | 05.49    | 0.0600         | 0.0058 | 0.0759 | 0.0082 | 0.04     |
|                         | CNN-LSTM | 0.0220          | 0.0009 | 0.0293 | 0.0030 | 10.37    | 0.0237         | 0.0010 | 0.0317 | 0.0032 | 0.04     |
|                         | CNN-GRU  | 0.0259          | 0.0012 | 0.0345 | 0.0035 | 14.93    | 0.0269         | 0.0012 | 0.0349 | 0.0037 | 0.04     |
| DO (ml/L)               | LSTM     | 0.0290          | 0.0014 | 0.0374 | 0.0055 | 24.64    | 0.0295         | 0.0015 | 0.0384 | 0.0056 | 0.05     |
|                         | GRU      | 0.0331          | 0.0017 | 0.0414 | 0.0062 | 25.49    | 0.0319         | 0.0017 | 0.0411 | 0.0060 | 0.05     |
|                         | CNN      | 0.1072          | 0.0151 | 0.1229 | 0.0201 | 06.13    | 0.0974         | 0.0129 | 0.1135 | 0.0185 | 0.04     |
|                         | CNN-LSTM | 0.0226          | 0.0009 | 0.0292 | 0.0042 | 10.68    | 0.0250         | 0.0010 | 0.0321 | 0.0047 | 0.04     |
|                         | CNN-GRU  | 0.0258          | 0.0012 | 0.0346 | 0.0049 | 14.21    | 0.0262         | 0.0011 | 0.0334 | 0.0050 | 0.04     |
| Temperature(°C)         | LSTM     | 0.5552          | 0.4080 | 0.6388 | 0.0222 | 24.40    | 0.6071         | 0.4633 | 0.6807 | 0.0240 | 0.05     |
|                         | GRU      | 0.2852          | 0.1357 | 0.3684 | 0.0114 | 24.88    | 0.2951         | 0.1444 | 0.3800 | 0.0116 | 0.05     |
|                         | CNN      | 0.5346          | 0.5422 | 0.7363 | 0.0217 | 05.51    | 0.5229         | 0.5295 | 0.7276 | 0.0207 | 0.04     |
|                         | CNN-LSTM | 0.2252          | 0.0846 | 0.2908 | 0.0091 | 10.60    | 0.2471         | 0.1036 | 0.3219 | 0.0098 | 0.04     |
|                         | CNN-GRU  | 0.2560          | 0.1162 | 0.3409 | 0.0103 | 14.08    | 0.2410         | 0.0938 | 0.3063 | 0.0096 | 0.04     |

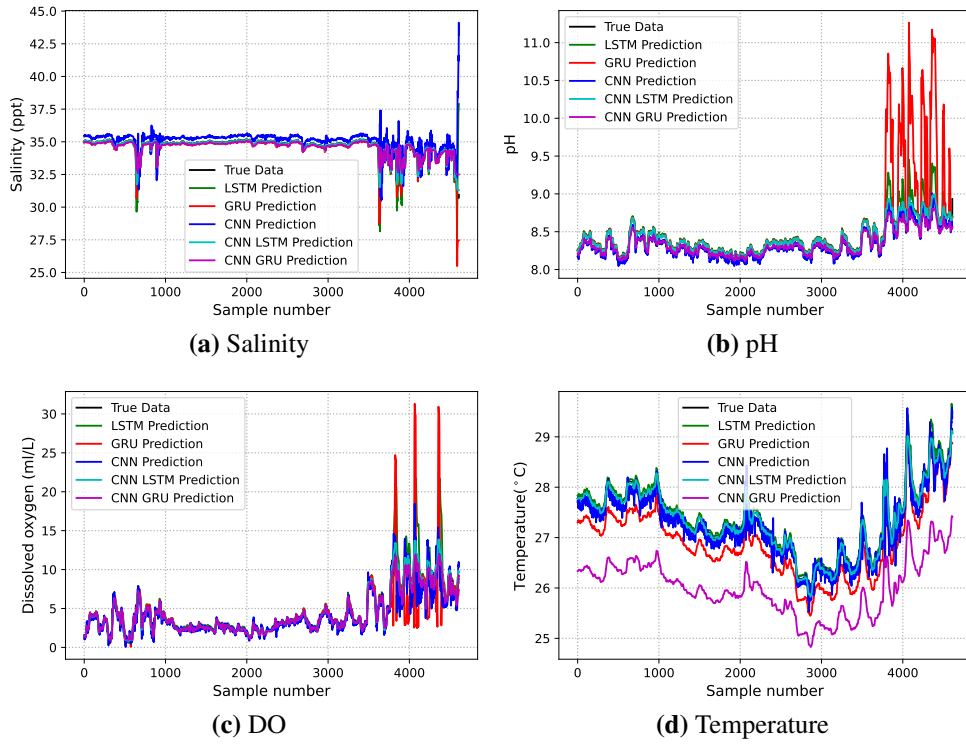
$h_p$  :  $L_r = 0.0008$ ,  $E_p = 100$ ,  $W_z = 30$ ,  $B_z = 32$ , and Adam optimizer.

LSTM have outperformed CNN in prediction accuracy. However, it is noticeable that the computation time required for training CNN is just 5.55s which is only 23.64% the computation time required by LSTM, which requires 23.72s. The LSTM model has the ability to learn from long-term dependencies, and CNN can learn features from a large dataset. However, the hybrid CNN-LSTM requires a computation time of 10.27s only for training the model, which is 43.75% of the computation time of LSTM saving 56.24% in computation time and outperforming all other models in terms of prediction accuracy.

For pH also, the CNN-LSTM model (MAE = 0.0237, MSE = 0.0010, RMSE = 0.0317 and MAPE = 0.0032) outperforms all other baseline DL models in terms of prediction accuracy. The same performance is repeated for DO (MAE = 0.0250 ml/L, MSE = 0.0010 ml/L, RMSE = 0.0321 ml/L, and MAPE = 0.0047 ml/L) and temperature (MAE = 0.2471 °C, MSE = 0.1036 °C, RMSE = 0.3219 °C, and MAPE = 0.0098 °C) prediction as well. Also, the computation time required for CNN-LSTM is around 45% of the computation time needed by LSTM and GRU models. CNN-LSTM outperforms CNN with prediction accuracy even though the CNN model requires lesser computational time when compared to CNN-LSTM. Hence, it can be concluded that CNN-LSTM is the DL model providing better performance for ADAK WQ data.

#### 5.4.2 MAC water quality dataset

Figure 5.14 compares the prediction of all DL models with the true data on the MAC WQ dataset. All these models are trained using 80% of the ADAK WQ dataset and



**Figure 5.14:** Comparison of predicted values using LSTM, GRU, CNN, CNN-LSTM, and CNN-GRU predictions and MAC water quality observed data.(a) Salinity, (b) pH, (c) DO, (d) Temperature

tested on the 20% values. The predictions are compared with the true data. Here a set of  $h_p$  (epochs = 50, learning rate = 0.0008, batch size = 64, window size = 30 and Adam optimizer) is fixed. The models are evaluated using MAE, MSE, RMSE, and MAPE in the training and testing periods. The computation time required for each of the models is also calculated.

For the MAC WQ dataset, the prediction performance of all models is shown in Table 5.2. From the table, it is clear that the proposed CNN-LSTM model (MAE = 0.0381 ppt, MSE = 0.0068 ppt, RMSE = 0.0827 ppt and MAPE = 0.0011 ppt) outperforms all baseline DL models in the training and testing period. For the salinity prediction, the LSTM model (MAE = 0.0535 ppt, MSE = 0.0860 ppt, RMSE = 0.2933 ppt and MAPE = 0.016 ppt) outperforms the CNN model (MAE = 0.4783 ppt, MSE = 0.7089 ppt, RMSE = 0.8240 ppt and MAPE = 0.0141 ppt). But it is noticeable that the computation time required for training CNN is just 21.88s which is only 19.57% the computation time required by LSTM, which requires 111.75s. The hybrid CNN-LSTM requires a computation time of 36.83s only for training the model, which is 32.95% of LSTM saving 67.04% in computation time and outperforming all other models in terms of prediction accuracy.

For pH also, the CNN-LSTM model (MAE = 0.0042, MSE = 0.00004, RMSE =

**Table 5.2:** Prediction accuracy of LSTM, GRU, CNN, CNN-LSTM and CNN-GRU for MAC water quality dataset

| Water Quality Parameter | Model    | Training Period |         |        |        |          | Testing Period |         |        |        |          |
|-------------------------|----------|-----------------|---------|--------|--------|----------|----------------|---------|--------|--------|----------|
|                         |          | MAE             | MSE     | RMSE   | MAPE   | Time (s) | MAE            | MSE     | RMSE   | MAPE   | Time (s) |
| Salinity (ppt)          | LSTM     | 0.0173          | 0.0010  | 0.0320 | 0.0005 | 111.75   | 0.0535         | 0.0860  | 0.2933 | 0.0016 | 0.35     |
|                         | GRU      | 0.0210          | 0.0007  | 0.0258 | 0.0006 | 114.13   | 0.0571         | 0.0975  | 0.3123 | 0.0018 | 0.31     |
|                         | CNN      | 0.3612          | 0.1455  | 0.3815 | 0.0104 | 021.88   | 0.4783         | 0.7089  | 0.8420 | 0.0141 | 0.11     |
|                         | CNN-LSTM | 0.0241          | 0.0011  | 0.0338 | 0.0007 | 36.83    | 0.0381         | 0.0068  | 0.0827 | 0.0011 | 0.17     |
|                         | CNN-GRU  | 0.0989          | 0.0159  | 0.1259 | 0.0029 | 055.74   | 0.1460         | 0.0736  | 0.2714 | 0.0043 | 0.19     |
| pH                      | LSTM     | 0.0166          | 0.0003  | 0.0175 | 0.0020 | 116.86   | 0.0339         | 0.0047  | 0.0689 | 0.0039 | 0.35     |
|                         | GRU      | 0.0041          | 0.0001  | 0.0101 | 0.0005 | 115.93   | 0.1358         | 0.1839  | 0.4288 | 0.0154 | 0.31     |
|                         | CNN      | 0.0611          | 0.0042  | 0.0645 | 0.0074 | 022.56   | 0.0696         | 0.0055  | 0.0744 | 0.0083 | 0.11     |
|                         | CNN-LSTM | 0.0031          | 0.00002 | 0.0045 | 0.0004 | 039.07   | 0.0042         | 0.00004 | 0.0063 | 0.0005 | 0.17     |
|                         | CNN-GRU  | 0.0429          | 0.0024  | 0.0492 | 0.0051 | 058.74   | 0.0673         | 0.0063  | 0.0792 | 0.0079 | 0.19     |
| DO (ml/L)               | LSTM     | 0.3148          | 0.1310  | 0.3620 | 0.0803 | 116.31   | 0.4767         | 1.3341  | 1.1550 | 0.0873 | 0.36     |
|                         | GRU      | 0.2235          | 0.0762  | 0.2760 | 0.0569 | 119.47   | 0.6789         | 4.8710  | 2.2070 | 0.0785 | 0.32     |
|                         | CNN      | 0.3300          | 0.1454  | 0.3813 | 0.0866 | 023.20   | 0.4492         | 0.6705  | 0.8188 | 0.0952 | 0.11     |
|                         | CNN-LSTM | 0.0467          | 0.0033  | 0.0572 | 0.0129 | 040.76   | 0.0566         | 0.0065  | 0.0804 | 0.0145 | 0.18     |
|                         | CNN-GRU  | 0.1094          | 0.0263  | 0.1623 | 0.0346 | 060.21   | 0.2439         | 0.1961  | 0.4428 | 0.0509 | 0.20     |
| Temperature(°C)         | LSTM     | 0.0663          | 0.0053  | 0.0728 | 0.0026 | 118.71   | 0.0672         | 0.0067  | 0.0818 | 0.0025 | 0.36     |
|                         | GRU      | 0.3977          | 0.1607  | 0.4009 | 0.0157 | 120.67   | 0.4373         | 0.1943  | 0.4408 | 0.0160 | 0.32     |
|                         | CNN      | 0.1138          | 0.0333  | 0.1825 | 0.0046 | 023.76   | 0.1441         | 0.0349  | 0.1867 | 0.0052 | 0.12     |
|                         | CNN-LSTM | 0.0166          | 0.0006  | 0.0246 | 0.0007 | 043.21   | 0.0216         | 0.0009  | 0.0303 | 0.0008 | 0.18     |
|                         | CNN-GRU  | 0.8879          | 0.9603  | 0.9799 | 0.0340 | 062.96   | 1.3416         | 1.8252  | 1.3510 | 0.0489 | 0.20     |

$h_p$  :  $L_r = 0.0008$ ,  $E_p = 50$ ,  $W_z = 30$ ,  $B_z = 32$ , and Adam optimizer.

0.0063 and MAPE = 0.0005) outperforms all other models in terms of prediction accuracy. The same performance is repeated for DO (MAE = 0.0566 ml/L, MSE = 0.0065 ml/L, RMSE = 0.0804 ml/L, and MAPE = 0.0145 ml/L) and temperature (MAE = 0.0216 °C, MSE = 0.0009 °C, RMSE = 0.0303 °C, and MAPE = 0.0008 °C) prediction as well. Also, the computation time required for CNN-LSTM is around 33% of the computation time needed by LSTM and GRU models. CNN model has lesser computational time when compared to CNN-LSTM. Hence, it can be concluded that CNN-LSTM is the DL model providing better performance for MAC WQ data.

In the evaluation of WQP performance of different DL-RNN models on both the ADAK and MAC water quality datasets the hybrid CNN models outperformed other modes. For the ADAK dataset, the hybrid CNN-LSTM model consistently outperformed all baseline models, achieving lower MAE, MSE, RMSE, and MAPE values. The CNN-LSTM model also demonstrated efficient training times compared to LSTM, making it a superior choice for salinity, pH, DO, and temperature prediction. Similarly, on the MAC dataset, the CNN-LSTM model excelled in prediction accuracy and computation time efficiency, establishing its superiority over LSTM, GRU, CNN, and CNN-GRU for salinity, pH, DO, and temperature forecasting.

## 5.5 SUMMARY

In this chapter, hybrid DL models, CNN-LSTM and CNN-GRU are proposed for aquaculture WQP. The developed hybrid prediction models were trained and tested on two distinct datasets. The impact of varying the  $h_p$  is extensively analysed. For perfor-

mance comparison and further analysis, optimal  $h_p$  were used. The performance of these hybrid DL models (CNN-LSTM and CNN-GRU) is compared with baseline DL models (LSTM, GRU, and CNN) in terms of MAE, MSE, RMSE, and MAPE. Results show that the CNN-LSTM hybrid model provides significant improvement in prediction accuracy as well as computation time compared to the baseline DL models, offering a realistic solution for predicting WQ parameters in smart aquaculture.

## CHAPTER 6

### ATTENTION MECHANISM BASED DEEP LEARNING MODELS FOR IMPROVING THE ACCURACY OF WATER QUALITY PREDICTION IN AQUACULTURE

This chapter investigates the development and implementation of attention mechanism-based DL models for enhancing the accuracy of WQP in aquaculture. To address this, ABi-LSTM and ABi-GRU DL-RNN models are proposed, specifically tailored for aquaculture WQP. The proposed models excel in their fusion of two essential elements: the attention mechanism's weighting feature and the non-linear mapping capability of Bi-LSTM and Bi-GRU. This unique combination allows the proposed DL-RNN models to offer enhanced accuracy and efficiency in WQP, making them a promising solution to the limitations faced by traditional methods. Section 6.1 of this chapter serves as an introductory section, outlining the key contributions of this work. Section 6.2 of this chapter explains the proposed ABD-RNN models. A critical aspect of building an accurate DL WQP model is the identification of the  $h_p$ . Section 6.3 presents an extensive study analysing the impact of various  $h_p$  variations on the performance of the proposed ABi-LSTM and ABi-GRU DL-RNN models. To facilitate a comprehensive comparison, two different WQ datasets are utilized for the experimental evaluations, considering existing models such as LSTM, GRU, Bi-LSTM, Bi-GRU, and A-LSTM and A-GRU. Section 6.4 delves deeper into the performance analysis of all the DL-RNN models for a selected set of  $h_p$ , ultimately presenting the final results and comparisons. The chapter is summarized in Section 6.5.

#### 6.1 INTRODUCTION

The proliferation of DL technologies has led to more sophisticated mechanisms to parse and interpret complex patterns in data. A prime example is the attention mechanism, a feature central to this chapter, which focuses on its deployment within WQP. With proven efficacy in various DL tasks, the attention mechanism assigns differential attention or weights to distinct parts of the input data, thereby directing the model to focus on the most salient features.

This functionality proves instrumental when managing time series data, such as

those in WQP. It enables the model to determine the significance of different time steps in the prediction process, capturing essential temporal patterns within the data. Attention mechanism based RNN models, specifically attention-based LSTM and GRU models, are introduced to leverage these advantages. These models are further optimized by incorporating bidirectional LSTM/GRU layers with the attention mechanism, thus improving WQP. The efficacy of these models is critically assessed on the ADAK and MAC datasets, highlighting the significance of attention mechanisms in enhancing both model accuracy and computational efficiency.

Attention-based DL-RNN models have proven valuable in various time-series prediction tasks. For instance, in [124], a self-attention-based time-varying prediction model with a two-stage training method is proposed to improve performance compared to baseline DL-RNN models. Similarly, [125] introduces an integrated multiple directed attention (IMDA) DL-RNN model for air pollution prediction, demonstrating superior performance over other DL models. Furthermore, in [126], a self-attention-based forecasting model for regional influenza prediction (SAIFlu-Net) outperforms baseline DL-RNN models.

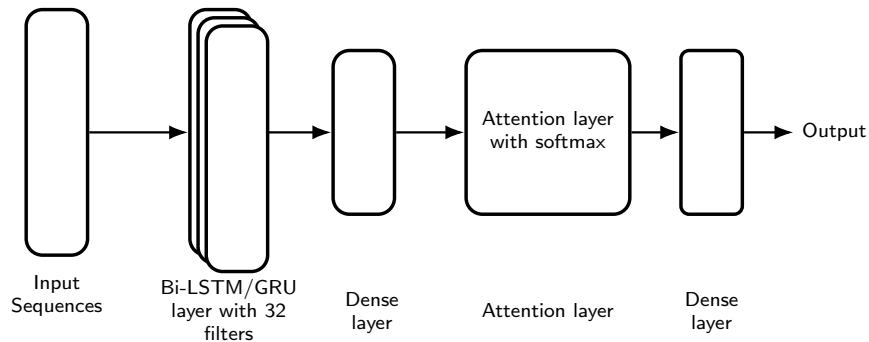
In order to meet the requirement of high prediction accuracy for high-intensity aquaculture, ABD-RNN models are proposed in this chapter. Bi-LSTM and Bi-GRU incorporate additional training layers that allow input data to be trained twice to improve the accuracy of predictions. In this work, the prediction accuracy and training time of the proposed ABi-LSTM and ABi-GRU models are compared with LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM and A-GRU DL-RNN models for WQP in aquaculture. The significant contributions of this work are as follows:

1. ABD-RNN (ABi-LSTM and ABi-GRU) models are proposed. The bi-directional models process the data in both forward and backward directions, enabling the model to capture more patterns than the unidirectional model along with the attention mechanism.
2. Attention mechanism is implemented with bidirectional models to take advantage of the weighing feature of the attention mechanism.
3. Extensive study of the impact of  $h_p$  on the performance of the proposed ABD-RNN models using the two different WQ datasets is presented. Determining the right set of hyper-parameters has a significant role in building an accurate DL-RNN WQP model.
4. The performance of ABD-RNN (ABi-LSTM and ABi-GRU) models are compared with baseline (LSTM and GRU), bidirectional (Bi-LSTM and Bi-GRU),

and attention-based (A-LSTM and A-GRU) DL-RNN models. Results show that proposed models significantly improve aquaculture WQP accuracy and are computationally efficient.

## 6.2 PROPOSED ABD-RNN MODELS

WQP is a typical time series problem, which can be considered to be a sequential machine learning regression problem. For training the model, this sequential time series data is converted into a set of feature values and the corresponding target value using the sliding window procedure since the regression is a supervised learning problem. The predicted output depends on the current input and the entire length of the time sequence with different levels of importance or weightage. Many prediction models map relationships with the input and output data at a particular time, ignoring the features in the lengthy sequence. To capture the features from this whole sequence data, bidirectional RNNs are used in the DL layer. This ensures the utilization of the information from the sequence data in the forward and backward directions. In the bidirectional NNs, with their non-linear mapping of the sequence, the DL layer accurately captures the sequence correlation of data to provide more precise prediction results. The DL layer is followed by an attention mechanism, which selectively focuses on and extracts the critical features from the sequence by giving the suitable weightage, thus improving the performance of the WQP model. The overview of the proposed ABD-RNN models is shown in Fig. 6.1.

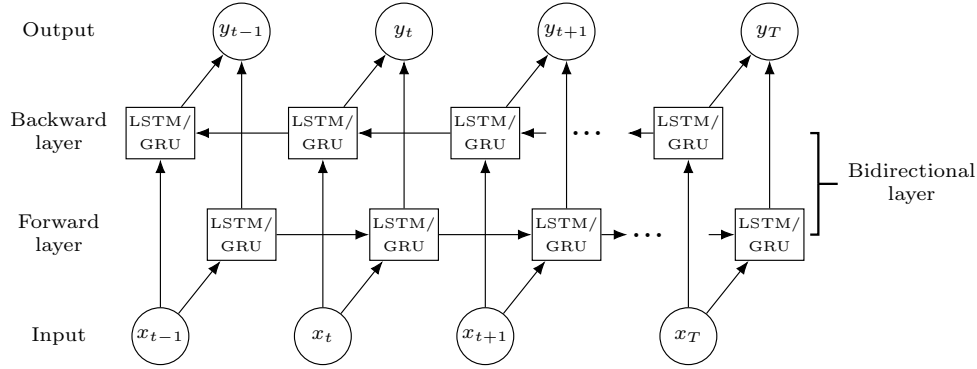


**Figure 6.1:** The proposed ABD-RNN model architecture

### 6.2.1 Bidirectional LSTM/GRU DL-RNN layer

In this DL layer, bi-directional DL-RNN (LSTM/GRU) is used. LSTM and GRU are two types of DL-RNN models used for sequence modeling tasks. LSTM was introduced to enhance long sequence learning and avoid the vanishing-gradient issue of RNN [138]. And GRU DL-RNN was proposed by authors [139] in 2014. It has per-

formance comparable with LSTM in terms of accuracy and lower demand for processing resources. Both models are designed to address the vanishing gradient problem in traditional RNNs, where the gradients of the error function can become infinitesimally small as they are propagated back through time.



**Figure 6.2:** The Bi-LSTM/GRU structure

The bi-directional DL-RNN (LSTM/GRU) structure is shown in Fig. 6.2. The bidirectional DL models consist of two processes in both forward and backward directions, as depicted in Fig. 6.2. The processing in both directions helps the model capture patterns that the unidirectional LSTM and GRU models cannot capture. The bi-directional DL-RNNs are an extension of LSTM and GRU. Bi-directional DL-RNNs can fit data in both forward and backward directions and concatenate the prediction, while LSTM/GRU can only fit data in one direction [143]. A Bi-LSTM/Bi-GRU model first feeds data into one of the LSTM/GRU models for training the given data sequence, i.e., feedback layer and re-training with another LSTM/GRU model, but with the opposite order of the input data sequence. Bi-LSTM/Bi-GRU model can capture patterns ignored by LSTM/GRU because it has added a reverse directional layer. Double LSTM/GRU application improves long-term study and thereby improves accuracy [144]. Bi-directional networks are the extension of the LSTM/GRU model, which can learn about the relation of past and future factors of load and current load, which helps in extracting the in-depth features of loaded data [145].

### 6.2.2 Attention layer

After the Bi-LSTM/Bi-GRU layers extract the features in both directions, the output is fed to the attention layer [149]. The attention mechanism emphasizes the influence of critical segments by allocating distinct weights to hidden layer elements. The attention layer automatically understands the significance of each hidden state and assigns a weight to it based on its importance. The attention layer calculates the importance of

the input features by taking the dot product of the hidden state vector and a learned weight vector. The softmax function is then applied to ensure that the sum of all weights is 1, and the weights and the input features are multiplied and summed to get the output.

$$a_{tk} = \frac{\exp(e_{tk})}{\sum_{i=1}^n \exp(e_{ti})} \quad (6.1)$$

To implement the attention layer, first the attention weights  $a_k$  are calculated using the Eq. (6.1). Then multiply the attention weights by the hidden states and sum the result to get the output  $z_t$ . The learned variables  $v_e$ ,  $b_e$ ,  $W_e$ , and  $U_e$  are updated during training to improve the model's performance. First, for each time step  $t$  and each hidden state  $h_t$ , the attention layer calculates a score  $e_{tk}$  that represents the importance of  $h_t$  for predicting the value of the  $k$ -th input feature. This is done using the Eq. (6.2).

$$e_{tk} = v_{eT}(W_e[h_{t-1}, c_{t-1}] + U_e h_t + b_e) \quad (6.2)$$

In Eq. (6.2),  $v_{eT}$ ,  $W_e$ ,  $U_e$ , and  $b_e$  are learned parameters.  $v_{eT}$  is a row vector of size  $T$ , where  $T$  is the length of the input sequence.  $W_e$  is a matrix of size  $T \times m$ , where  $m$  is the number of neurons in the hidden layer.  $U_e$  is a matrix of size  $m \times m$ , and  $b_e$  is a bias vector of size  $T \times m$ . The vector  $[h_{t-1}, c_{t-1}]$  is the concatenation of the previous hidden state  $h_{t-1}$  and cell state  $c_{t-1}$ , which are output by the LSTM or GRU layers.

The scores  $e_{tk}$  are then used to calculate the attention weights  $a_{tk}$  for each time step  $t$  and each input feature  $k$ . In Eq. (6.1), the numerator calculates the exponential of the score  $e_{tk}$ , while the denominator calculates the exponential of all scores  $e_{ti}$  for the same input feature  $k$ . The resulting attention weights  $a_{tk}$  represent the relative importance of each time step  $t$  for predicting the  $k$ -th input feature. The attention layer then computes the output  $z_t$  by taking a weighted sum of the hidden states  $h_t$ , using the attention weights  $a_{tk}$ .

$$z_t = \sum_{i=t}^T a_{ti} h_i \quad (6.3)$$

The Eq. (6.3) sums the attention weights  $a_{tk}$  for all time steps  $t$  from the current time step  $t$  to the end of the sequence  $T$ , and multiplies each weight by its corresponding hidden state  $h_t$ . The resulting output  $z_t$  is a weighted combination of all the hidden states, with higher weights given to the more important time steps as determined by the attention mechanism. Overall, the attention layer allows the model to selectively focus on the most relevant features and time steps for predicting the target variable, leading to improved prediction accuracy.

### 6.2.3 Model implementation

After standardization, a dataset is scaled to unit variance, removing the mean from the dataset. The whole dataset is normalized using the train set before training the DL-RNN models. The StandardScaler module from the Scikit-Learn Python library is utilized [146] to implement it in these DL-RNN models. The WQ sequences are broken into smaller segments once the datasets have been standardized by moving a window of the same size depending on the specified window size ( $W_z$ ). By creating many short sequences from a single lengthy series, this step reduces the input sequence to a fixed-length sample and increases the sample size for training the model. This transformed data is used to train all the DL-RNN models.

## 6.3 EXPERIMENTS: HYPERPARAMETER TUNING

In this section, two WQ data sets are used to conduct experiments to analyse and evaluate the performance of the proposed attention-based bi-directional DL-RNN models (ABi-LSTM and ABi-GRU) models. Comparing the DL-RNN models with our proposed models in terms of prediction accuracy and computation time performance, the effectiveness of the proposed models can be validated. All experiments are conducted on Microsoft Azure Virtual Machines with specifications: Inter(R) Xeon (R) 8272CL CPU @2.60GHz, 32 GB RAM. The open source software used are Python and Keras based on Tensorflow.

The ABi-LSTM and ABi-GRU DL-RNN models are compared with LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, and A-GRU DL-RNN models. In the data set, 80% of the data is used to train the model, and the remaining 20% to test the model's prediction accuracy. The most crucial task in making the most effective and efficient DL-RNN model is setting  $h_p$  and optimizing them. In this work, the focus is to improve prediction accuracy without compromising the computation efficiency of the model. To find the model with the best prediction accuracy without compromising on computation efficiency, the performance of all the DL-RNN models are extensively studied through five sets of experiments varying each  $h_p$ . The fixed  $h_p$  of each model are dropout of 0.2 is applied after the primary layers to avoid over-fitting, and *ReLU* is applied as the activation function. The loss function used in all the models is MSE. The number of neurons of each hidden layer is set to 32.

The extensive study included the experiments with the tuning of following five  $h_p$  : optimizer ( $O_p$ ), learning rate ( $L_r$ ), epochs ( $E_p$ ), window size ( $W_z$ ) and batch size ( $B_z$ ). The  $h_p$  are plotted against RMSE. The  $h_p$  are plotted against the computation time as well. All the experiments are done on both datasets.

### 6.3.1 Optimizers ( $O_p$ )

$O_p$  are algorithms that update each layer's attributes, such as weights and  $L_r$ , to reduce the loss after each iteration. A suitable  $O_p$  determines the right weight and gives the best results [150]. The performance of DL-RNN models with various  $O_p$  is studied to find the ideal one for this work. The  $O_p$  used in this work are Adam, AdaGrad, AdaMAX, Nadam, Ftrl, and RMSprop. In Fig. 6.3 the RMSE bar plots for different  $O_p$  indicating the prediction accuracy of each DL-RNN model for various WQ parameters on the ADAK WQ dataset are shown. Compared to the proposed ABD-RNN models, it can be observed that the performance of the other DL-RNN models is inadequate.

Figure 6.4 shows the RMSE bar plots for different  $O_p$  to indicate the prediction accuracy of each DL-RNN model for various WQ parameters on the MAC WQ dataset. The bar plots show the ABD-RNN models have outperformed all other models. Adam produces the best results for each DL-RNN model. Hence Adam is chosen as the default  $O_p$  for all further experiments and the final comparison of the prediction performance of all DL-RNN models.

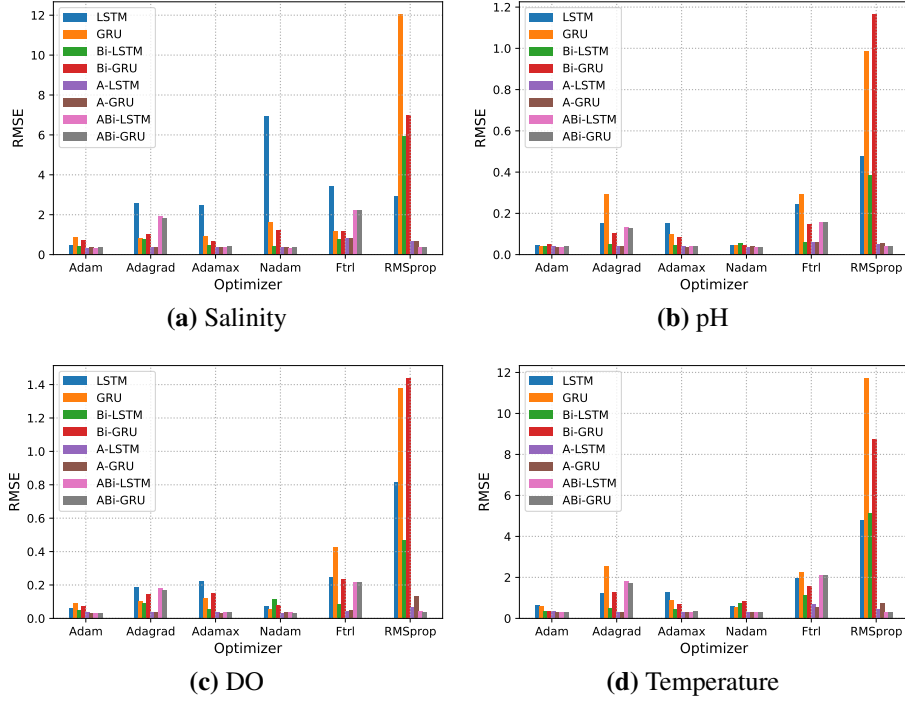
### 6.3.2 Learning rate ( $L_r$ )

This experiment investigate the prediction accuracy of each DL-RNN model for a defined set of  $L_r$   $\{0.01, 0.001, 0.0008, 0.0006, 0.0004, 0.0001\}$ . Figure 6.5 shows the bar plot RMSE for different DL-RNN models for the defined set of  $L_r$  while tested with the ADAK WQ dataset. The proposed ABD-RNN models perform better than other DL-RNN models.

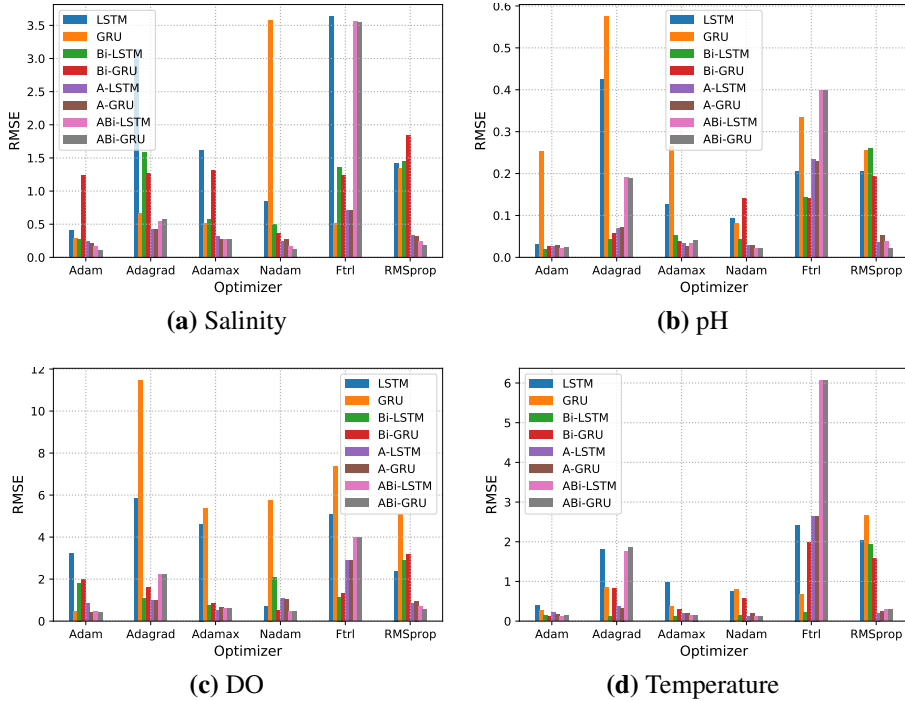
The same experiment is repeated with the MAC WQ dataset to review the performance of these DL-RNN models for a different dataset. Figure 6.6 shows the bar plots of RMSE for each DL-RNN model for various  $L_r$  with the MAC WQ dataset. Here, from Fig. 6.6 it can be observed that the bidirectional and attention-based models perform better when compared to baseline models. This experiment helps in selecting the default  $L_r$  for other experiments. From Fig. 6.6, it can be observed that all the DL-RNN models achieved their best prediction accuracy for each WQ parameter in both datasets when  $L_r = 0.001$ .

### 6.3.3 Epoch ( $E_p$ )

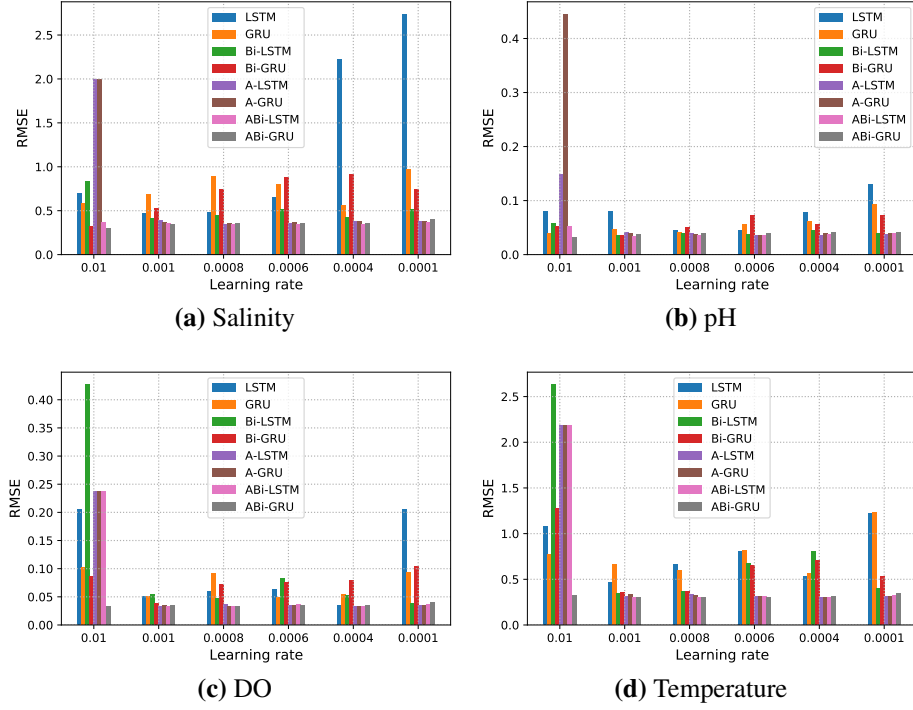
In this experiment, the outcome of choosing different  $E_p$   $\{10, 20, 30, 40, 50\}$  for each DL-RNN model is studied. Figure 6.7 shows the RMSE bar plots for various  $E_p$  for predicting each WQ parameter in the ADAK WQ dataset using different DL-RNN models. The proposed bidirectional DL-RNN models Bi-LSTM and Bi-GRU maintain



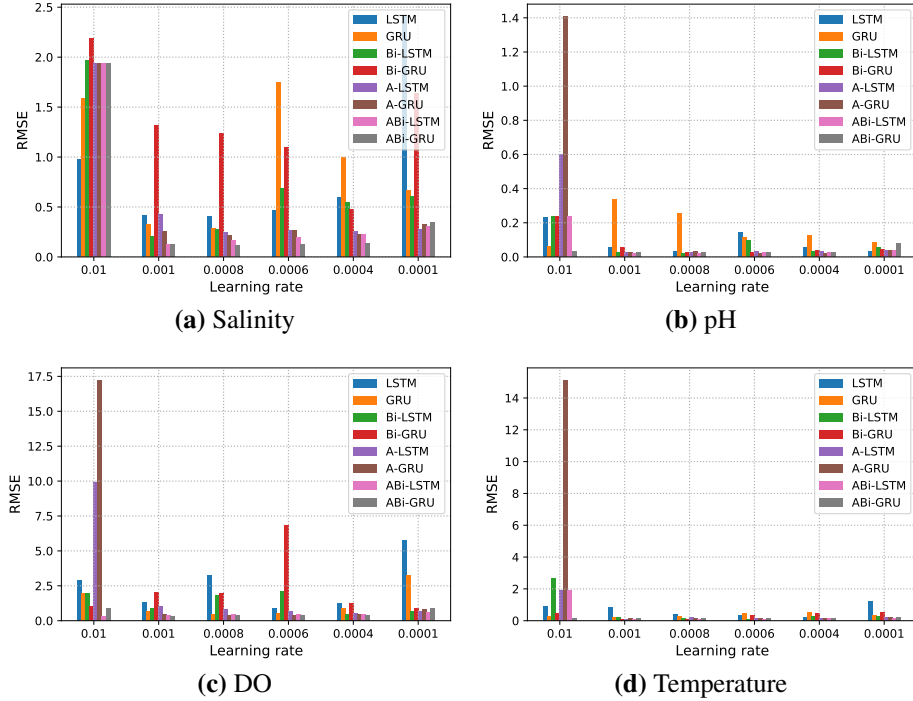
**Figure 6.3:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $O_p$  and fixed  $h_p$  ( $L_r = 0.001$ ,  $E_p = 10$ ,  $W_z = 30$ , and  $B_z = 64$ ) on the ADAK water quality dataset



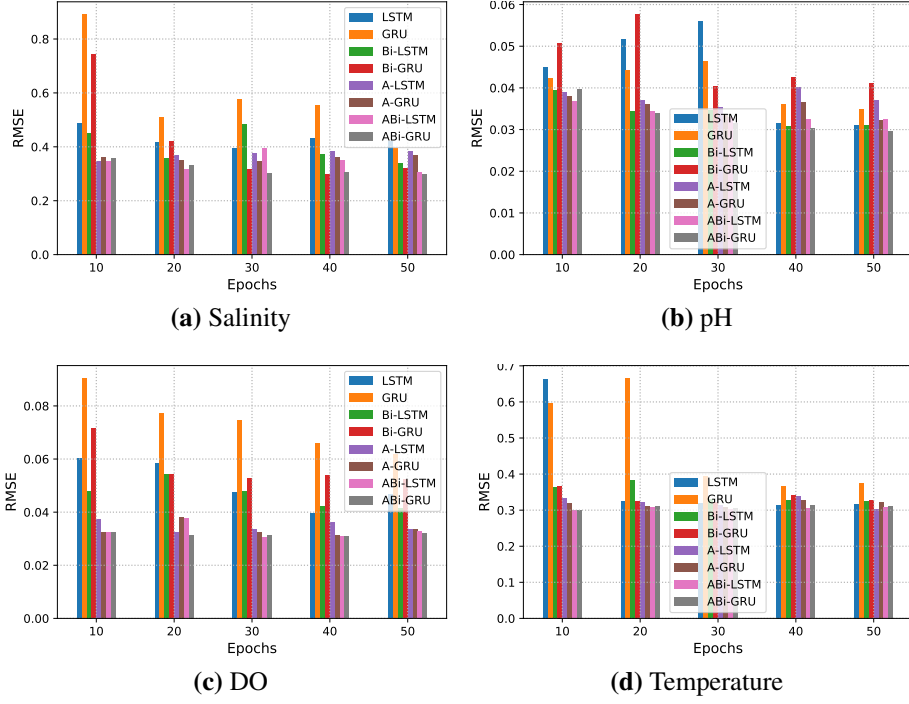
**Figure 6.4:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $O_p$  and fixed  $h_p$  ( $L_r = 0.001$ ,  $E_p = 10$ ,  $W_z = 30$ , and  $B_z = 64$ ) on the MAC water quality dataset



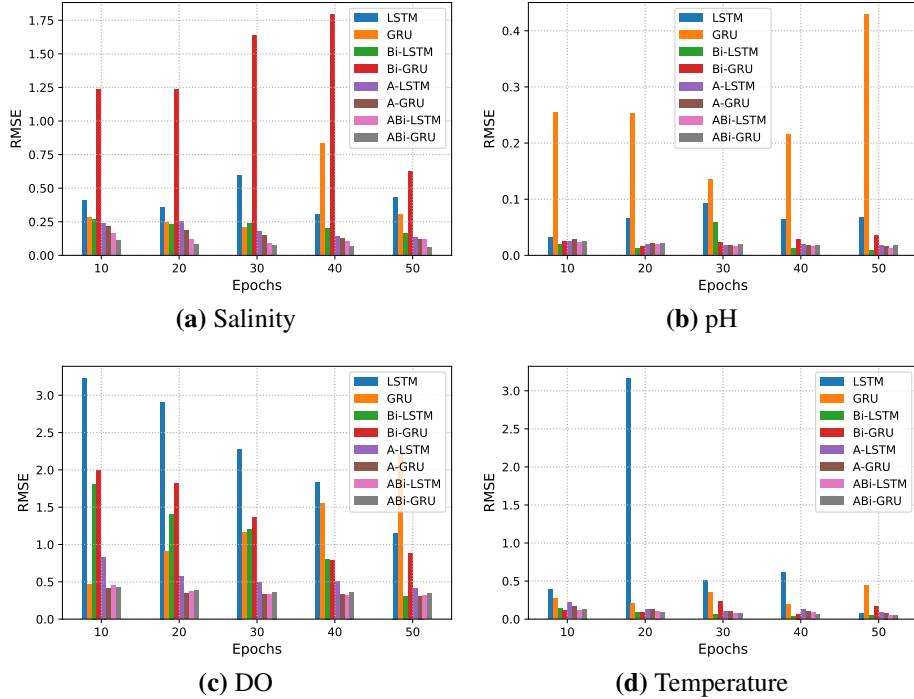
**Figure 6.5:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $L_r$  and fixed  $h_p$  ( $E_p = 10$ ,  $W_z = 30$ ,  $B_z = 64$  and  $O_p$  is Adam) on the ADAK water quality dataset



**Figure 6.6:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $L_r$  and fixed  $h_p$  ( $E_p = 10$ ,  $W_z = 30$ ,  $B_z = 64$  and  $O_p$  is Adam) on the MAC water quality dataset



**Figure 6.7:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $E_p$  and fixed  $h_p$  ( $L_r = 0.001$ ,  $W_z = 30$ ,  $B_z = 64$  and  $O_p$  is Adam) on the ADAK water quality dataset



**Figure 6.8:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $E_p$  and fixed  $h_p$  ( $L_r = 0.001$ ,  $W_z = 30$ ,  $B_z = 64$  and  $O_p$  is Adam) on the MAC water quality dataset

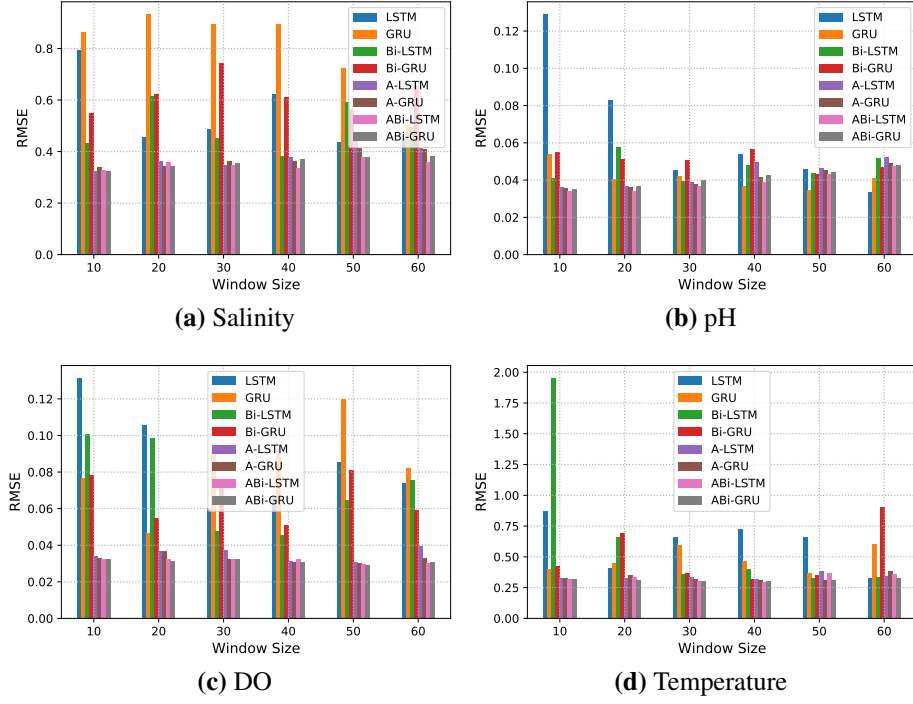
better performance than baseline DL-RNN models LSTM and GRU. Bidirectional DL-RNN and attention-based DL-RNN models achieve good performance within  $E_p = 10$  to 20, and the generalization is attained within  $E_p = 20$ . Prediction accuracy is maintained for the prediction performance of all four WQ parameters for  $E_p = 10$  to 50. Also, the bidirectional and attention-based DL-RNN models are higher than the baseline DL-RNN models. Baseline DL-RNN models require more  $E_p$  to achieve generalization and are under-fitting training data for fewer  $E_p$ . The baseline models need more computation resources to achieve the prediction accuracy of the proposed ABD-RNN (ABi-LSTM and ABi-GRU) models.

The same experiment is done with the MAC WQ dataset to evaluate the performance further. The RMSE bar plots for various  $E_p$  assess the prediction accuracy of four WQ parameters of the MAC dataset with all DL-RNN models are shown in Fig. 6.8. Here the proposed ABD-RNN models perform much better than the baseline DL-RNN models for predictions of WQ parameters. The ABD-RNN models attained good accuracy and generalization within  $E_p = 20$  in both datasets. Furthermore, the accuracy loss is minimal after increasing  $E_p$ . As the experiment is advanced, the  $E_p = 10$  is chosen for all models since all the proposed models achieved the best prediction accuracy for all WQ parameters in both datasets. In the results section, the results on both datasets is compared with  $E_p = 10$  and 50.

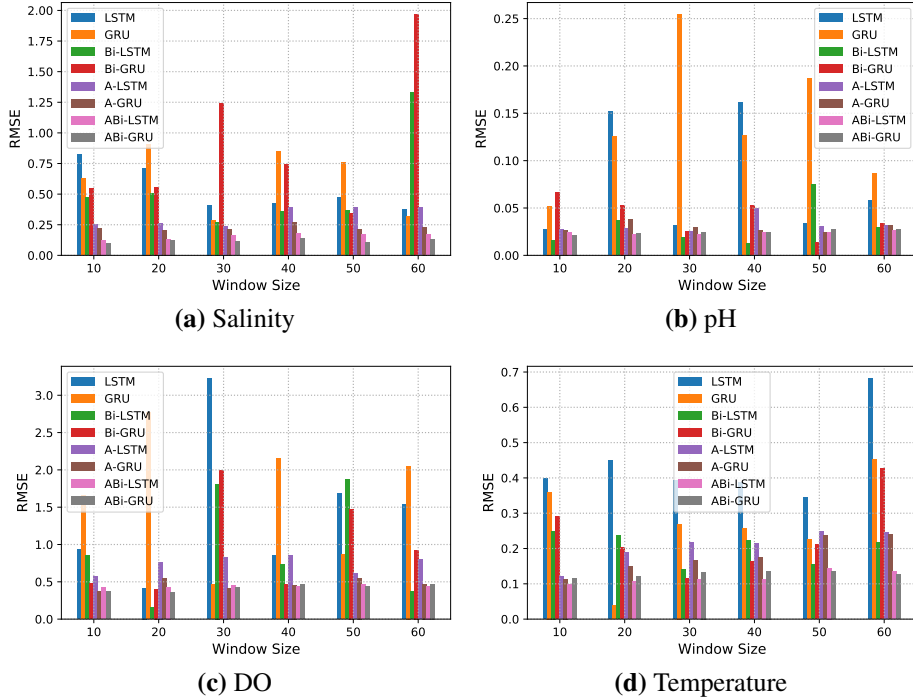
#### 6.3.4 Window size ( $W_z$ )

$W_z$  is also known as a look back.  $W_z$  is the number of samples used to predict one instance. In this experiment, the performance of each DL-RNN model while selecting different  $W_z$  is analysed. Figure 6.9 shows the RMSE bar plots for different  $W_z$  for predicting each WQ parameter on the ADAK WQ dataset using all DL-RNN models. The ABD-RNN models outperform all other models. The accuracy of the baseline models improves the  $W_z$  is increased, indicating baseline models require more resources for better prediction accuracy. In this experiment, baseline model accuracy is less than other DL-RNN models for different  $W_z$ .

The same experiment is done with the MAC WQ dataset to evaluate the performance further. The RMSE bar plots for various  $W_z$  assess the prediction accuracy of four WQ parameters of the MAC dataset with all DL-RNN models are shown in Fig. 6.8. Furthermore, it can be observed from Fig. 6.10 that the bidirectional models consistently perform well compared to the other models. As this experiment advances, the  $W_z$  can be chosen as 30 for all models as all the models achieved good prediction accuracy on both datasets. In addition, it can be observed from the experiments that the choice of  $W_z$  influences the prediction accuracy and computation time of all DL-RNN



**Figure 6.9:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $W_z$  and fixed  $h_p$  ( $L_r = 0.001$ ,  $E_p = 10$ ,  $B_z = 64$  and  $O_p$  is Adam) on the ADAK water quality dataset



**Figure 6.10:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $W_z$  and fixed  $h_p$  ( $L_r = 0.001$ ,  $E_p = 10$ ,  $B_z = 64$  and  $O_p$  is Adam) on the MAC water quality dataset

models. As the  $W_z$  increases required computation time also increases.

### 6.3.5 Batch size ( $B_z$ )

$B_z$  is the number of sub-samples that the NN uses to update the weights in each iteration and the right weight and gives the best results. The prediction performance of DL-RNN models with  $B_z \{16, 32, 64, 128, 256, 512\}$  is analysed. In Fig. 6.11, the RMSE bar plots indicating the prediction accuracy of each DL-RNN model with different  $B_z$  for four WQ parameters in the ADAK WQ dataset are shown. Compared to the proposed ABD-RNN models, it can be observed that the performance of the other DL-RNN models is inadequate.

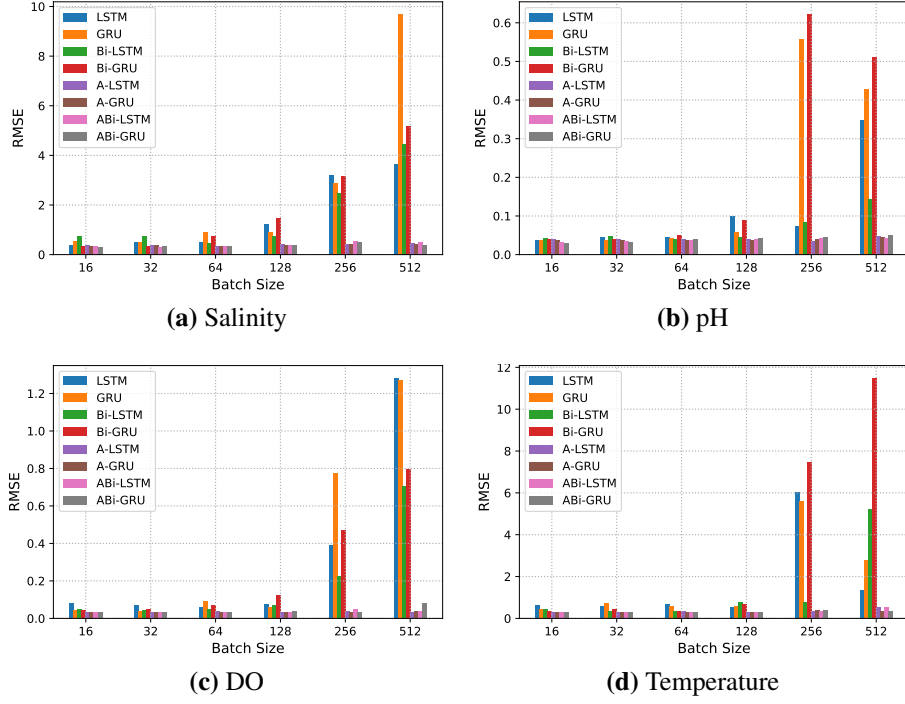
Figure 6.12 shows the RMSE bar plots indicating the prediction accuracy of each DL-RNN model with different  $B_z$  for four WQ parameters in the MAC WQ dataset. In this experiment, it can be observed that compared to ABD-RNN models, other DL-RNN models have less prediction accuracy for different  $B_z$ .

### 6.3.6 Computation time

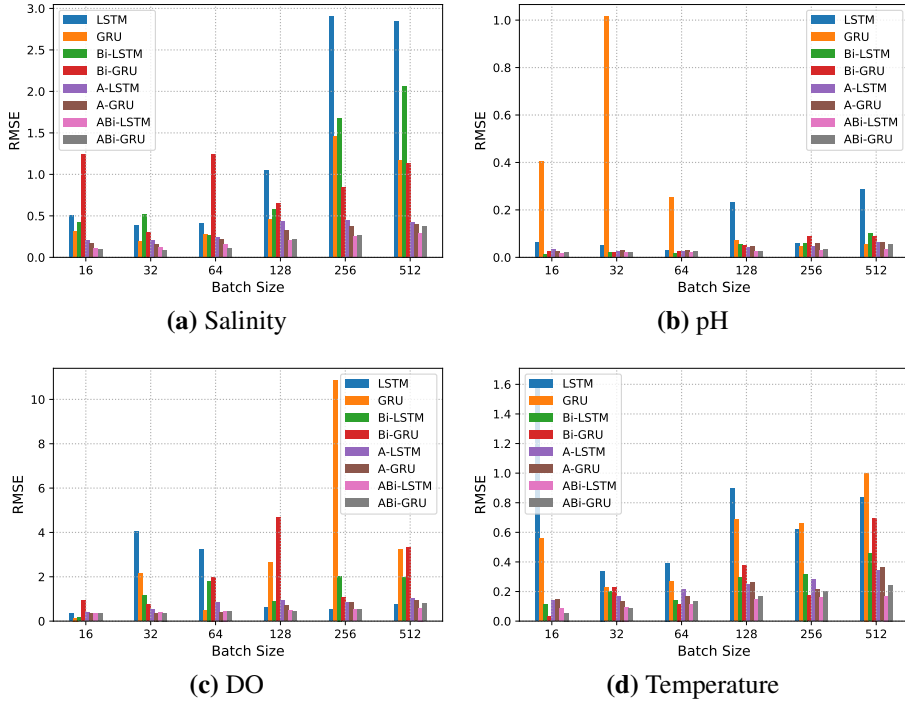
While studying the prediction performance by varying  $O_p$ ,  $L_r$ ,  $E_p$ ,  $W_z$ , and  $B_z$ , the computation time required for each model with each one of the  $h_p$  was also noted. Figure 6.13 and Fig. 6.14 show the bar plot for the computation time of each DL-RNN model for each set of  $h_p$  on the ADAK and MAC WQ dataset, respectively. From Fig. 6.13 (a) shows the computation time in seconds of each model for various  $L_r$ , it can be observed that the computation time is increasing with a decrease in  $L_r$ . From Fig. 6.13 (b), it can be observed that the computation time is increasing with an increase in  $E_p$ . A similar observation can be seen in Fig. 6.13 (c), where the computation time increases with an increase in  $W_z$ . In Fig. 6.13 (d), the computation time decreases with an increase in  $B_z$ . When considering  $O_p$ , it is evident that the computation time is the least for Adam  $O_p$ . This observation makes the selection of Adam a straightforward choice for all the experiments and in the results section. The bar plots in Fig. 6.14 of MAC data for computation time of various  $h_p$  help us conclude that the observation made in the ADAK data is valid.

## 6.4 RESULTS AND DISCUSSIONS

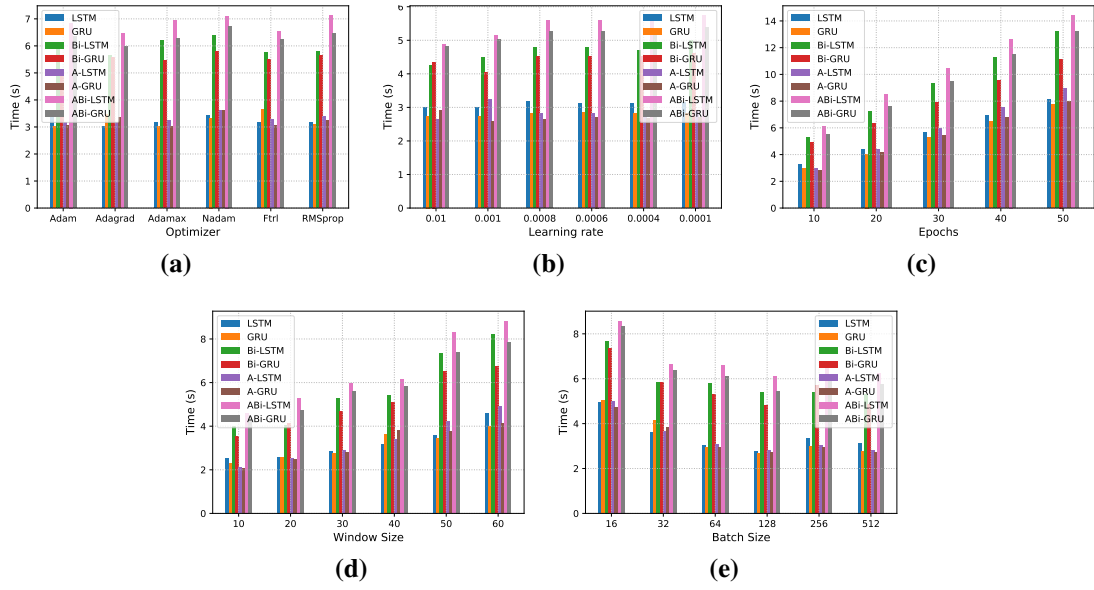
The objective is to finally analyse and report the prediction performance of the proposed ABD-RNN models and compare them with LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, and A-GRU DL-RNN models on both datasets. The performance comparison is made with a fixed set of  $h_p$  on the two WQ datasets. Each of the  $h_p$  is selected by studying the prediction performance and computation efficiency of the models for



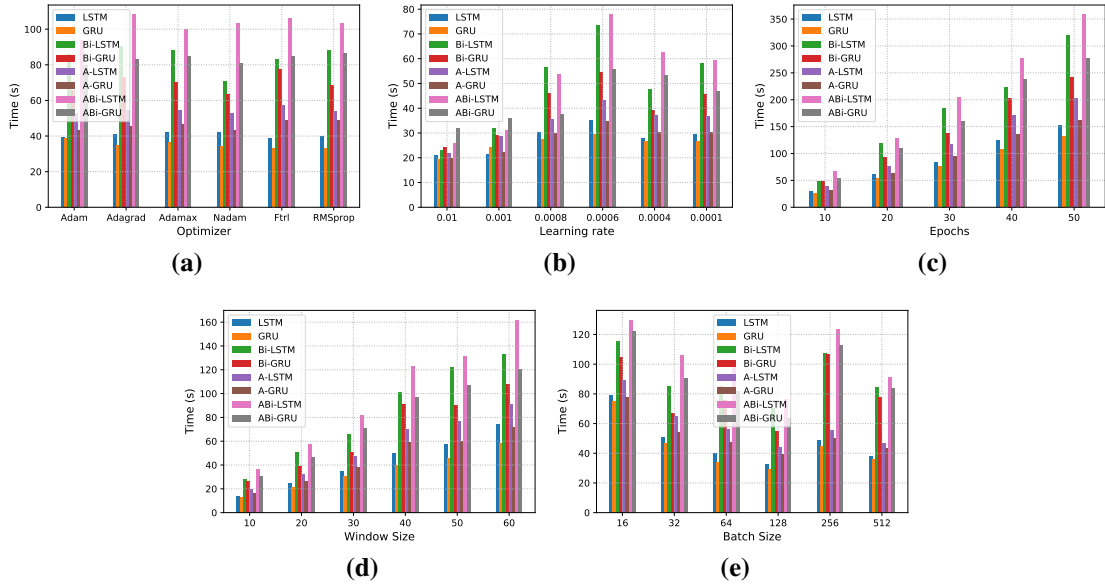
**Figure 6.11:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $B_z$  and fixed  $h_p$  ( $L_r = 0.001$ ,  $E_p = 10$ ,  $W_z = 30$ , and  $O_p$  is Adam) on the ADAK water quality dataset



**Figure 6.12:** Comparison of prediction accuracy (RMSE) of DL-RNN models for various  $L_r$  and fixed  $h_p$  ( $E_p = 10$ ,  $W_z = 30$ ,  $B_z = 64$  and  $O_p$  is Adam) on the ADAK water quality dataset



**Figure 6.13:** Comparison of computation time for prediction using all DL-RNN models. (a)  $O_p = \{\text{Adam, AdaGrad, AdaMAX, Nadam, Ftrl, RMSprop}\}$ , (b)  $L_r = \{0.01, 0.001, 0.0008, 0.0006, 0.0004, 0.0001\}$ , (c)  $E_p = \{10, 20, 30, 40, 50\}$ , (d)  $W_z = \{10, 20, 30, 40, 50, 60\}$ , and (e)  $B_z = \{16, 32, 64, 128, 256, 512\}$  on ADAM water quality dataset



**Figure 6.14:** Comparison of computation time for prediction using all DL-RNN models. (a)  $O_p = \{\text{Adam, AdaGrad, AdaMAX, Nadam, Ftrl, RMSprop}\}$ , (b)  $L_r = \{0.01, 0.001, 0.0008, 0.0006, 0.0004, 0.0001\}$ , (c)  $E_p = \{10, 20, 30, 40, 50\}$ , (d)  $W_z = \{10, 20, 30, 40, 50, 60\}$ , and (e)  $B_z = \{16, 32, 64, 128, 256, 512\}$  on MAC water quality dataset

each  $h_p$  ( $O_p$ ,  $L_r$ ,  $E_p$ ,  $W_z$  and  $B_z$ ) in the experiment section. This selected optimal set of  $h_p$  is used for comparing the performance of all the DL-RNN models in this section.

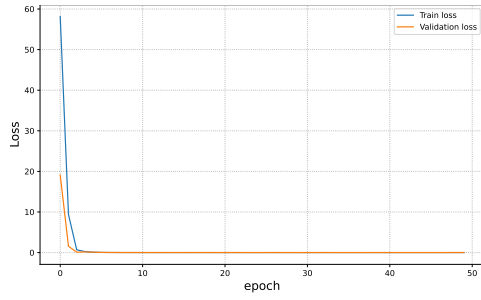
Figure 6.15 and Fig. 6.16 show the loss plots for convergence analysis of DL-RNN models. The  $h_p$  utilized in the training process include are  $E_p = 50$ ,  $L_r = 0.001$ ,  $W_z = 30$ ,  $B_z = 64$ , and  $O_p$  is Adam on both the ADAK and MAC water quality datasets. Examining the learning curves for each model, it is evident that the training loss progressively decreases and stabilizes. Additionally, the validation loss exhibits a similar trend of decreasing and stabilizing, with a slight gap with the training loss. The combined loss plots suggest that the DL-RNN models are in a balance between being overfitted and underfitted, signifying their effective ability to generalize across the data.

#### 6.4.1 ADAK water quality dataset

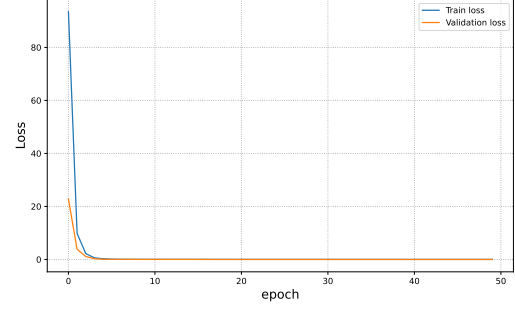
The prediction result of the ADAK WQ dataset is reported in Table 6.1, which gives MAE, MSE, RMSE and MAPE comparative analysis of the baseline DL-RNN models LSTM, GRU, bidirectional DL-RNN models Bi-LSTM, Bi-GRU, attention-based DL-RNN models A-LSTM, A-GRU and the proposed ABD-RNN models ABi-LSTM and ABi-GRU. Table 6.1 also reports the computation time required for training each model. Here, a fixed set of  $h_p$  is considered ( $E_p = 10, 50$ ), along with a learning rate of 0.001,  $B_z = 64$ ,  $W_z = 30$ , and  $O_p = \text{Adam}$ .

From Table 6.1, it is clear that the proposed ABD-RNN models have outperformed all other DL-RNN models in terms of prediction accuracy for all four WQ parameters. When comparing the training time required by all DL-RNN models, it becomes evident that the baseline DL-RNN models require less training time for the same epoch. But you can note that baseline DL-RNN models take more computation time to achieve the prediction accuracy of the proposed ABD-RNN models. Table 6.1 compares the results of all DL-RNN models for  $E_p = 10$  and 50 to substantiate this argument.

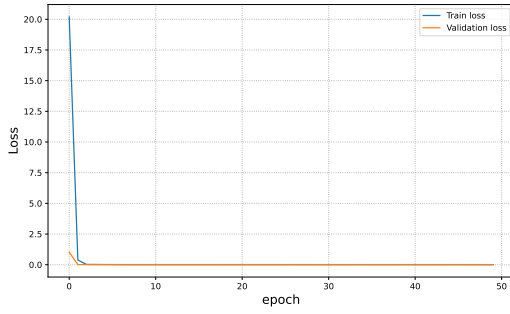
The prediction performance of salinity can be compared for  $E_p = 10$  and 50. In this comparison, the prediction results of the proposed DL-RNN models with  $E_p = 10$  are evaluated against all other DL-RNN models with  $E_p = 50$ . It is evident that for  $E_p = 50$ , the Bi-LSTM model with MAPE = 0.0146 demonstrates higher accuracy than the proposed DL-RNN models with  $E_p = 10$ . But the training time of the Bi-LSTM model with  $E_p = 50$  is 8.91s, which is almost double the training time of the proposed DL-RNN models with  $E_p = 10$ . The proposed ABi-LSTM achieves MAPE = 0.0180 with a much lesser training time of 5.05s. And the proposed ABi-GRU achieves MAPE = 0.0170 with a training time of 5.29s, saving more than 40% computational resources required for training the model.



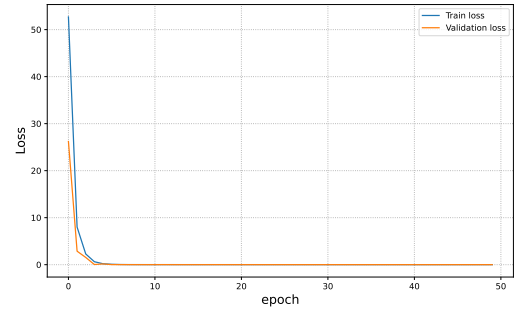
(a) LSTM



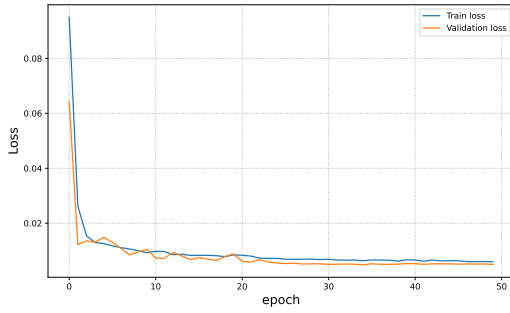
(b) GRU



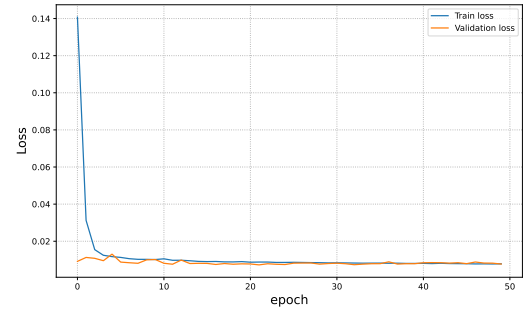
(c) Bi-LSTM



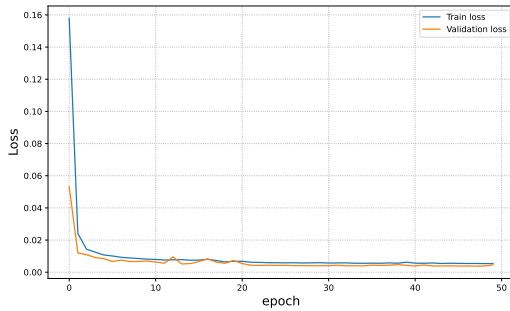
(d) Bi-GRU



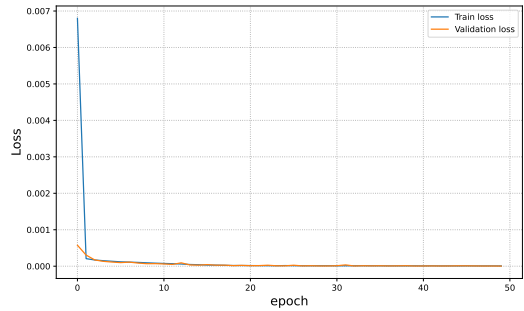
(e) A-LSTM



(f) A-GRU

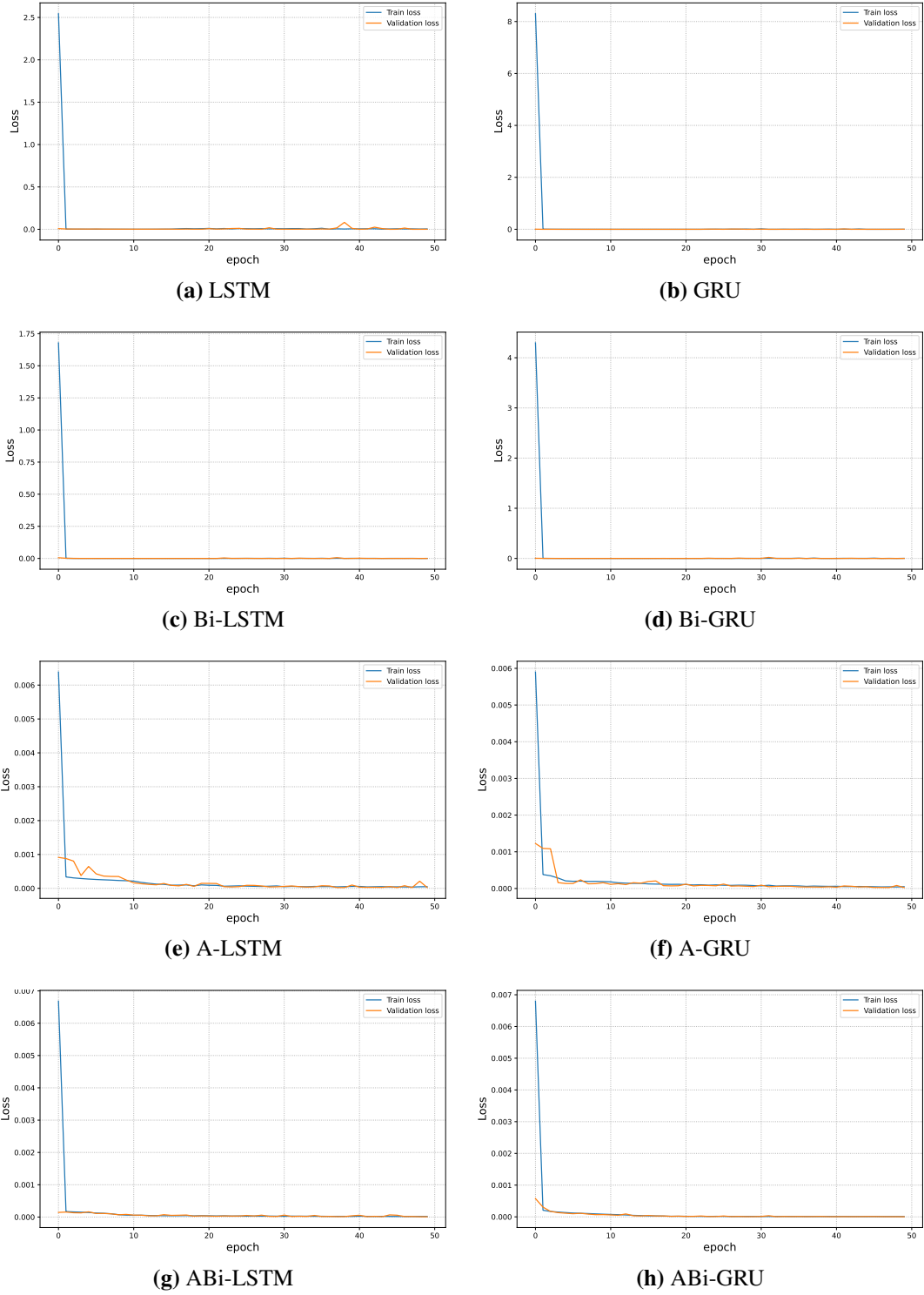


(g) ABi-LSTM



(h) ABi-GRU

**Figure 6.15:** Loss plots for DL-RNN convergence analysis using  $h_p$  ( $E_p = 50$ ,  $L_r = 0.001$ ,  $W_z = 30$ ,  $B_z = 64$ , and  $O_p$  with Adam) on the ADAK water quality dataset.



**Figure 6.16:** Loss plots for DL-RNN convergence analysis using  $h_p$  ( $E_p = 50$ ,  $L_r = 0.001$ ,  $W_z = 30$ ,  $B_z = 64$ , and  $O_p$  with Adam) on the MAC water quality dataset.

**Table 6.1:** Prediction accuracy of LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, A-GRU, ABi-LSTM and ABi-GRU DL models for ADAK water quality dataset

| Water Quality Parameter | Model    | Epochs = 10 |        |        |        |          | Epochs = 50 |        |        |        |          |
|-------------------------|----------|-------------|--------|--------|--------|----------|-------------|--------|--------|--------|----------|
|                         |          | MAE         | MSE    | RMSE   | MAPE   | Time (s) | MAE         | MSE    | RMSE   | MAPE   | Time (s) |
| Salinity(ppt)           | LSTM     | 0.3827      | 0.2238 | 0.4730 | 0.0247 | 2.82     | 0.3289      | 0.1756 | 0.4191 | 0.0212 | 7.01     |
|                         | GRU      | 0.5383      | 0.4695 | 0.6852 | 0.0350 | 2.87     | 0.2710      | 0.1180 | 0.3435 | 0.0175 | 6.80     |
|                         | Bi-LSTM  | 0.3157      | 0.1717 | 0.4144 | 0.0204 | 5.13     | 0.2257      | 0.0854 | 0.2923 | 0.0146 | 8.91     |
|                         | Bi-GRU   | 0.3816      | 0.2816 | 0.5307 | 0.0245 | 4.35     | 0.2665      | 0.1138 | 0.3374 | 0.0171 | 8.75     |
|                         | A-LSTM   | 0.2894      | 0.1542 | 0.3927 | 0.0186 | 2.83     | 0.2339      | 0.0932 | 0.3053 | 0.0151 | 6.71     |
|                         | A-GRU    | 0.2705      | 0.1334 | 0.3652 | 0.0175 | 3.19     | 0.2580      | 0.1137 | 0.3372 | 0.0167 | 7.50     |
|                         | ABi-LSTM | 0.2759      | 0.1247 | 0.3532 | 0.0180 | 5.05     | 0.2336      | 0.0939 | 0.3064 | 0.0151 | 10.02    |
| pH                      | ABi-GRU  | 0.2625      | 0.1245 | 0.3528 | 0.0170 | 5.29     | 0.2334      | 0.0923 | 0.3038 | 0.0152 | 10.31    |
|                         | LSTM     | 0.0628      | 0.0064 | 0.0800 | 0.0085 | 2.89     | 0.0246      | 0.0011 | 0.0328 | 0.0034 | 6.93     |
|                         | GRU      | 0.0355      | 0.0023 | 0.0479 | 0.0048 | 2.82     | 0.0330      | 0.0017 | 0.0410 | 0.0045 | 6.93     |
|                         | Bi-LSTM  | 0.0285      | 0.0013 | 0.0355 | 0.0039 | 4.48     | 0.0269      | 0.0012 | 0.0348 | 0.0037 | 9.65     |
|                         | Bi-GRU   | 0.0285      | 0.0014 | 0.0368 | 0.0039 | 4.52     | 0.0281      | 0.0013 | 0.0365 | 0.0038 | 9.18     |
|                         | A-LSTM   | 0.0342      | 0.0017 | 0.0416 | 0.0047 | 2.71     | 0.0245      | 0.0010 | 0.0317 | 0.0033 | 7.58     |
|                         | A-GRU    | 0.0318      | 0.0015 | 0.0389 | 0.0043 | 2.64     | 0.0242      | 0.0010 | 0.0318 | 0.0033 | 7.09     |
| DO(ml/L)                | ABi-LSTM | 0.0272      | 0.0012 | 0.0346 | 0.0037 | 5.05     | 0.0255      | 0.0011 | 0.0325 | 0.0035 | 9.92     |
|                         | ABi-GRU  | 0.0321      | 0.0015 | 0.0388 | 0.0044 | 5.09     | 0.0221      | 0.0009 | 0.0297 | 0.0030 | 9.88     |
|                         | LSTM     | 0.0395      | 0.0025 | 0.0505 | 0.0075 | 2.87     | 0.0315      | 0.0017 | 0.0417 | 0.0060 | 6.87     |
|                         | GRU      | 0.0379      | 0.0025 | 0.0498 | 0.0072 | 3.36     | 0.0248      | 0.0010 | 0.0318 | 0.0047 | 7.72     |
|                         | Bi-LSTM  | 0.0403      | 0.0028 | 0.0534 | 0.0076 | 4.39     | 0.0313      | 0.0016 | 0.0399 | 0.0059 | 9.20     |
|                         | Bi-GRU   | 0.0299      | 0.0014 | 0.0379 | 0.0057 | 4.57     | 0.0239      | 0.0010 | 0.0310 | 0.0045 | 9.22     |
|                         | A-LSTM   | 0.0266      | 0.0011 | 0.0336 | 0.0051 | 2.74     | 0.0266      | 0.0011 | 0.0331 | 0.0051 | 6.88     |
| Temperature(°C)         | A-GRU    | 0.0285      | 0.0012 | 0.0353 | 0.0054 | 2.61     | 0.0246      | 0.0010 | 0.0321 | 0.0047 | 6.72     |
|                         | ABi-LSTM | 0.0268      | 0.0011 | 0.0334 | 0.0051 | 5.61     | 0.0240      | 0.0010 | 0.0312 | 0.0046 | 11.47    |
|                         | ABi-GRU  | 0.0286      | 0.0012 | 0.0353 | 0.0054 | 5.17     | 0.0246      | 0.0010 | 0.0315 | 0.0047 | 10.23    |
|                         | LSTM     | 0.3536      | 0.2130 | 0.4615 | 0.0141 | 3.32     | 0.3893      | 0.2640 | 0.5138 | 0.0153 | 7.94     |
|                         | GRU      | 0.5352      | 0.4440 | 0.6663 | 0.0213 | 2.86     | 0.3217      | 0.1881 | 0.4337 | 0.0127 | 7.42     |
|                         | Bi-LSTM  | 0.2752      | 0.1187 | 0.3445 | 0.0109 | 4.23     | 0.2699      | 0.1161 | 0.3408 | 0.0107 | 10.68    |
|                         | Bi-GRU   | 0.2814      | 0.1270 | 0.3564 | 0.0112 | 4.72     | 0.2431      | 0.0976 | 0.3124 | 0.0096 | 10.46    |
|                         | A-LSTM   | 0.2509      | 0.0990 | 0.3147 | 0.0100 | 2.83     | 0.2437      | 0.0939 | 0.3065 | 0.0097 | 8.72     |
|                         | A-GRU    | 0.2550      | 0.1088 | 0.3299 | 0.0101 | 2.70     | 0.2427      | 0.0962 | 0.3101 | 0.0096 | 7.79     |
|                         | ABi-LSTM | 0.2376      | 0.0889 | 0.2982 | 0.0094 | 5.59     | 0.2432      | 0.0958 | 0.3095 | 0.0096 | 10.94    |
|                         | ABi-GRU  | 0.2410      | 0.0938 | 0.3062 | 0.0095 | 5.12     | 0.2467      | 0.0992 | 0.3150 | 0.0098 | 10.55    |

$h_p : E_p = \{10, 50\}$ ,  $L_r = 0.001$ ,  $W_z = 30$ ,  $B_z = 64$ , and Adam optimizer.

Further, the prediction performance of pH for  $E_p = 10$  and 50 is compared. It can be observed that for  $E_p = 50$ , the A-LSTM DL-RNN model with MAPE = 0.0033 has better accuracy than the proposed DL-RNN models with  $E_p = 10$ . But with a training time of 7.58s. The proposed ABi-LSTM achieves MAPE = 0.0037 with a much lesser training time of 5.05s. And the proposed ABi-GRU achieves MAPE = 0.0044 with a training time of 5.09s, saving almost 25% of the computational resource required for training the model.

When making a similar comparison for DO, it is observed that the Bi-GRU model with  $E_p = 50$  exhibits better prediction performance compared to the proposed models with  $E_p = 10$ . The Bi-GRU model achieves a MAPE of 0.0045, whereas the proposed model achieves a MAPE of 0.0051 for  $E_p = 10$ , with a computation time of only 5.61s. However, the computation time for the Bi-GRU DL-RNN model is 9.22s, resulting in a 40% reduction in computational resources required for training the model. Furthermore, when comparing the prediction performance for temperature, it is observed that the A-GRU model with  $E_p = 50$  closely matches the performance of the proposed DL-RNN models with  $E_p = 10$ . The A-GRU models achieve a MAPE of 0.0096 with

a training time of 7.79s. In comparison, the proposed models achieve a slightly better MAPE of 0.0094 with a training time of 5.59s, thereby saving almost 30% of computational resources. Based on the information provided in Table 6.1, it can be concluded that the proposed ABD-RNN models outperform other DL-RNN models in terms of prediction accuracy while requiring fewer computational resources.

#### 6.4.2 MAC water quality dataset

The prediction performance of the MAC WQ dataset is reported in Table 6.2, which gives MAE, MSE, RMSE and MAPE comparative analysis of the proposed ABD-RNN models (ABi-LSTM and ABi-GRU) with baseline DL-RNN models (LSTM and GRU), bidirectional DL-RNN models (Bi-LSTM and Bi-GRU), and attention-based DL-RNN models (A-LSTM and A-GRU). Table 6.2 also reports the computation time required for training each model in seconds. Here, a fixed set of  $h_p$  is considered ( $E_p = 10, 50$ ), along with a learning rate of 0.001,  $B_z = 64$ ,  $W_z = 30$ , and  $O_p = \text{Adam}$ .

**Table 6.2:** Prediction accuracy of LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, A-GRU, ABi-LSTM and ABi-GRU DL models for MAC water quality dataset

| Water Quality Parameter           | Model    | Epochs = 10 |        |        |        |          | Epochs = 50 |        |        |        |          |
|-----------------------------------|----------|-------------|--------|--------|--------|----------|-------------|--------|--------|--------|----------|
|                                   |          | MAE         | MSE    | RMSE   | MAPE   | Time (s) | MAE         | MSE    | RMSE   | MAPE   | Time (s) |
| Salinity(ppt)                     | LSTM     | 0.1578      | 0.1737 | 0.4168 | 0.0047 | 24.62    | 0.1167      | 0.1081 | 0.3288 | 0.0035 | 108.87   |
|                                   | GRU      | 0.1011      | 0.1057 | 0.3251 | 0.0030 | 33.30    | 0.3251      | 0.6111 | 0.7817 | 0.0096 | 103.87   |
|                                   | Bi-LSTM  | 0.0984      | 0.0402 | 0.2005 | 0.0029 | 62.73    | 0.0358      | 0.0077 | 0.0876 | 0.0011 | 143.78   |
|                                   | Bi-GRU   | 0.2677      | 1.7427 | 1.3201 | 0.0083 | 52.17    | 0.2514      | 0.4023 | 0.6343 | 0.0075 | 163.78   |
|                                   | A-LSTM   | 0.1898      | 0.1822 | 0.4268 | 0.0057 | 37.63    | 0.1010      | 0.0314 | 0.1771 | 0.0030 | 147.60   |
|                                   | A-GRU    | 0.1223      | 0.0651 | 0.2551 | 0.0036 | 30.71    | 0.0773      | 0.0255 | 0.1598 | 0.0023 | 117.75   |
|                                   | ABi-LSTM | 0.0625      | 0.0153 | 0.1237 | 0.0018 | 60.70    | 0.0263      | 0.0078 | 0.0882 | 0.0008 | 163.99   |
|                                   | ABi-GRU  | 0.0709      | 0.0151 | 0.1228 | 0.0021 | 50.08    | 0.0223      | 0.0038 | 0.0617 | 0.0007 | 149.01   |
| pH                                | LSTM     | 0.0258      | 0.0033 | 0.0574 | 0.0030 | 30.80    | 0.0067      | 0.0001 | 0.0102 | 0.0008 | 109.21   |
|                                   | GRU      | 0.1119      | 0.1150 | 0.3392 | 0.0127 | 28.23    | 0.1871      | 0.1310 | 0.3619 | 0.0217 | 103.39   |
|                                   | Bi-LSTM  | 0.0174      | 0.0007 | 0.0256 | 0.0020 | 53.47    | 0.0122      | 0.0004 | 0.0205 | 0.0014 | 254.55   |
|                                   | Bi-GRU   | 0.0221      | 0.0034 | 0.0581 | 0.0025 | 45.57    | 0.0341      | 0.0123 | 0.1111 | 0.0039 | 215.89   |
|                                   | A-LSTM   | 0.0151      | 0.0006 | 0.0244 | 0.0018 | 32.96    | 0.0111      | 0.0005 | 0.0216 | 0.0013 | 166.99   |
|                                   | A-GRU    | 0.0153      | 0.0007 | 0.0269 | 0.0018 | 27.60    | 0.0104      | 0.0004 | 0.0210 | 0.0012 | 148.13   |
|                                   | ABi-LSTM | 0.0120      | 0.0005 | 0.0226 | 0.0014 | 55.51    | 0.0089      | 0.0004 | 0.0198 | 0.0010 | 243.47   |
|                                   | ABi-GRU  | 0.0117      | 0.0007 | 0.0262 | 0.0014 | 35.14    | 0.0087      | 0.0005 | 0.0232 | 0.0010 | 158.24   |
| DO(ml/L)                          | LSTM     | 0.8591      | 1.7852 | 1.3361 | 0.2111 | 23.52    | 0.1778      | 0.0544 | 0.2332 | 0.0496 | 110.41   |
|                                   | GRU      | 0.2744      | 0.4710 | 0.6863 | 0.0503 | 22.57    | 0.1829      | 0.1879 | 0.4335 | 0.0384 | 101.92   |
|                                   | Bi-LSTM  | 0.3547      | 0.7464 | 0.8639 | 0.0601 | 43.22    | 0.1138      | 0.0308 | 0.1754 | 0.0271 | 248.23   |
|                                   | Bi-GRU   | 0.6842      | 4.1303 | 2.0323 | 0.0920 | 34.57    | 0.4389      | 1.9743 | 1.4051 | 0.0524 | 176.40   |
|                                   | A-LSTM   | 0.6402      | 1.0682 | 1.0336 | 0.1486 | 30.34    | 0.1875      | 0.1154 | 0.3397 | 0.0419 | 158.22   |
|                                   | A-GRU    | 0.2174      | 0.2372 | 0.4870 | 0.0369 | 30.88    | 0.1642      | 0.1405 | 0.3749 | 0.0296 | 114.72   |
|                                   | ABi-LSTM | 0.2088      | 0.1667 | 0.4083 | 0.0435 | 69.19    | 0.1577      | 0.0739 | 0.2719 | 0.0404 | 150.57   |
|                                   | ABi-GRU  | 0.1335      | 0.1081 | 0.3288 | 0.0237 | 59.26    | 0.1062      | 0.1090 | 0.3301 | 0.0157 | 178.27   |
| Temperature( $^{\circ}\text{C}$ ) | LSTM     | 0.7791      | 0.7409 | 0.8608 | 0.0284 | 32.62    | 1.0359      | 1.0853 | 1.0418 | 0.0379 | 116.69   |
|                                   | GRU      | 0.1330      | 0.0390 | 0.1974 | 0.0048 | 27.19    | 0.4753      | 0.2287 | 0.4782 | 0.0174 | 119.78   |
|                                   | Bi-LSTM  | 0.1445      | 0.0489 | 0.2212 | 0.0053 | 56.01    | 0.0885      | 0.0087 | 0.0934 | 0.0032 | 246.85   |
|                                   | Bi-GRU   | 0.0776      | 0.0122 | 0.1103 | 0.0028 | 45.06    | 0.0614      | 0.0042 | 0.0651 | 0.0022 | 193.02   |
|                                   | A-LSTM   | 0.0800      | 0.0143 | 0.1194 | 0.0029 | 35.81    | 0.0532      | 0.0070 | 0.0839 | 0.0019 | 130.40   |
|                                   | A-GRU    | 0.1082      | 0.0294 | 0.1716 | 0.0039 | 30.23    | 0.1042      | 0.0176 | 0.1325 | 0.0038 | 108.78   |
|                                   | ABi-LSTM | 0.0774      | 0.0152 | 0.1232 | 0.0028 | 56.36    | 0.0297      | 0.0032 | 0.0565 | 0.0011 | 200.36   |
|                                   | ABi-GRU  | 0.0803      | 0.0163 | 0.1278 | 0.0029 | 50.13    | 0.0204      | 0.0025 | 0.0497 | 0.0007 | 167.30   |

$h_p : E_p = \{10, 50\}$ ,  $L_r = 0.001$ ,  $W_z = 30$ ,  $B_z = 64$ , and Adam optimizer.

From Table 6.2, it is clear that the proposed ABD-RNN models have outperformed all other DL-RNN models in terms of prediction accuracy for all four WQ parameters. When comparing the training time required by all DL-RNN models, it becomes evident

that the baseline DL-RNN models need less training time for the same epoch. It can be noted that baseline DL-RNN models take more computation time to achieve the prediction accuracy of the proposed ABD-RNN models. Table 6.2 compares the results of all DL-RNN models for  $E_p$  10 and 50 to substantiate this argument.

The prediction performance of salinity for  $E_p = 10$  and 50 is compared. In this comparison, the prediction results of proposed DL-RNN models with  $E_p = 10$  and the prediction results of all other DL-RNN models with  $E_p = 50$ . It is evident that for  $E_p = 50$ , the Bi-LSTM model with MAPE = 0.0011 has better accuracy than the proposed DL-RNN models with  $E_p = 10$ . But the training time of the Bi-LSTM model with  $E_p = 50$  is 143.78s, which is more than double the training time of the proposed DL-RNN models with  $E_p = 10$ . The proposed ABi- LSTM achieves MAPE = 0.0018 with a much lesser training time of 60.70s. And the proposed ABi-GRU achieves MAPE = 0.0021 with an even lesser training time of 50.08s, saving more than 50% computational resources required for training the model.

Further, the prediction performance of pH for  $E_p = 10$  and 50 is compared. it can be observed that for  $E_p = 50$ , the LSTM DL-RNN model with MAPE = 0.0008 has better accuracy than the proposed DL-RNN models with  $E_p = 10$  and 50. But with a training time of 109.21s, almost double the training time of the proposed DL-RNN models for  $E_p = 10$ . Another DL-RNN model is A-GRU achieves a MAPE = 0.0012 but requires a training time of 148.13s. The proposed ABi- LSTM achieves MAPE = 0.0014 with a much lesser training time of 55.51s. And the proposed ABi-GRU achieves MAPE = 0.0014 with an even lesser training time of 35.14s, saving more than 65% of the computational resource required for training the model.

When making a similar comparison for DO, it can be observed that Bi-LSTM with  $E_p = 50$  is the only model that comes close to the prediction performance of proposed models for  $E_p = 10$ . Here Bi-LSTM has a MAPE = 0.0271, whereas the proposed model achieves a MAPE = 0.0237 for  $E_p = 10$  and with just a computation time of 59.26s. But the computation time of the Bi-LSTM DL-RNN model is 248.23s which is almost four times the computation time required by proposed models, saving 75% of computational resources for training the model. When comparing the prediction performance for temperature, it is observed that the Bi-GRU model with  $E_p = 50$  comes close to the prediction performance of the proposed DL-RNN models with  $E_p = 10$ . The Bi-GRU models achieve MAPE = 0.0022 but with a training time of 193.02s. In comparison, the proposed models achieve MAPE = 0.0029 with a training time of just 56.36s saving almost 70% computational resources. From Table 6.2, it can be concluded that the proposed ABD-RNN models have outperformed all other DL-RNN models in prediction accuracy and use fewer computational resources.

In the evaluation of WQP models on the ADAK and MAC datasets, the proposed ABD-RNN models, namely ABi-LSTM and ABi-GRU, consistently outperformed all other baseline and attention-based DL-RNN models in terms of prediction accuracy for various water quality parameters, including salinity, pH, DO, and temperature. The ABD-RNN models achieved lower MAE, MSE, RMSE, and MAPE values, indicating their superior predictive capabilities. Despite requiring slightly longer training times, the ABD-RNN models demonstrated significant advantages in computational efficiency compared to some baseline models, making them an attractive choice for accurate water quality prediction tasks. This performance improvement signifies the potential of ABD-RNN models in enhancing water quality monitoring and management systems.

## 6.5 SUMMARY

This chapter introduces the ABD-RNN approach (ABD-RNN) with Adam optimization, aimed at enhancing the accuracy of aquaculture WQP. The proposed models incorporate bidirectional DL-RNN layers to capture the sequence correlation of the data effectively, while the attention mechanism assigns varying weights to focus on crucial data points. A comprehensive investigation is conducted on various DL-RNN models to assess the impact of different hyperparameters. Based on this study, optimal values for  $h_p$  are determined and utilized for performance comparison and further analysis in the results section. The outcomes demonstrate that the proposed ABD-RNN models, namely ABi-LSTM and ABi-GRU, surpass other DL-RNN models, including LSTM, GRU, Bi-LSTM, Bi-GRU, A-LSTM, and A-GRU, in terms of both prediction accuracy and computational efficiency. These findings emphasize the effectiveness of the ABD-RNN approach for aquaculture WQP tasks.

## CHAPTER 7

### CONCLUSIONS

This Chapter is organized as follows: Section 7.1 presents the summary of this thesis. Section 7.2 explains the applications of the proposed methodologies and Section 7.2 presents future research directions for this research work.

#### 7.1 SUMMARY OF THESIS

The research work presented here provides a series of novel approaches for WQM and WQP systems. The use of virtual sensors combined with DL models has been thoroughly explored, resulting in a set of models that replace the traditional WQM systems. The first part of the research work introduced the DL Enhanced Virtual Sensing (DLEVS) mechanism, a novel approach for monitoring DO levels that combines the benefits of LSTM DL-RNN with the strengths of Virtual Sensing (VS) techniques. The model proved effective in providing accurate and reliable estimations of DO levels, demonstrating the promising potential of DL models in virtual sensing, especially in WQM. In the second part of the work, a detailed evaluation of standalone DL models was carried out, identifying their strengths and limitations. Through the evaluation, it was found that none of the models significantly outperformed the others across all WQ parameters for different datasets, highlighting the need for more sophisticated models. This laid the groundwork for the introduction of hybrid models and attention based models, which leverage the strengths of individual models to achieve superior predictive performance.

The third part of the research provides an introduction of hybrid CNN-LSTM and CNN-GRU models, which combined the strengths of LSTM and GRU models with the robust feature extraction capabilities of Convolutional Neural Networks (CNNs). The hybrid models demonstrated superior performance in terms of prediction accuracy and computation efficiency, proving their efficacy in the realm of WQP. Finally, in the fourth part of the research, attention-based models, namely ABi-LSTM and ABi-GRU, were introduced and tested. These models integrated the advantages of LSTM, GRU, and bidirectional models with the ability of an attention mechanism to focus on crucial data by assigning different weights. The attention-based models achieved optimal results in WQP tasks, significantly enhancing the accuracy of prediction and computation efficiency. The research has successfully introduced and validated several

novel models for WQM and WQP. These models could be used as a tool to enhance the efficiency and reliability of monitoring systems in such environments.

## **7.2 FUTURE WORK**

The research presented in this thesis opens up several potential avenues for future work in the field of WQ management. Despite the fact that the offered methodologies have demonstrated improvement over prior benchmarks, there is still scope for improvement based on these new viewpoints to attain more efficiency and reliability. The following are some future research directions that can be pursued to further enhance the methodologies and applications discussed in this thesis:

- Investigate the application of transfer learning techniques in the context of WQM and WQP. Transfer learning can leverage pre-trained DL models on large-scale datasets to improve the performance of models in scenarios with limited labeled data. By fine-tuning pre-trained models or using them as feature extractors, it may be possible to enhance the accuracy and generalization capabilities of the proposed methodologies.
- Explore the integration of other environmental parameters and factors into the DL models. WQ in aquaculture systems is influenced by various factors such as pH, turbidity, ammonia levels, and nutrient concentrations. Investigate the feasibility of incorporating these additional parameters into the DL models to provide a more comprehensive and holistic assessment of WQ.
- Conduct further research on data augmentation techniques specifically tailored for WQ data. Data augmentation can help alleviate data scarcity issues and improve the robustness of DL models. Investigate methods to generate realistic synthetic data that capture the characteristics of different WQ conditions, thereby expanding the training dataset and enhancing model performance.

Exploring these future research directions will help in advancing the field of DL-based WQM and WQP, ultimately contributing to more efficient, accurate, and sustainable aquaculture WQ management practices.

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## LIST OF PAPERS BASED ON THESIS

### Patents

1. Harigovindan V. P. and Rasheed Abdul Haq K. P., “VIRTUAL SENSING SYSTEM FOR WATER QUALITY MONITORING AND METHOD THEREOF”, Indian Patent 397595, June 10,2021. ([Granted on: May 25, 2022](#))
2. Harigovindan V. P. and Rasheed Abdul Haq K. P., “HYBRID DEEP LEARNING BASED METHOD AND SYSTEM FOR AQUACULTURE WATER QUALITY PREDICTION”, Indian Patent Application No.202341015286 A. ([Published on: March 17, 2023](#))
3. Harigovindan V. P. and Rasheed Abdul Haq K. P., “DEEP LEARNING BASED VIRTUAL SENSING METHOD AND SYSTEM FOR LOW-COST AQUACULTURE WATER QUALITY MONITORING”, Indian Patent Application No.202341015467 A. ([Published on: May 5, 2023](#))

### Referred Journals

1. Rasheed Abdul Haq K. P. and Harigovindan V. P., “Water Quality Prediction for Smart Aquaculture Using Hybrid Deep Learning Models”, *IEEE Access*, vol. 10, pp. 60078-60098, June 2022. (SCI Indexed IF: 3.476)  
<https://doi.org/10.1109/ACCESS.2022.3180482>
2. Rasheed Abdul Haq K. P. and Harigovindan V. P., “Water Quality Prediction System based on Adam Optimised LSTM Neural Network for Aquaculture: A Case Study in Kerala, India”, *J. Inst. Eng. India Ser. B, Springer*, 103, 2177–2188, October 2022. (Scopus Indexed)  
<https://doi.org/10.1007/s40031-022-00806-7>

### Journals under review

1. Rasheed Abdul Haq K. P. and Harigovindan V. P. “Deep Learning Enhanced Virtual Sensing: A Cost-Effective Approach for Water Quality Monitoring in Aquaculture” (IEEE Transactions on Instrumentation and Measurement, Status: Submitted)

2. Rasheed Abdul Haq K. P., Harigovindan V. P. and Manoj P. Samuel, “ABD-RNN: Attention based Bi-directional Deep Learning Recurrent Neural Networks for Aquaculture Water Quality Prediction” (Expert Systems with Applications, Elsevier, Status: Required Reviews completed)

### **Book Chapter**

1. Rasheed Abdul Haq K. P. and Harigovindan V. P., “DHRNN: Deep Hybrid Recurrent Neural Networks for Aquaculture Water Quality Prediction”, *Advances in Communication, Devices and Networking*. Springer, December 2022.  
<https://doi.org/10.1007/978-981-99-1983-3>

### **Presentations in Conferences**

1. Rasheed Abdul Haq K. P. and Harigovindan V. P., “Review on Water Quality Monitoring Systems for Aquaculture”, *Emerging Trends in Computing and Expert Technology. COMET 2019*. Lecture Notes on Data Engineering and Communications Technologies, vol 35, Springer, Cham, November 2019.  
[https://doi.org/10.1007/978-3-030-32150-5\\_71](https://doi.org/10.1007/978-3-030-32150-5_71)
2. Rasheed Abdul Haq K. P. and Harigovindan V. P., “A review on WSN based energy-efficient water quality monitoring and reporting systems for aquaculture”, *Proceedings of National Conference on Communication Systems (NCSOCS-2019)*, NIT Puducherry, September 2019.
3. Rasheed Abdul Haq K. P. and Harigovindan V. P., “Scheduling based Energy Efficient Water Quality Monitoring System for Aquaculture”, In: Dave, M., Garg, R., Dua, M., Hussien, J. (eds) *Proceedings of the International Conference on Paradigms of Computing, Communication and Data Sciences. Algorithms for Intelligent Systems*. Springer, February 2021.  
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4. Rasheed Abdul Haq K. P. and Harigovindan V. P., “Water Quality Monitoring and Reporting System for Aquaculture with Solar Energy Harvesting”, *2020 International Conference on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE)*, Bengaluru, India, pp. 66-70, July 2020.  
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## **BRIEF CURRICULUM VITAE**

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